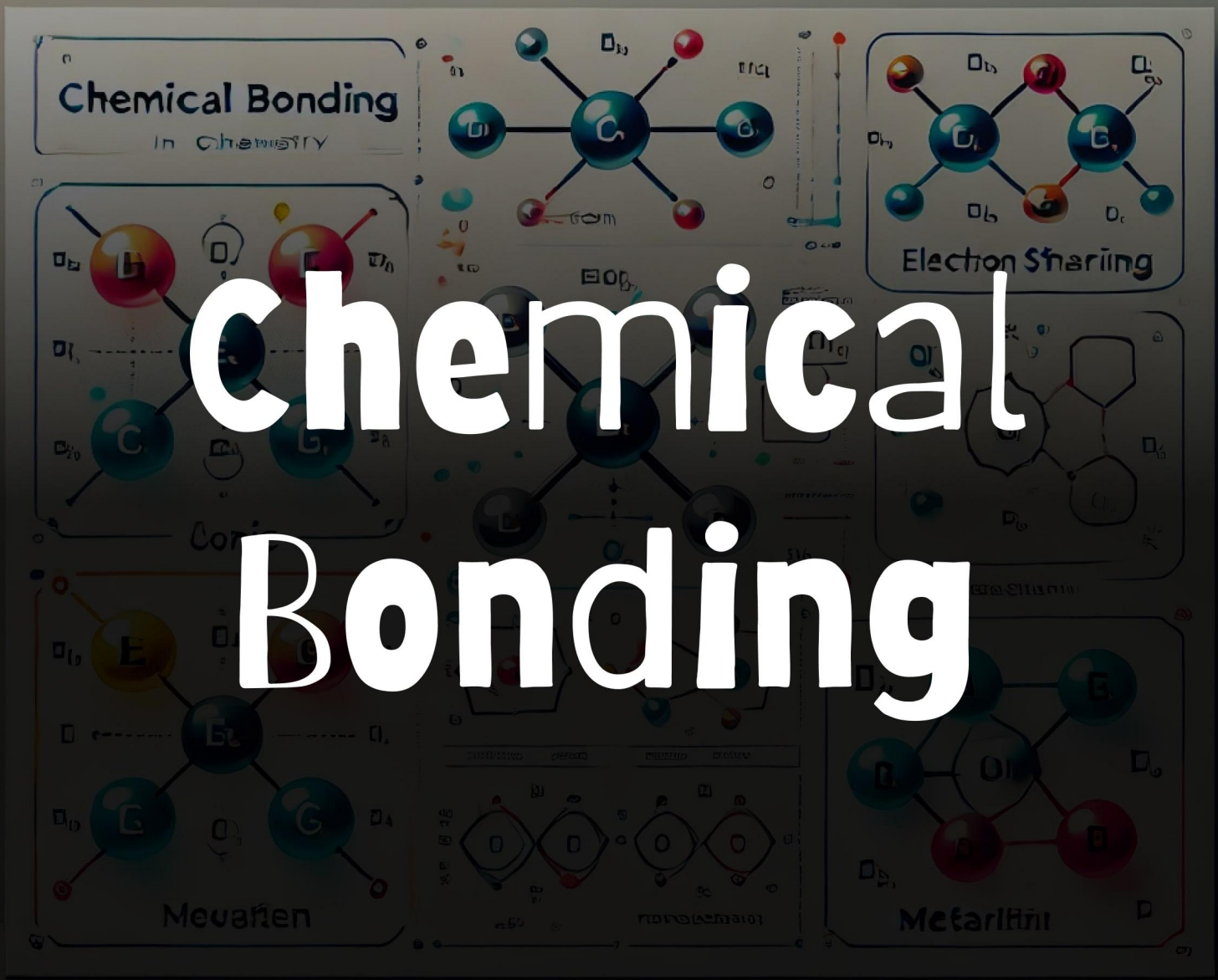


Chemical Bonding

CHEMICAL BONDING

{LET'S PLAY BOND-BOND}

What is chemical Bond?

Force of attraction between 2 atoms which hold them together in a molecule is known as chemical bond.

Types of chemical Bond

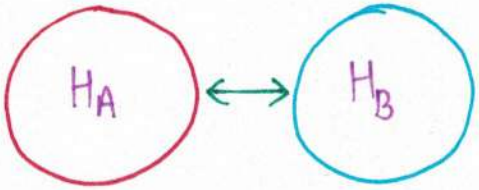
- chemical bond
  - Strong Bond
    - Electrovalent/ionic bond
    - Covalent bond
    - Co-ordinate bond
    - Metallic bond
  - Weak Bond
    - Hydrogen bond
    - Vander Waals forces

Condition Required to form bonds

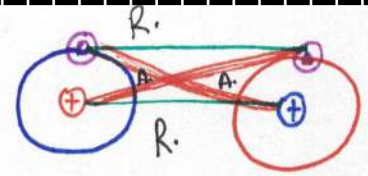
Condition to require chemical bonding of elements :-

a. to acquire stability or to get inert gas configuration

b. to acquire lower energy state.



H_2

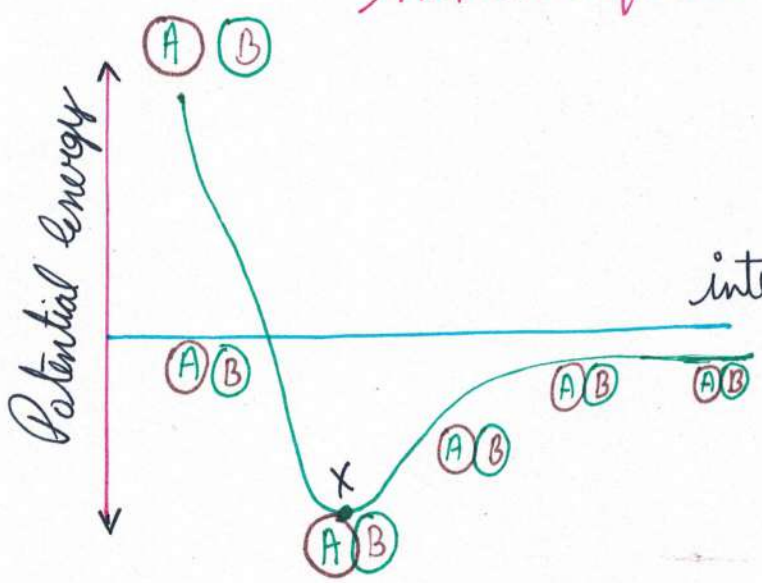


Repulsion ← [Nucleus A - Nucleus B
Electron A - Electron B

Attraction ← [Nucleus A - Electron B
Nucleus B - Electron A

Chemical bond is formed when
Attractive force > Repulsion force

Energy Diagram



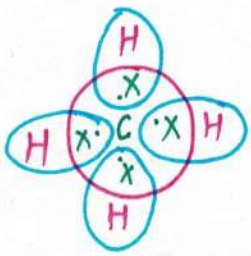
at pt. X bond is formed b/w atom A & B.

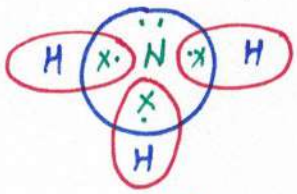
LEVI'S OCTAL LAW

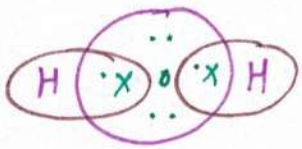
According to Levi's, every element want to get stable electronic configuration [inert gas confi] because noble gases are stable in nature. They are inert to chemical reaction or bonding to achieve inert gas confi element gains electron, loses electron or shares electron.

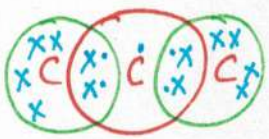
1. Transfer of e⁻s b/w two atom → Electrovalent / Ionic Bond
2. Sharing of electron → Covelent Bond

@ Equal sharing → Covelent Bond
ⓑ Unequal sharing → Co-ordinate Bond


CH methane


NH_3 ammonia

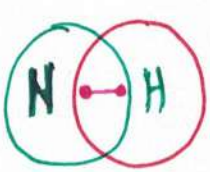

H_2O Water

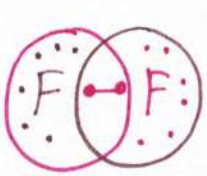

CO_2 Carbon dioxide

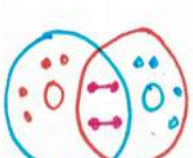
Lewis-Dot-Structure

In this structure, the no. of valence shell electrons in an element are represented by dots or crosses.

Covelent Bond

duplet 
H_2 single bond
H-F sharing of $2e^-$

Octel 
F_2 single bond
F-F

Octel 
O_2 double bond
O=O sharing of $4e^-$

Octel
N_2 triple bond
N≡N sharing of $6e^-$

Types of Electronic Configuration

A. INERT GAS CONFIGURATION

noble gas → ns^2np^6 [except He = $1s^2$]

The cation or anions which completes their octet or duplet are said to have achieved inert gas configuration.

Li^+ Be^{+2} H^- = [He]

Na^+ Mg^{+2} Al^{+3} = [Ne]

elements which follow octet :- Na^+, K^+, Rb^+, Cs^+, Mg^{+2}, Ca^{+2}, Sa^{+2}, Ba^{+2}, F^-, O^{-2}, Cl^-, Br^-, I^-, S^{-2}, N^{-3}, P^{-3}, Se^{-2}, Sc^{+3}, Ti^{+4}, V^{+5}, Cr^{+6}, Mn^{+}

B. PSEUDO-INERT GAS CONFIGURATION

Stability of elements possessing this configuration is like inert gas but configuration is not like inert gas. Cations having 9 to 18 e⁻'s, in valance shell then the config. They get is known as pseudo inert gas configuration.

$_{30}Zn^{+2} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^{10}}_{18e^-}$

$_{29}Cu^{+2} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^9}_{17e^-}$

$_{25}Mn^{+2} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^5}_{13e^-}$

$_{23}V^{+3} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^3}_{11e^-}$

$_{22}Ti^{+3} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^2}_{10e^-}$

$_{31}Ga^{+3} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^{10}}_{10e^-}$

$_{47}[Ag]^{+1} = 18[Ar]\ 3d^{10}.\ \underbrace{4s^2\ 4p^6\ 4d^{10}}_{18e^-}\ 5s^1$

PSUEDO-INERT GAS [ns^2] CONFIGURATION

$(n-1)\ s^2\ p^6 d^{10} + ns^2$

$_{31}Ga^{+1} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^{10}}_{18}\ \underbrace{4s^2}_{2}$

$_{81}Tl^{+1} = 54[Xe]\ 4f^{14}\ \underline{5d^{10}}\ \underline{6s^2}\ 6p^0$ [inert-pair effect]

$_{31}Ge^{+1} = 1s^2\ 2s^2\ 2p^6\ \underline{3s^2\ 3p^6\ 3d^{10}}\ \underline{4s^2}$

Limitation of Octel Law

electron precise compound → $8e^-$ present in vallence shell of central atom after bonding.

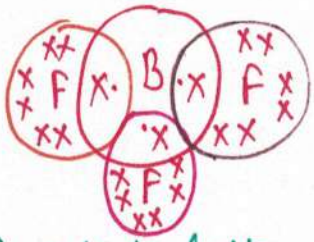
CO_2, H_2O, PCl_3, NH_3

1. HYPOVALENT COMPOUND

Compound in which less than $8e^-$ present in valence shell of central atom after bonding.

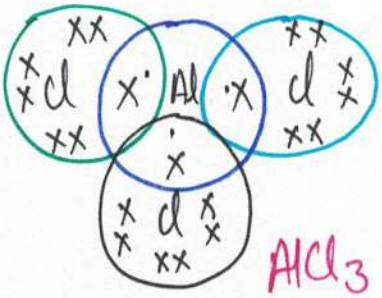
e.g:- III A group elements form hypovalent comp.

BF_3, BCl_3, BBr_3, BI_3



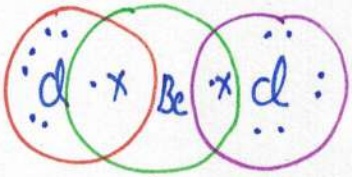
B → Central Atom

F → Bounded Atom



$AlCl_3$

6e⁻ in central atom

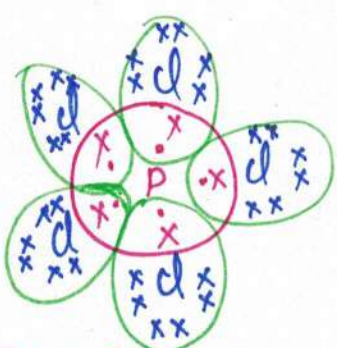


$BeCl_2$

4e⁺ in central atom

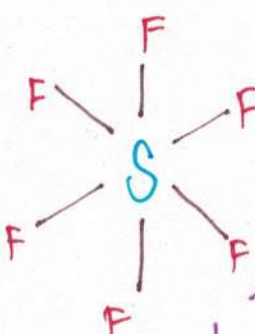
2. HYPERVALENT COMPOUNDS

Compounds in which more than 8e⁻ present in valence shell of central atom after bonding.



$[PCl_5]$ [Phosphorus pentachloride]

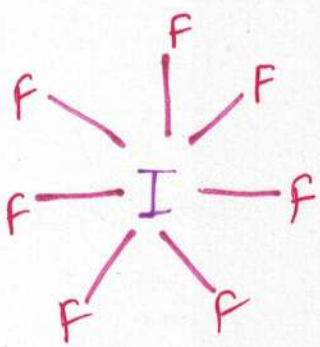
10e⁻ in central atom



$[SF_6]$

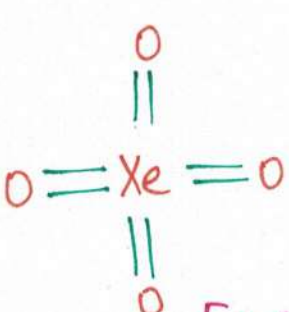
Sulphur hexafloride

12e⁻ in central Atom



$[IF_7]$

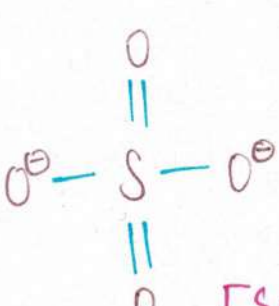
14e⁻ in central atom



$[XeO_4]$

Xenon Tetraoxide

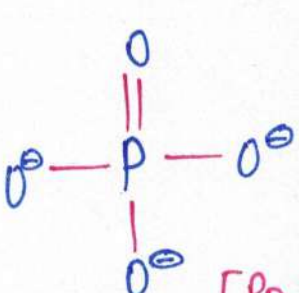
Xe = 8×2 = 16e⁻



$[SO_4^{2-}]$

sulphate

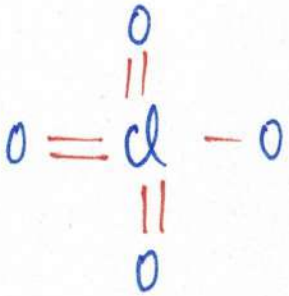
Central atom 6×2 = 12 e⁻



$[PO_4]^{3-}$

Phosphat

C.A. = 10 e⁻



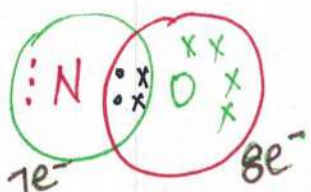
Perchlorate

$7 \times 2 = 14e^-$

3. ODD ELECTRON SPECIES

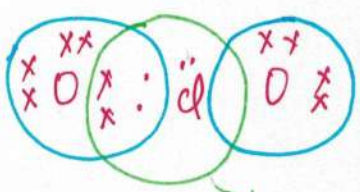
Odd no. of e^- is present in central atom after bonding is known as odd electron species.

eg. [Na] nitric oxide



Odd no. of $e^\ominus$

[ClO_2] chlorine dioxide



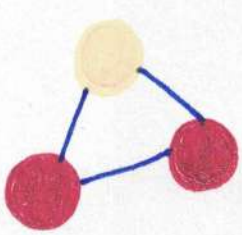
Odd no. $11e^\ominus$

4. PSUEDO INERT GAS CONFIGURATION

P, d and f-block element metals forms stable compound in psuedo inert gas configuration.

e.g:- in. $CuSO_4$ O.S. of. Cu is +2

$_{29}Cu^{+2} = 1s^2\ 2s^2\ 2p^6\ \underbrace{3s^2\ 3p^6\ 3d^9}_{19e^\ominus}$



$PbCl_2$

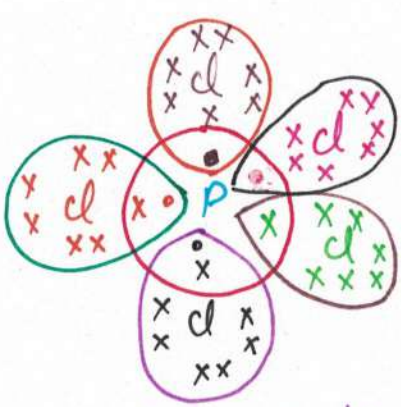
$_{82}Pb^{+2} = 54[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2$

Limitations of Octel Law can be explained by:-

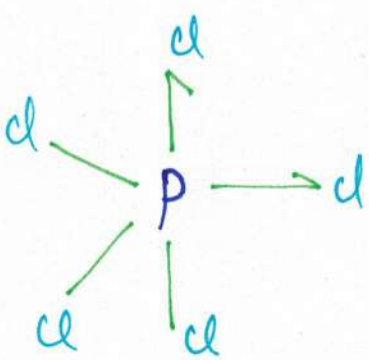
1. Sudgen's Single electron Concept
2. Sidgwick's Maximum Covalency Concept
3. Overlapping of vacant Orbitals

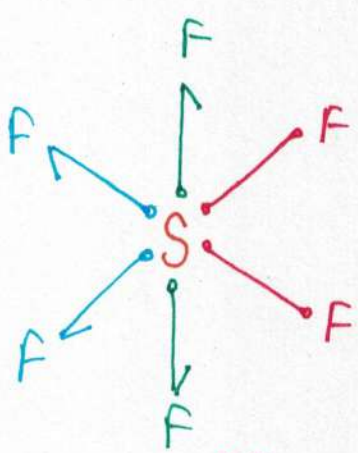
1. Sudgen's Single electron Concept

According to Sudgen, maximum 8 electron are present in valence shell of central atom after bonding. If central atom forms more no. of bonds, bond formed between central atom and bonded atoms is through one $e^{\ominus}$ or single electron bond.



PCl_5 Phosphorus Pentachloride





SF_6 Sulphur hexa-floride

no. of single e⁻ bond = no. of valence shell e⁻ — no. of $e^{\ominus}$s required to complete octel

$S = N - A$

Single band = e⁻ present in valence shell − e⁻ req" to comp. octel

1. PCl_5 ⇒ no. of single e⁻ band = 5 − 3 = 2
2. SF_6 $S = 6 - 2 = 4$
3. IF_7 $S = 7 - 1 = 6$
4. XeO_4 $S = 8 - 0 = 8$

2. Sigdwick's Maximum Covalency Concept

According to sigdwick maximum covalency of each period (no. of bond) is :-

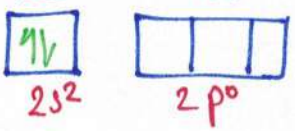
1st period = 1
2nd period = 4
3rd period = 6
4th period = 6
5th, 6th, 7th period = 8

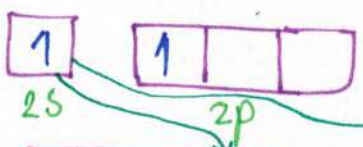
> Beryllium

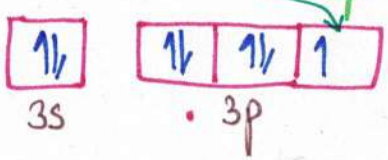
Be = $1s^2\ 2s^2\ 2p^0$

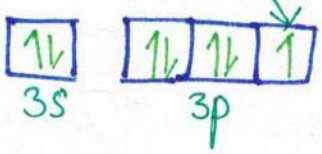
o) $BeCl_2$

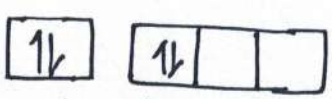
Cl — Be — Cl

Be g.s. = 

Be e.s. = 

Cl = 

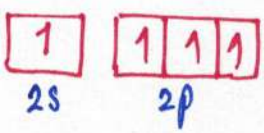


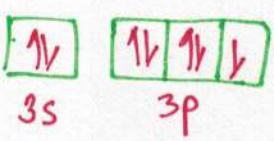
Be in $BeCl_2$ = 

(hypovalent)

o) $K_2[BeCl_4]$

$K_2[BeCl_4] \rightarrow 2K^+ + [BeCl_4]^{2-}$

Be g.s. = 

Cl = 

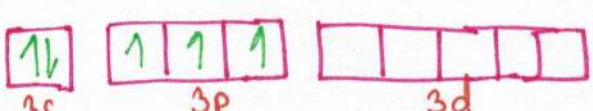
Be in $Be[Cl_4]^{2-}$ =

(e⁻ precise)

maximum covelency of Be = 4

> P = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

o) PCl_3

P g.s. = 

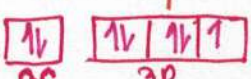
3Cl =

Pin PCl_3 =

($e^{\ominus}$ precise)

o) PCl_5

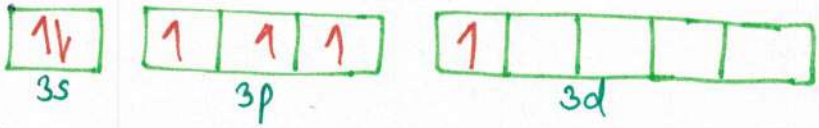
P e.s

5Cl = 

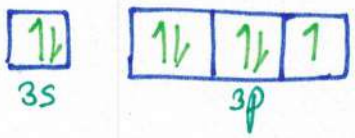
Pin PCl_5

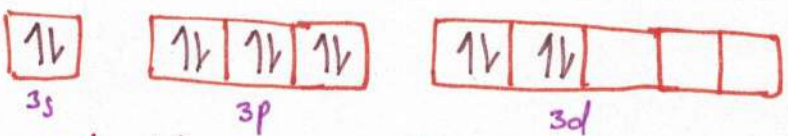
(Hypovalent)

c) PCl_6^-

P g.s = 

P e.s =

6 Cl = 

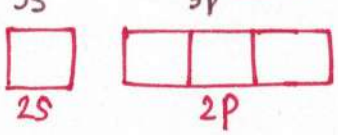
P in PCl_6^- = 

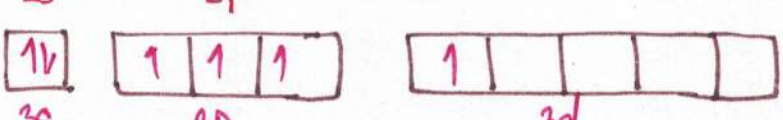
(Hypovalent) 12e⁻ after bonding

Maximum covalency of P = 6

c) SF_4

S g.s. =

4F = 

Se.s = 

S after bonding in SF_4 =

10e⁻

[Hypervolent]

Applications of Maximum Covalency Concept

1. Existence of Molecule

a. NCl_5 does not exist but PCl_5 does exist (Nitrogen Pentachloride) (Phosphorus Pentachloride)

Reason:- Maximum covalency of N is 4 but max. covalency of P is 6.

b. OF_4 & OF_6 dosnot exist but SF_4 & SF_6 exist.

2. Hydrolysis

Carbon tetrachloride [CCl_4] does not hydralyse but $SiCl_4$ hydrolysis.

Carbon is II[nd] period element. It's maximum covalency is 4. Silicon is III[rd] period element it's maximum covalency is 6. Two vacant orbitals available in valence shell of silicon in $SiCl_4$. Lone Pair of oxygen atom of H_2 forms coordinate bond with vacant orbital of Si. For hydrolysis, vacant orbitals are necessary and covalency should be concerned.

> CCl_4

C_{gs} = ↿⇂ (2s) ↿ ↿ ☐ (2p)

C_{es} = ↿ (2s) ↿ ↿ ↿ (2p)

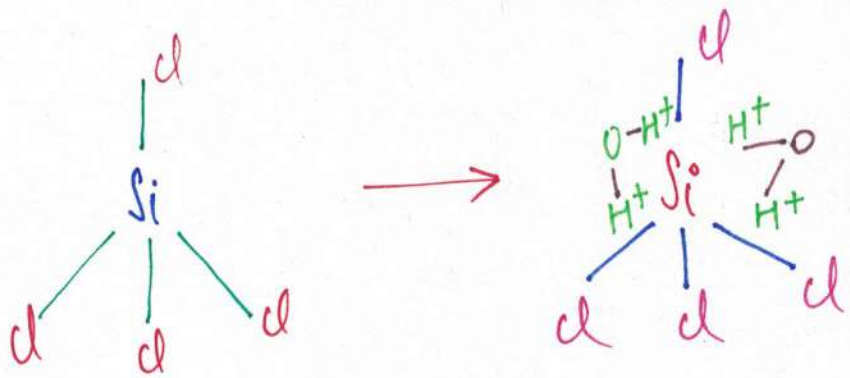
[no vacant orbital
no hydrolysis]

> $SiCl_4$

in $SiCl_4$ → Si e.s = ↿⇂ ↿⇂ ↿⇂ ↿⇂ ☐☐☐☐☐

on dissolving in H_2O

Si = ↿⇂ ↿⇂ ↿⇂ ↿⇂ OH_2 OH_2 ☐☐☐



$SiCl_2(OH)_2 = +2HCl$ (more than 3 hydroxide group are unstable)

$\downarrow +2H_2O$

$2HCl + Si(OH)_4$

$\downarrow$

$H_2SiO_3 + H_2O \rightarrow SiO_2 + H_2O$

SiO_2 is silica

ELECTROVALENT BOND / IONIC BOND

Electrostatic force of attraction between metal and non-metal ion is known as ionic bond.

Ionic bond has non-directional character.

Condition required to form Ionic Bond:-

- Electronegativity difference between metal and non-metal should be more than 2. strong ionic bonds are formed b/w IA & VII A group elements.

1. Ionisation Potential of Metal atom should be low.

$Li < Na < K < Rb < Cs$

Down the group as IP decreases. reactivity increase

Reactivity $Na > Mg > Al$

2. Electron affinity of Non-metal atom should be high

ex:- $Cl > F > Br > I$

ex:- $F > O > N$

3. Lattice energy of Ionic Compound should be high.

$M + IE \longrightarrow M^+ + e^\ominus$ (endothermic) — (I)

(Metal)

$X + e^- \longrightarrow X^- + EA$ (exothermic) — (II)

(non-metal)

$M^\oplus + X^\ominus \longrightarrow MX + O$ (exothermic) — (III)

(lattice energy)

$I < II + III$

IP of metal < EA of nonmetal + lattice energy

$$V = K \frac{Z^+ Z^-}{r_c + r_A}$$

V = lattice energy
Z^+ = charge on cation
Z^- = charge on anion
r_c = radius of cation
r_A = radius of anion

Que:- Write the correct order of increasing lattice energy of Al_2O_3, $MgBr_2$, CaO, $NaCl$.

Soln:- $Z^+ Z^-$

$Al_2O_3 = 3 \times 2 = 6$ $\quad$ $CaO = 2 \times 2 = 4$

$MgBr_2 = 2 \times 1 = 2$ $\quad$ $NaCl = 2 \times 1 = 2$

Order:- $NaCl < MgBr_2 < CaO < Al_2O_3$

Determination of Lattice Energy Born Haber's Cycle

Hess Law – Heat of formation of any compound remains constant either it form from single step or multiple steps.

$Na(s) + Cl_2(g) \longrightarrow NaCl(s)$ $\quad$ $\Delta_f H\ NaCl = x\ K\,cal\ mol^{-1}$

I^st step:- Sublimation Energy

energy required to convert 1 mol solid substance to 1 mol gaseous substance is known as sublimation energy

$M(s) + SE \longrightarrow M(g)$

(endothermic process)

II^nd step :- **Ionisation energy**

energy required to remove 1 e^⊖ from valence shell neutral isolated gaseous atom is known as ionisation potential.

$$M_{(g)} + I.E \longrightarrow M^+$$

(endothermic process)

III^rd step :- **Dissociation energy**

energy required to break bond between 2 atoms in a molecule is known as bond dissociation energy. It is an endothermic process.

$$X_{2(g)} + d.e \longrightarrow 2X_{(g)}$$

(endothermic process)

IV step :- **Electron Affinity**

energy released when addition of 1 e^- on valence shell of neutral isolated gaseous atom takes place is known as electron affinity.

$$X_{(g)} + e^- \longrightarrow X^- + EA$$

(exothermic process)

V step :- **Lattice energy**

energy released when 1 mole ionic compound forms from their constituent ions is known as lattice energy. It is exothermic process.

$$M^+ + X^- \longrightarrow MX + U$$

(exothermic process)

Heat of formation = sublimation energy + Ionisation energy + Dissociation energy + electron affinity + Lattice energy.

$$\Delta_f H = SE + IE + d.e + EA + U$$

Electronegativity & hydration energy cannot be measured by Born Hayber's cycle.

e.g-1

$$Na(s) + Cl_2(g) \longrightarrow NaCl(s)$$

$\downarrow +SE$ $\quad$ $\downarrow +\frac{1}{2}de$

$Na(g)$ $\quad$ $Cl(g)$

$+e^- \downarrow +IE$ $\quad$ $+e^- \downarrow -EA$

$$Na^+(g) + Cl^-(g) \xrightarrow{-U} NaCl(s)$$

$$\Delta_f H\ NaCl = SE + IE_1 + \frac{1}{2}de - EA - U$$

eg-2

$$Mg(s) + O_2 \longrightarrow MgO(s)$$

$\downarrow +SE$ $\quad$ $\downarrow +\frac{1}{2}de$

$Mg(g)$ $\quad$ $O(g)$

$-e^- \downarrow +IP_1$ $\quad$ $+e^- \downarrow -EA_1$

Mg^{+1} $\quad$ O^-

$-e^- \downarrow +IP_2$ $\quad$ $+e^- \downarrow -EA_2$

$$Mg^{+2} + O^{2-} \xrightarrow{-O} MgO(s)$$

$$\Delta_f H\ MgO = SE + (IE_1 + IE_2) + \frac{1}{2}de - (EA_1 + EA_2) - U$$

eg-3

$$Mg(s) + Cl_2 \longrightarrow MgCl_2$$

$\downarrow +SE$ $\quad$ $\downarrow +doe$

$Mg(g)$ $\quad$ $2Cl(g)$

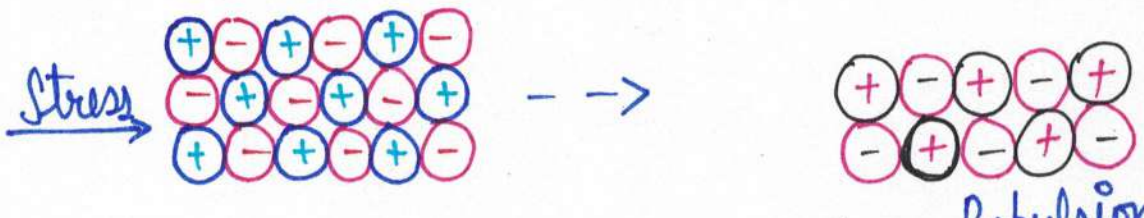
$-2e^- \downarrow +(IE_1+IE_2)$ $\quad$ $\downarrow +2e^-/-2EA$

$$Mg^{+2}(g) + 2Cl^- \xrightarrow{-U} MgCl_2$$

$$\Delta_f H\ MgCl_2 = SE + de + (IE_1 + IE_2) - 2EA - U$$

PROPERTIES OF IONIC COMPOUNDS

- Ionic compounds are hard and brittle in nature Hardness $\propto$ Lattice energy.



- Ionic Compounds have non-directional characters.
- Ionic Compounds do not show isomerism.
- Ionic Compounds show isomorphism

If isoelectronic species forms ionic compound, their crystalline structure remains same.

A. Isoelectronic species

NaF and MgO are isomorphons.

Na^+ F^- Mg^{+2} O^{-2}

$10e^-$ $10e^-$ $10e^-$ $10e^-$

KCl and CaS are isomorphones

K^+ Cl^- Ca^{+2} S^{-2}

$18e^-$ $18e^-$ $18e^-$ $18e^-$

B. If oxidation state of central atom or metal atome remains same, crystalline water also remains same in a compound, their crystalline structure is same.

$Mg^{+2}SO_4 \cdot 7H_2O$ $Fe^{+2}SO_4 \cdot 7H_2O$ $Zn^{+2}SO_4 \cdot 7H_2O$

epsum salt green vitrial white Vitrial

$K_2SO_4 \cdot Al_2^{+3}(SO_4)_3 \cdot 24H_2O$ alum

$K_2SO_4 \cdot Fe_2^{+3}(SO_4)_3 \cdot 24H_2O$ ferric alum

$K_2SO_4 \cdot Cr_2^{+3}(SO_4)_3 \cdot 24H_2O$ chrome alum

These are called Double Salt.

Ionic compounds conduct electricity as they dissociate into ions in water.

SOLUBILITY

Ionic compounds are soluble in polar solvents & insoluble in non-polar solvents/ organic solvents.

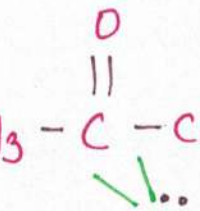
Solubility of ionic compounds depends 4 factors :-

A. Dielectric constant of solvent (epsilon ϵ) capability of solvent to neutralize charge of solvent particles in known as dielectric constant of solvent.

Order of electric constant

$H_2O > CH_3OH > CH_3CH_2OH >$

$\epsilon = 80$ Water $\epsilon = 32$ methanol ethanol



acetone

dimethyle ether

Non polar → Benzene

Que. Solubility of KCl is max in :-

a. CH_3OH b. CH_3CH_2OH c. C_6H_6 d. $CH_3-O-CCl_3$

Ans. a CH_3OH (E of CH_3OH is maximum)

minimum ability in C_6H_6

Dielectric constant of H_2O_2 & H_2SO_4 is more than H_2O but both are not used as solvent

H_2O_2 & H_2SO_4 are oxidising agents, so, they'll oxidise the ionic compound.

$H_2O_2 \longrightarrow H_2O + (O)$

solute + (O) $\longrightarrow$ oxidised

$H_2SO_4 + H_2O + SO_2 + (O)$

oxidising agent

solubility & dielectric constant of solvent.

b.

LATTICE ENERGY

Energy acquired to break bond between ions in the ionic compound is known as lattice energy.

$M^+X^- + U \longrightarrow M^+ + X^-$

$U \propto Z^+ Z^-$

$$U \propto \frac{1}{r_c + r_a}$$

$Z^+ \rightarrow$ charge of cation

$Z^- \rightarrow$ charge of anion

$r_c \rightarrow$ size of cation

$r_a \rightarrow$ size of anion

$r_c + r_a \longrightarrow$ interionic distance

solubility $\propto \frac{1}{U}$

2. HYDRATION ENERGY

Solubility ∝ hydration energy

Energy released when one mole ionic comp. is dissolved in H_2O is known as hydration energy.

When **hydration energy > lattice energy**, compound will be completly soluble in H_2O.

When **hydration energy < lattice energy**, compound will be insoluble in water.

When **hydration energy = lattice energy**, compound will be strangly a partially soluble in H_2O.

When charge of cation & anion remain constant, as size difference b/w cation & anion increases, solubility of compound also increases.

$M^+ X^-$

$r_{M^+} = 10\ A^\circ \quad r_{X^-} = 100\ A^\circ$

$$V = \frac{KZ^+Z^-}{r_c + r_a}$$

$$V = K\frac{1\times1}{10+100} = \frac{K}{110} = 0.09K$$

$$\text{Hydration energy} = KZ^+Z^-\left[\frac{1}{r_c} + \frac{1}{r_A}\right] = K(1\times1)\left(\frac{1}{10} + \frac{1}{100}\right) = 0.11$$

H.E. > V, solubility ↑

e.g. (1.) $LiF < LiCl < LiBr < LiI$ [solubility]

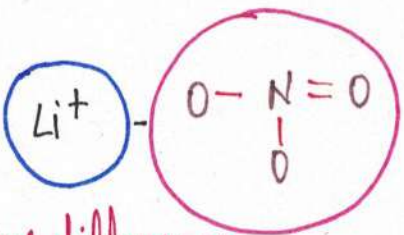
Although some covalent character is found in LiI, it is completely soluble in H_2O because size difference of cation & anion is more.

2. LiCl < NaCl < KCl < RbCl < CsCl

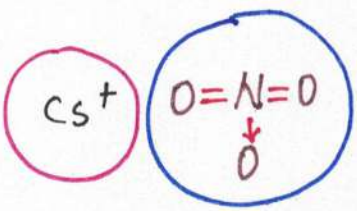
3. LiF < NaF < KF < RbF < CsF

4. LiOH < NaOH < KOH < RbOH < CsOH

5. $LiNO_3$ > $NaNO_3$ > KNO_3 > $RbNO_3$ > $CsNO_3$



size difference is more



6. $LiClO_4$ > $NaClO_4$ > $KClO_4$ > $RbClO_4$ > $CsClO_4$

7. $BeCO_3$ > $MgCO_3$ > $CaCO_3$ > $SrCO_3$ > $BaCO_3$

soluble — white ppt — insoluble

8. $BeSO_4$ > $MgSO_4$ > $CaSO_4$ > $SrSO_4$ > $BaSO_4$

soluble — insoluble

9. BeC_2O_4 < MgC_2O_4 < CaC_2O_4 < SrC_2O_4 < BaC_2O_4

size of C_2O_4 is bigger

10. $BeCrO_4$ > $MgCrO_4$ > $CaCrO_4$ > $SrCrO_4$ > $BaCrO_4$

solubility of alkaline earth metal oxalate down the group increases.

when one cation is attached with two anions & two cation attached with on anion, solubility down the group increases.

1. Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_2

2. Li_2SO_4 < Na_2SO_4 < K_2SO_4 < Rb_2SO_4 < Cs_2SO_4

3. $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

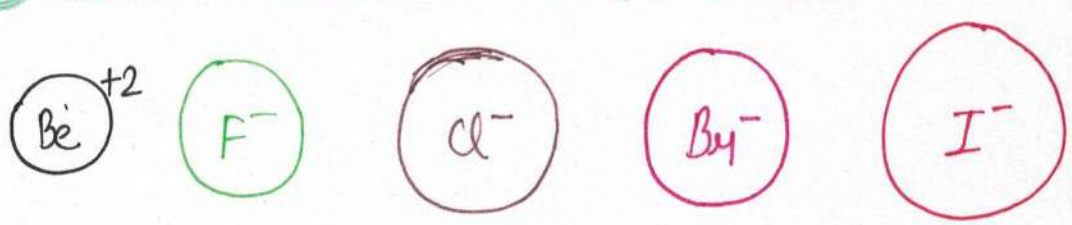
4. $LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$

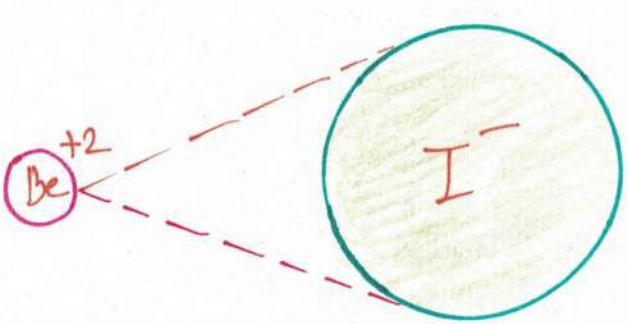
solubility of alkali metal components down the group increases.

d. **COVALENT CHARACTER**

$$\text{solubility} \propto \frac{1}{\text{covalent character}}$$

1. $BeF_2 > BeCl_2 > BeBr_2 > BeI_2$





Shifting of electron cloud of I^- towards Be^{+2}

↓ Polarization

covalent electron $\propto$ Polarization

2. $AgF > AgCl > AgBr > AgI$

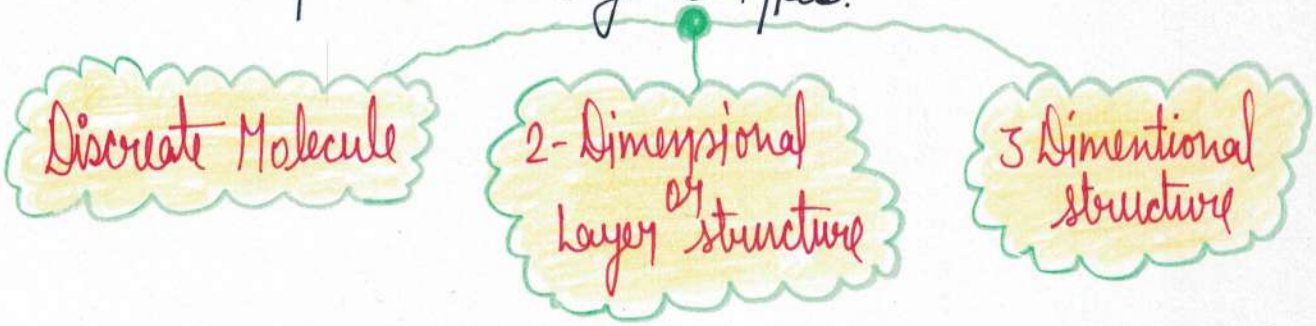
COVALENT BOND

Equal sharing of e^- takes places b/w Two non-metal atom gives rise to covalent bond.

PROPERTIES OF COVALENT COMPONENTS

1. PHYSICAL STATE

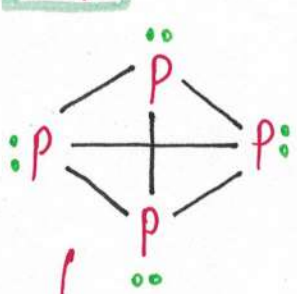
- Covalent compounds exist in solid liquid or gas phase.
- Covalent bond has directional character.
- Covalent compounds are of 3 types.



1. DISCRETE MOLECULE

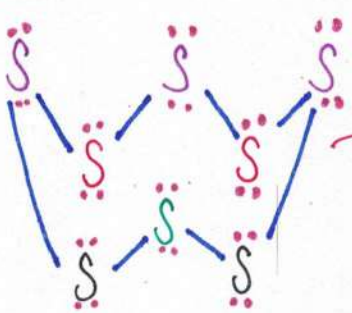
Some covalent compounds exist in discrete units.

e.g P_4



Red phosphorous hota hai matchstick me.

S_8 (crown Sulphur)

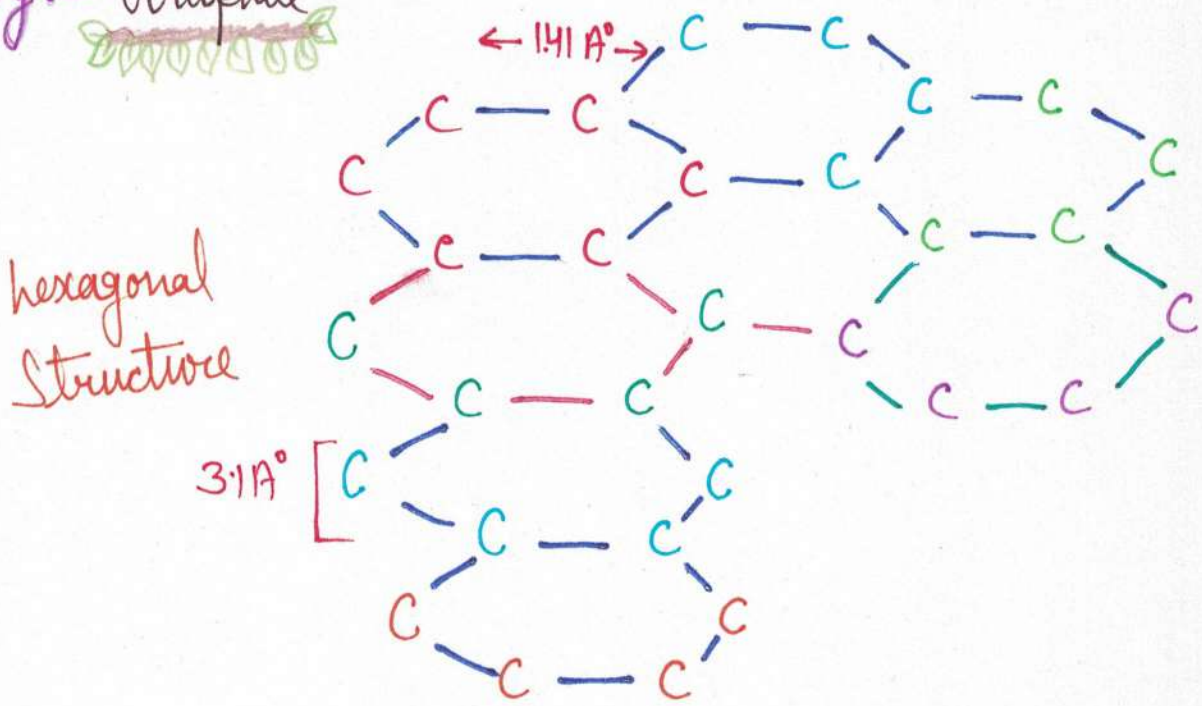


→ iska shape crown jaisa hota hai!

2. 2-DIMENSIONAL STRUCTURE

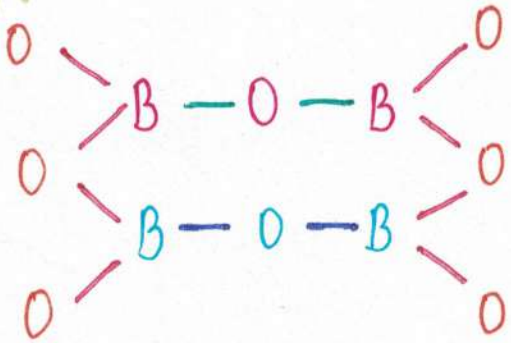
Some covalent compound exist in 2-D or Layer structure:

e.g:- Graphite



hexagonal Structure

Metaboric acid

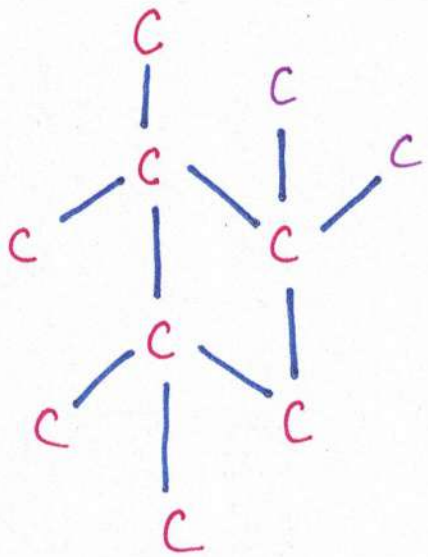


Mica

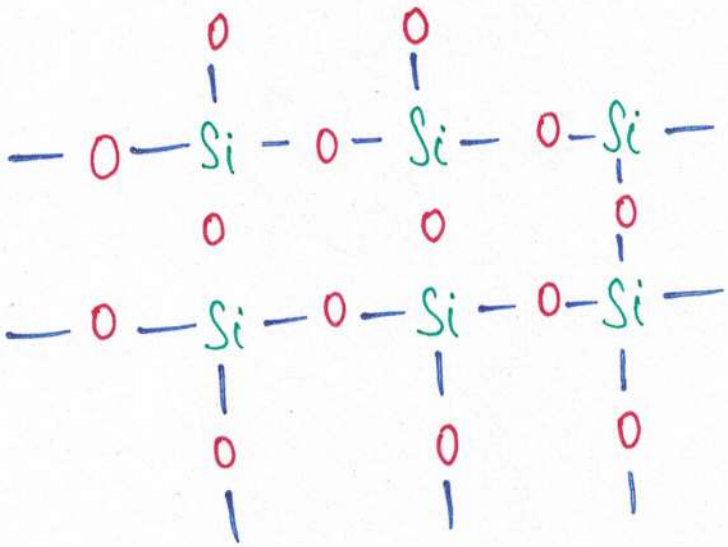
3. 3 DIMENSIONAL STRUCTURE/ COVALENT SOLID

Some covalent compounds exist in 3-D structure.

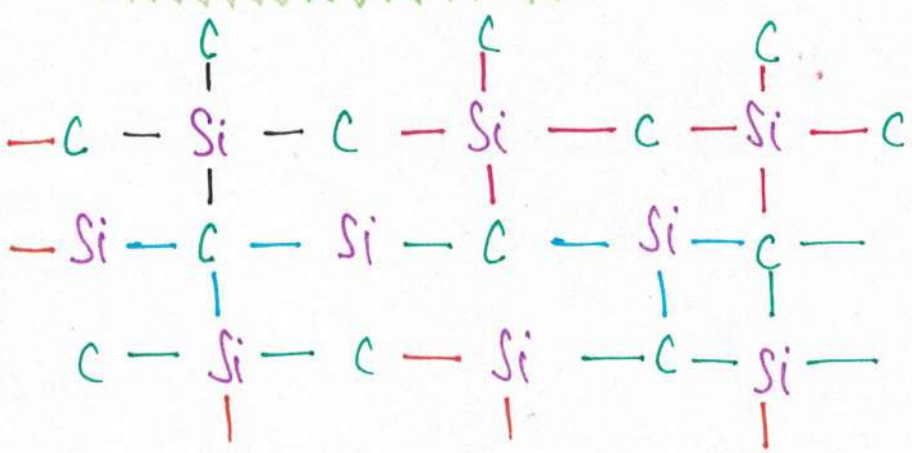
Diamond → one C-atom attached with 4-C atom in tetrahedral manner.



Silica $(SiO_2)_n$



Carborundum (SiC)

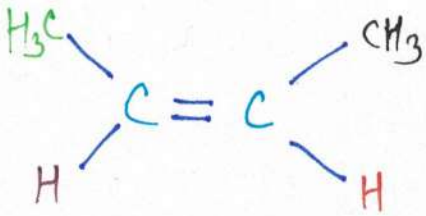


one C-atom attached with 4-silicon atom
&
1-silicon atom attached with 4-C atom.

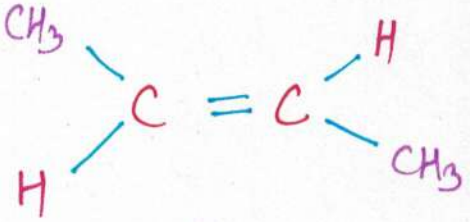
- This type of covalent compound have high melting & boiling point.
- All are hard substances.

4. COVALENT COMPOUND SHOWS ISOMERISM

e.g:- 2-butene



cis



trans

5. MELTING POINT AND BOILING POINTS

Covalent compounds have lower melting pt. and boiling pt. than ionic compounds.

6. COVALENT COMPOUNDS DO NOT CONDUCT ELECTRICITY

Covalent compounds do not conduct electricity. But graphite and Hcl (aq.) conduct electricity.

graphite $\longrightarrow$ one free e^-

$Hcl(aq) \rightleftharpoons H_3O^+ + Cl^-$

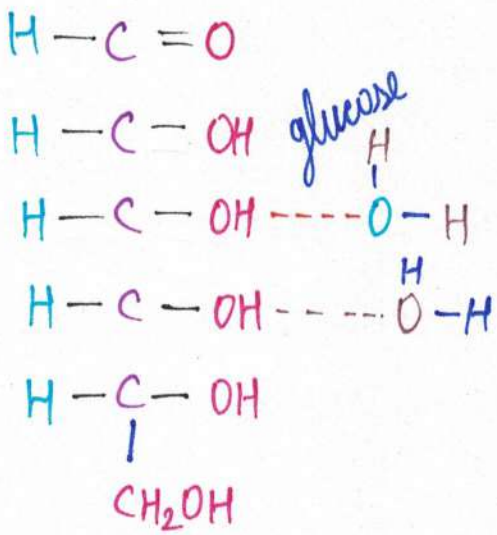
Ionic compounds are bad conductor of electricity in solid state. But in molten state or aqueous phase they conduct electricity.

solid < molten < aqueous

7. SOLUBILITY OF COVALENT COMPOUNDS

These are soluble in non-polar solvents. But some covalent compounds are soluble in water.

Example:- Glucose ($C_6H_{12}O_6$), Sucrose ($C_{11}H_{22}O_6$)



Reason:- It forms Hydrogen bond with H_2O. and some covalent compounds ionises in H_2O.

Ionic Reaction are faster than molecular reaction.

FAJAN'S RULE

Covalent character in ionic compound can be explained by Fajan's Rule. It depends on 2 factors.

1. Polarising power of cation
2. Polarisability of anion

POLARISATION

Small size of cation polarises bigger size of anion (small size of cation attracts electron of anion). Therefore, shape of anion is distorted. This phenominous is known as Polarisation.

It depends on 2 factors :-

1. POLARISING POWER OF CATION

Distortion ability of cation to anion is known as polarising power of cation.

It depends on 3 factors :-

Polarising power of cation (ϕ) = $\frac{\text{charge of cation}}{\text{size of cation}}$

i. Small size of cation
ii. Greater charge of cation
iii. pseudo- inert gas configuration

i SMALL SIZE OF CATION

Polarising power of cation $\propto \frac{1}{\text{size of cation}}$

eg:- $Li < Na^+ < K^+ < Rb^+ < Cs^+$ (size)

Li ↓ More p.p. → more covalent character develop in ionic compounds

Cs^+ ↓ Less p.p → ionic nature

i. $LiCl < NaCl < KCl < RbCl < CsCl$ [ionic nature]
down the group, covalent characters decrease & ionic charact increases.

ii. $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$ [Covalent nature]
small size of Be^{+2} polarises Cl^- more, therefore it has more polarising power. The more shifting of electron cloud towards cation takes place, the more polarisation it shows.

(ii) GREATER CHARGE OF CATION

P.P of cation $\propto$ charge of cation

(i) $NaCl < MgCl_2 < AlCl_3 < SiCl_4$

Covalent character $\propto$ P.P of cation

(ii) $LiCl < BeCl_2 < BCl_3$

(iii) $SnCl_2 < SnCl_4$ [Covalent character]

(iv) $Fe(OH)_2 < Fe(OH)_3$

(iii) PSUEDO INERT GAS CONFIGURATION

9-18 e^- present in valence shell of cations. More covalent character found in compounds in which cation having psuedo inert gas configuration.

P.P of cation $\propto$ psuedo-inert gas configuration

e.g:- CuCl and CaCl

$Na^+ \longrightarrow 1s^2 \underbrace{2s^2\ 2p^6}_{\text{octet}}\ 3s^0$

$Cu^+ \longrightarrow 1s^2\ 2s^2\ 2p^6 \underbrace{3s^2\ 3p^6\ 3d^{10}}_{\text{Psuedo-inert gas}}\ 4s^0$

$L_{eff}\ Cu^+ > L_{eff}\ Na^+$

size of Cu^+ < size of Na^+

Q. Why melting points of CCl is 480°C, melting point of NaCl is 850°C.

melting point is $\propto \frac{1}{\text{Covalent character}}$

CuCl covalent character ↑ M.P ↓

2. POLARISABILITY OF ANION

Small size of cation distorted shape of anion. Distortion ability of anion by cation is known as polarisability of anion.

Polarisability of anion depends on 2 factors :-

(a) bigger size of anion
(b) greater charge of anion

1. Size of anion $\propto$ P.P. of anion

$F^- < Cl^- < Br^- < I^-$ [covalent character]

- AgF < AgCl < AgBr < AgI
- $BeF_2 < BeCl_2 < BeBr_2 < BeI_2$
- LiF < LiCl < LiBr < LiI
- $O^{-2} < S^{-2}$
- PbO < PbS
- $N^{-3} < P^{-3}$

  $Li_3N < Li_3F$

2. Polarisability of anion $\propto$ charge of anion

- LiF < Li_2O < Li_3N
- $F^- < O^{-2} < N^{-3}$

APPLICATION'S OF FAJAN'S RULE

COVALENT CHARACTER

Covalent character ∝ Polarisation

Left to Right across the period, covalent character increase and down the group, ionic character increases.

1. NaCl, KCl, $CaCl_2$, $CuCl_2$

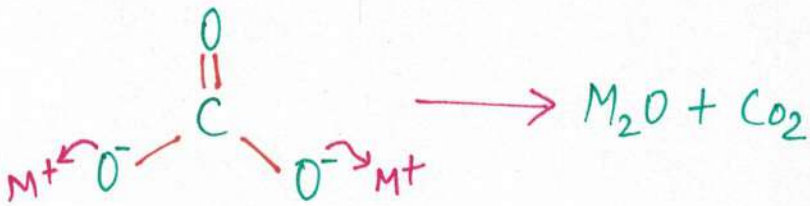
   $CuCl_2 < CaCl_2 < NaCl < KCl$ (ionic nature)

2. NaCl, AgCl, CuCl

   NaCl < AgCl < CuCl (covalent nature)

STABILITY OF CARBONATES

Small size of metal cation easily polarises O^{-2} of carbonates. Therefore metal oxygen bond becomes stronger & oxygen carbon bond become weaker. On heating, such type of carbonates decomposes metal oxide and carbon dioxide.



Thermal stability of alkali metal carbonate down the group increases.

$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$

$Li_2CO_3 \xrightarrow{\Delta} H_2O + CO_2$

$Na_2CO_3, K_2CO_3, Rb_2CO_3, Cs_2CO_3 \xrightarrow{\Delta}$ No reaxn

THERMAL STABILITY OF METAL HALIDES

M^+ F^- Cl^- Br^- I^-

$LiF < LiCl < LiBr < LiI$ [Covalent nature]

Thermal stability $\propto$ ionic character

$\propto \frac{1}{\text{covalent character}}$

$MF > MCl > MBr > MI$ (Thermal Stability)

e.g:- $BeF_2 > BeCl_2 > BrBr_2 > BeI_2$

NATURE OF OXIDES

Left to right across the period, as ϕ increases covalent character increases.

Down the group, as ϕ decreases, covalent character decreases & ionic character increases.

P.P of cation (ϕ) Δ $\frac{\text{charge}}{\text{size}}$

if $\sqrt{\phi} = 2.2$ to $3.2 \longrightarrow$ Amphoteric oxide

$\sqrt{\phi} < 2.2 \longrightarrow$ Basic oxide

$\sqrt{\phi} > 3.2 \longrightarrow$ Acidic oxide

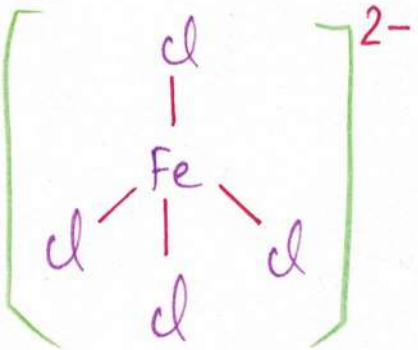
COMPLEX FORMING TENDENCY OF METAL IONS

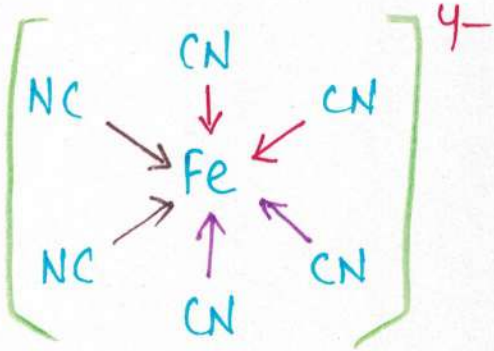
Generally d-block elements and Be, Mg forms complex compounds because size of d-block cation is very small & They have vacant orbital, small size of cation easily polar-ises e^- cloud of anion a co-ordinate bond is formed b/w vacant orbitals of metal ion and lone pair of non-metal ions.

Complex forming tendency $\propto$ polarising power of cation.

e.g:- $[Fe(CN)_6]^{4-}$ $[Cu(NH_3)_4]^{2+}$

$[BeCl_4]^{2-}$ $[Zn(CN)_4]^{2-}$ $[AlCl_6]^{3-}$





COLOUR INTENSITY

1. Colour intensity $\propto$ covalent character
2. Covalent character $\propto$ polarisability of anion

➢ $AgF < AgCl < AgBr < AgI$ [Colour intensity]

Colourless soluable in H_2O — white ppt. — Yellow ppt. — dark Yellow

- $Fe(OH)_2$ < $Fe(OH)_3$ [colour intensity]
  light green — rust brown
- $HgCl_2$ — HgI_2 [$Cl^- < I^-$]
  light yellow — scarlett red
- $SnCl_2$ — SnI_2 [$Cl^- < I^-$]
  White — Red
- PbO — PbS [$O^{-2} < S^{-2}$]
  Yellow — black

DIAGONAL RELATION

If cations have same ionic potential, cations show same chemical properties.

1. Be & Al
2. Li & Mg
3. B & Si

COVALENT BOND

VALENCE BOND THEORY [VBT]

- Proposed by Heitler and London, Modified by Pouling and Slater.
- According to VBT,

- Only valence shell e⁻s participate in bonding.
- Only unpaired e⁻s participate in bonding.
- e⁻ does not participate in bonding but atomic orbital participate in bond formation.
  (the one having unpaired e⁻)
- When two half filled atomic orbitals come closer, overlapping between 2 orbitals takes place.
- Bond strength depends on extent of overlapping.

  Bond strength ∝ extent of overlapping
- When atom forms more bond than unpaired e⁻ it is having then. (i) paired e⁻ promoted to vacant orbital of central atom. paired e⁻ becomes unpaired and no. of unpaired e⁻ increases.

TYPES OF OVERLAPPING

1. Co-axial overlapping
2. Co-lateral overlapping

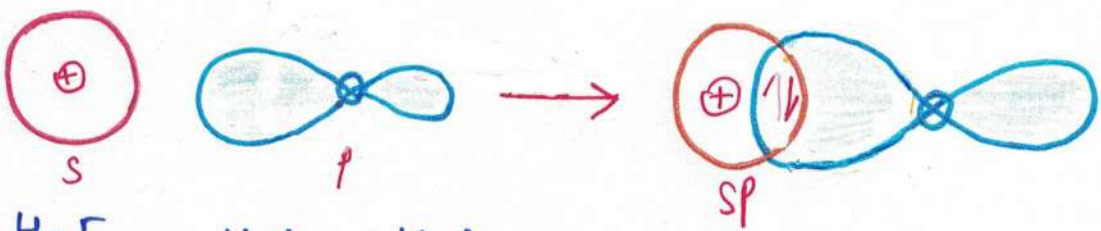
1. CO-AXIAL OVERLAPPING

When 2-half filled atomic orbitals overlap each other at internuclear axis, overlapping is known as Co-axial over-lapping.

(i) S-S OVERLAPPING

H → [↑] [↓]
1s 1s

ii S-P OVERLAPPING



e.g:- H-F , HCl , H-Br , HI

H → [↑↓] (1s)

F = [↑↓] (2s) [↑↓] [↑↓] [↑] (2p)

iii P-P OVERLAPPING

F_A = [↑↓] [↑↓] [↑↓] [↓]

F_B = [↑↓] [↑↓] [↑↓] [↑]

e.g:- F_2 , Cl_2 , Br_2 , I_2 , O_2 , N_2

σ (sigma) bond

iv OVERLAPPING OF HYBRID ORBITALS

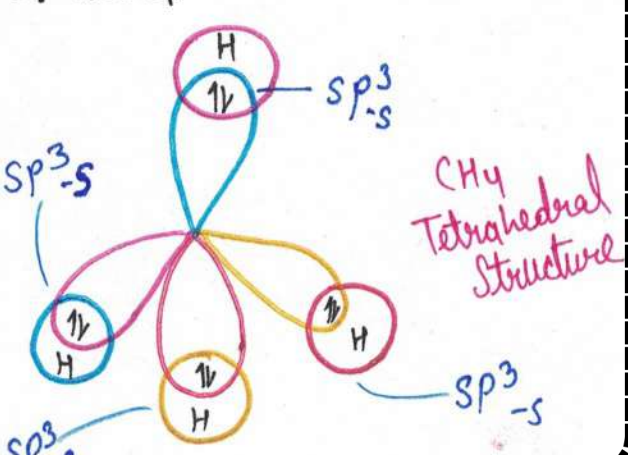
Hybrid orbitals always form sigma bond.

CH_4

C = [↑↓] (2s) [↑] [↑] (2p)

Ces = [↑] (2s) [↑] [↑] [↑] (2p)

hybrid → [↑] [↑] [↑] [↑] (sp^3)



CH_4 Tetrahedral Structure

Extent of Overlapping depends on 2 factors :-

(i) Principal quantum no. "n" $\propto \frac{1}{n}$

1s - 1s > 2s - 2s

2p - 2p > 2p - 3p > 2p - 4p > 2p - 5p

2p - 2p > 3p - 3p > 4p - 4p > 5p - 5p

(ii) Directional orbitals forms strong bonds

2s - 2s > 2s - 2p > 2p - 2p

extend of overlapping depends on head on collision or side-by-side overlapping.

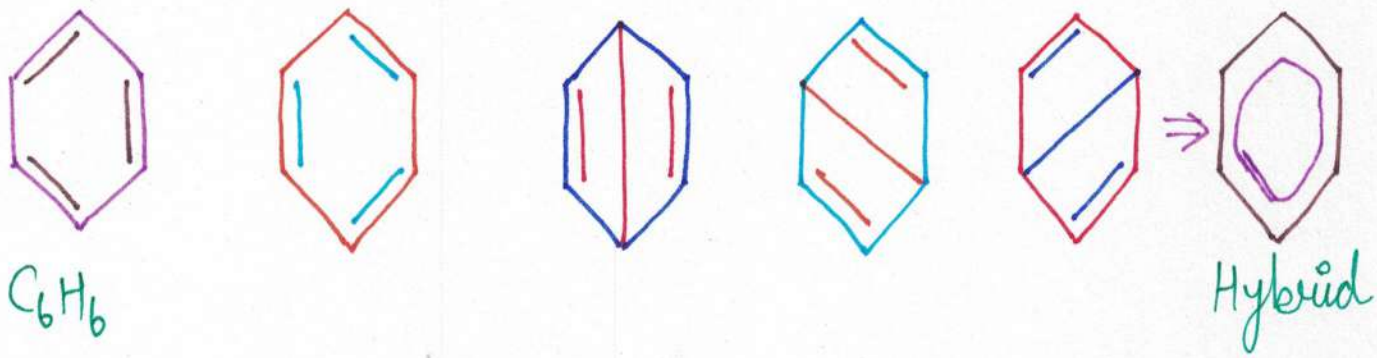
$P_x - P_x$ (co-axial) > $P_x - P_x$ (Colateral)

SIGMA-BOND

- When 2 half filled atomic orbitals overlap each other at internuclear axis, sigma bond is formed.
- Sigma bond formed by s-s, s-p, p-p, co-axial overlapping.
- Hybrid orbitals forms only σ (sigma) bond.
- Sigma bond is stronger bond.
- It does not takes part in Resonance.

Free rotation of atoms along sigma-bond takes place.

Resonatic structure of Benzene



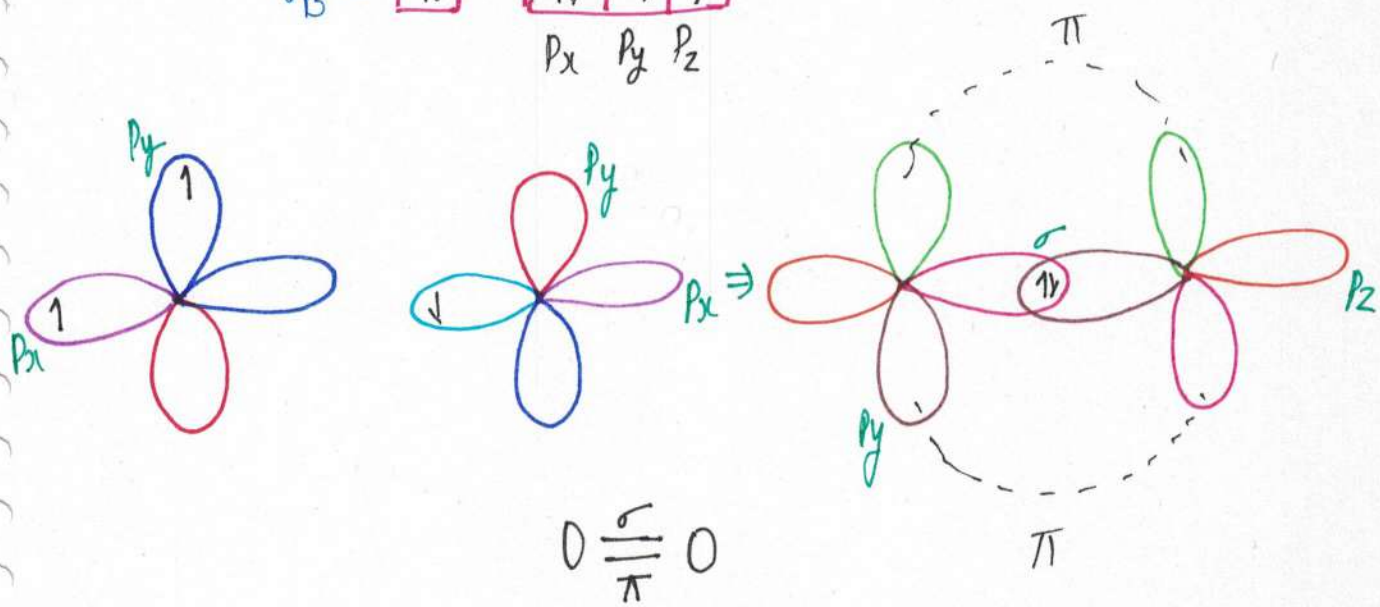
2. CO-LATERAL OVERLAPPING

When 2-half filled atomic orbitals overlap each-other at perpendicular to internuclear axis, pie bond will be formed.

e.g:- O_2 molecule

O_A = ↑↓ | ↑↓ | ↑ | ↑ — P_x P_y P_z

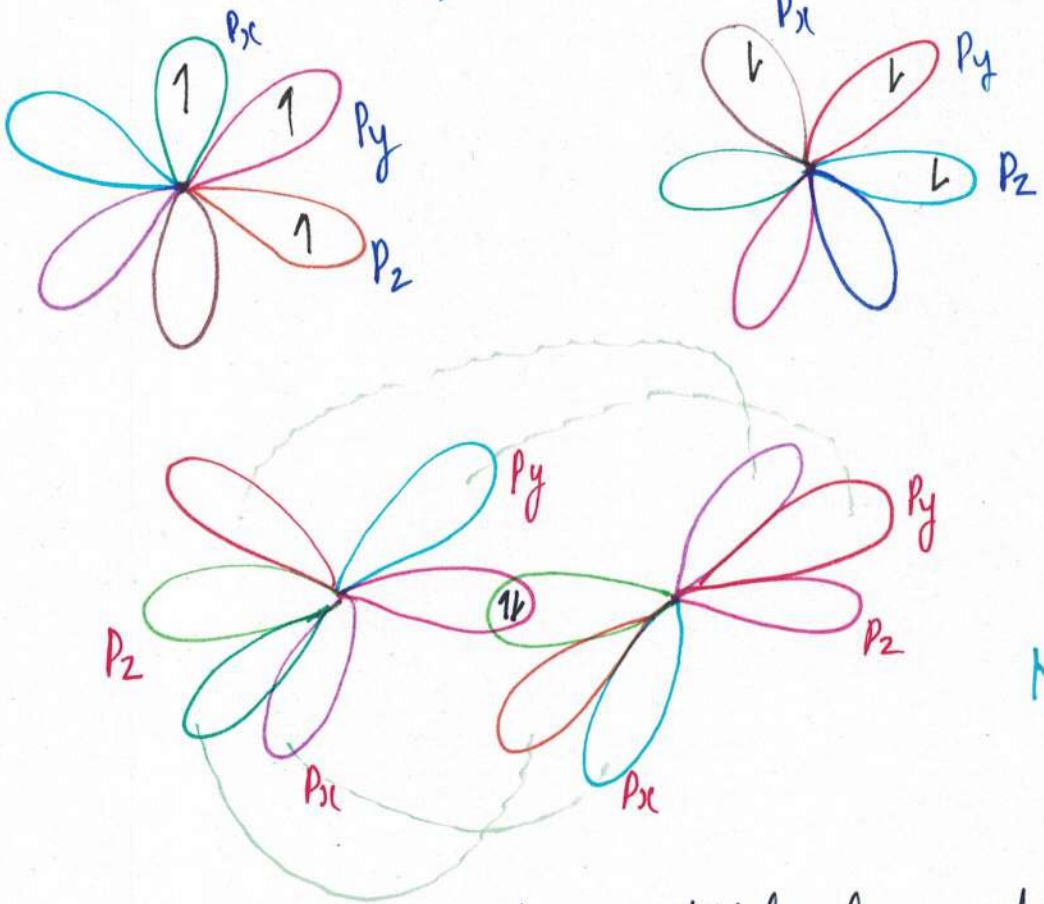
O_B = ↑↓ | ↑↓ | ↑↓ | ↑ — P_x P_y P_z



$O \overset{\sigma}{\underset{\pi}{=}} O$

e.g:- N_2 Molecule

N = | 1↓ | | 1 | 1 | 1 |
2s 2p



$N \equiv \overset{\sigma}{\underset{\pi}{}} N$

- Only unhybrid or pure orbitals form pie bond.
- Pie bond is weaker than sigma bond.
- Pie e⁻ takes part in Resonance.
- Free rotation not describe along pie bond. only restricted rotation possible.

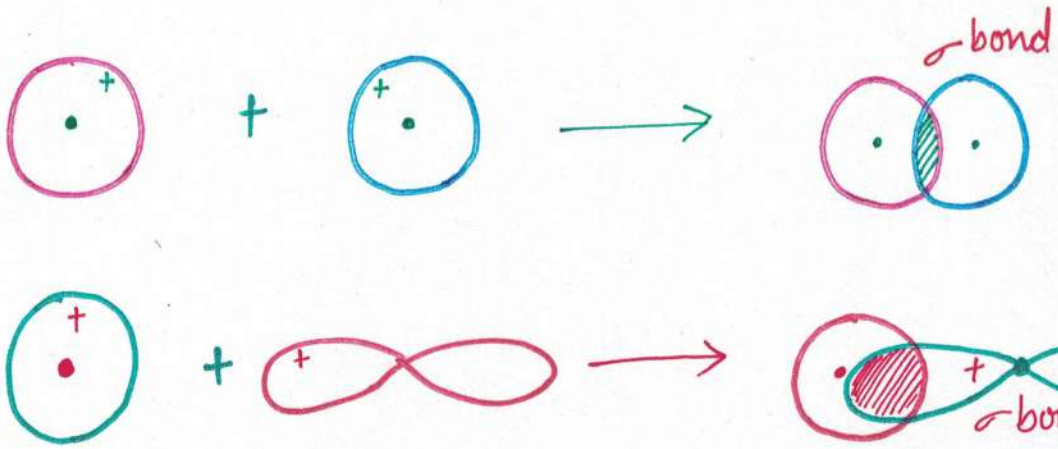
H/W Difference between Sigma & pie bond.

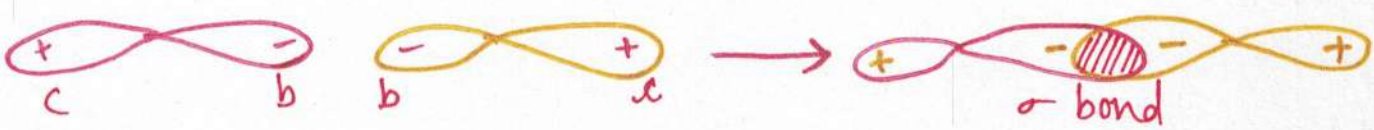
SIGMA BOND	PIE-BOND
& formed by the overlapping of 2 half filled atomic orbitals at internuclear axis.	& formed by the over-lapping of 2 half filled atomic orbitals perpendicular to internuclear axis

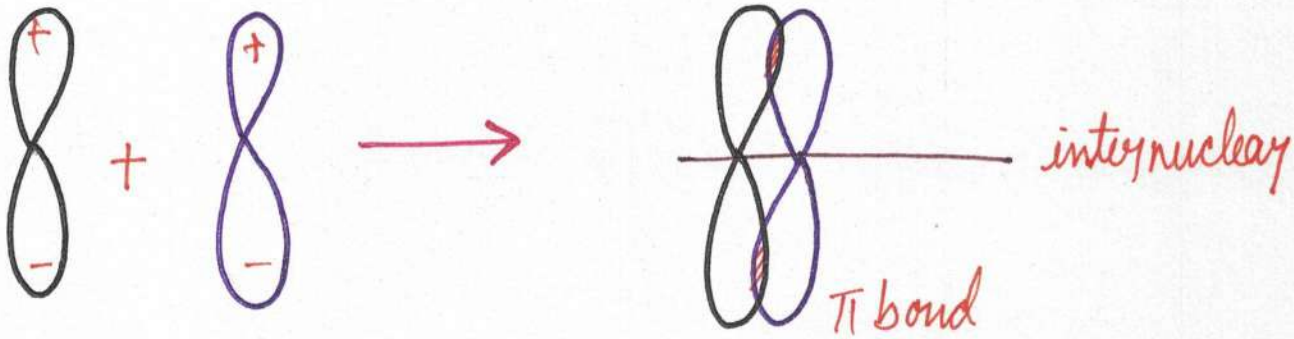
SIGMA - BOND	PIE - BOND
formed as a result of co-axial over-lapping	formed as a result of co-lateral over-lapping.
Strong Bond	Weak Bond
Sigma e^- do not take part in resonance	pie e^- takes part in resonance
Free rotation of atoms is possible	Free rotation of atoms is not possible

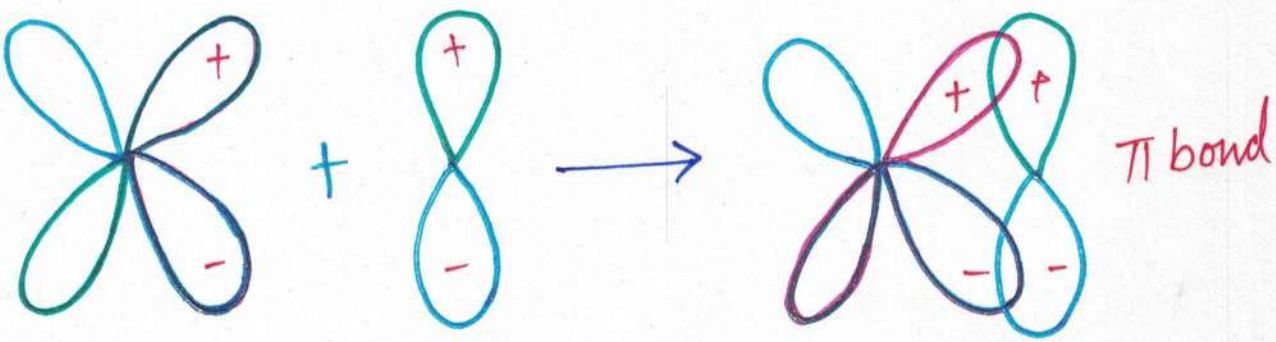
TYPES OF OVERLAPPING

1. POSITIVE OVERLAPPING

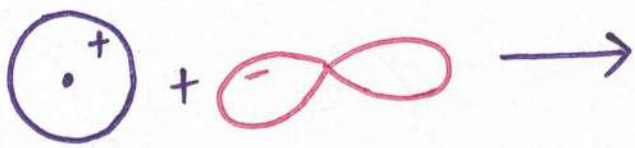


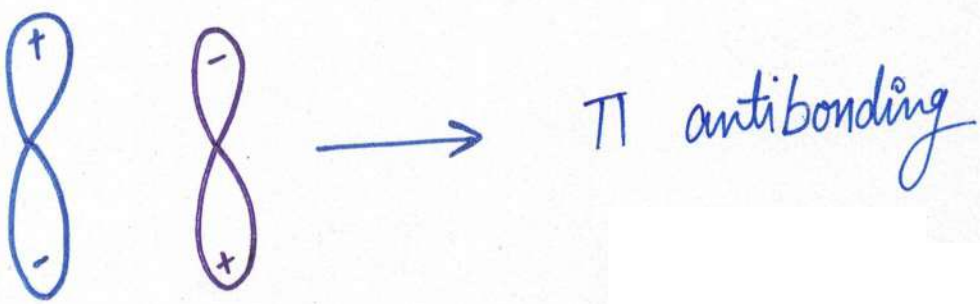


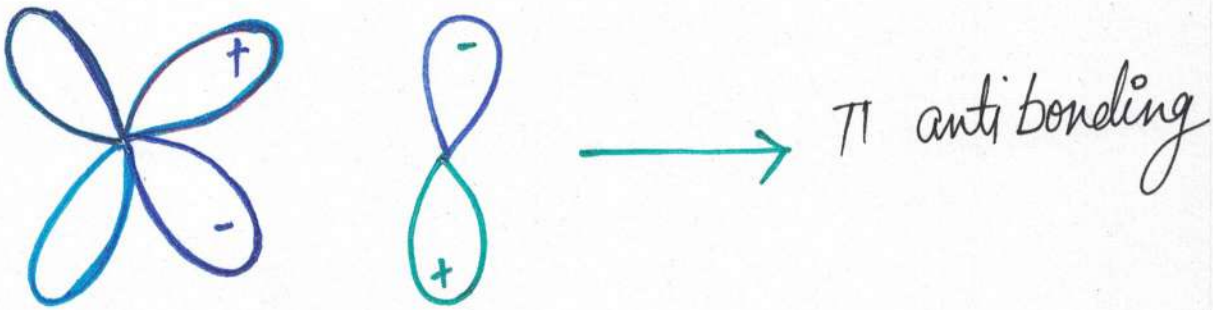




2. NEGATIVE OVERLAPPING

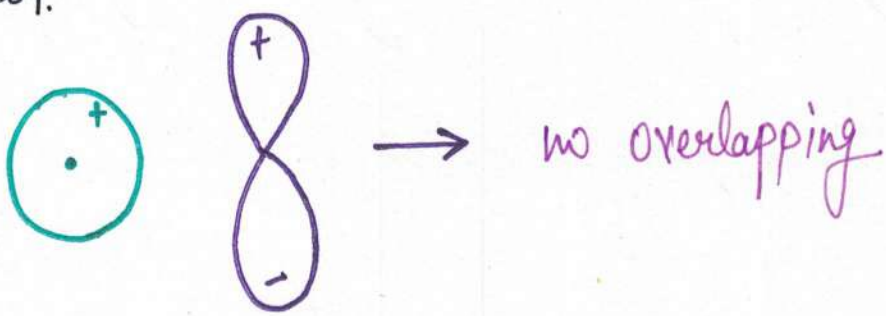


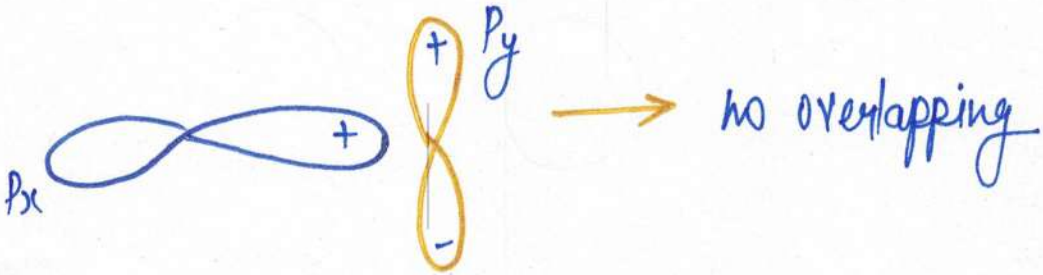




3. ZERO OVERLAPPING

Orientation of orbitals is not proper. Overlapping does not occur.





Structure-of-Covalent-Bond

- Total number of valence shell $e^{\ominus} = x$
  $e^{\oplus}$ required to complete octal / duet $= y$

  no. of bonds $= \frac{y-x}{2}$

- Total no. of valence e⁻s = V.e$^{\ominus}$ of C + V.e. of H X4
  $= 4 + 4 \times 1 = 8$

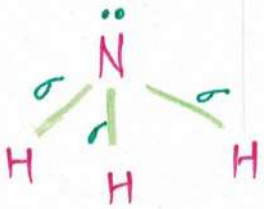
2. NH_3

Total no. of valence e^-s = V. e^- of N + V. e^- of H×3

$= 5 + 1 \times 3 = 8$

Total e^-s required = $8 + 2 \times 3 = 14$

no. of bonds = $\frac{14-8}{2} = \frac{6}{2} = 3$



3 σ bond and
1 lone pair on N atom.

3. H_2O

Valence e^- = 6+2 = 8

e^- needed = 8+2×2 = 8+4 = 12

no. of bonds = $\frac{12-8}{2} = \frac{4}{2} = 2$

2 sigma, 2 lone pair

4. HF

Valence e^- = 1+7 = 8

e^- needed = 2+8 = 10

no. of bonds = $\frac{10-8}{2} = \frac{2}{2} = 1$

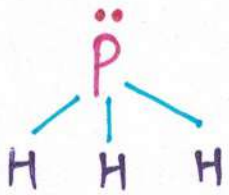
3 lone pair
1 sigma bond

5. PH_3 [Phosphine]

Valence e^- = 5+3 = 8

e^- needed = 8+3(2) = 14

no. of bonds = $\frac{14-8}{2} = \frac{6}{2} = 3$



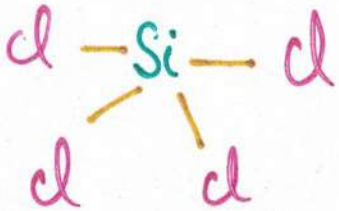
1 lone pair
3 σ bond

6. $SiCl_4$

Valence e^- = 4+7(4) = 28+4 = 32

e^- needed = 8+8(4) = 8+32 = 40

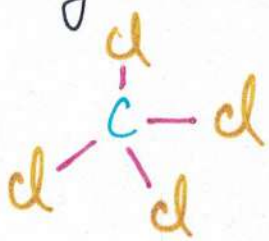
no. of bond = $\frac{40-32}{2} = 4$



4 σ bond 0 lone pair

7. CCl_4

no. of bonds = 4



4 σ bond
0 lone pair

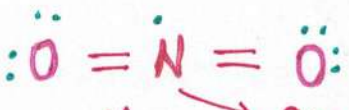
II. Gen Oxidation state of central atom = no. of bond

1. NITRATE OXIDE (NO)

- oxidation state of N = +2 = no. of bonds

2. NITROGEN DIOXIDE (NO_2)

one co-ordinate bond is formed b/w N & O.



it cannot have ce in its valence shell

3. SULPHUR DIOXIDE (SO_2)

O.S of S = 4

2 σ 2 π
one lone pair on S

4. SULPHUR TRIOXIDE (SO_3)

3 σ, 3 π
0 lone pair on S

5. SULPHUROUS ACID (H_2SO_3)

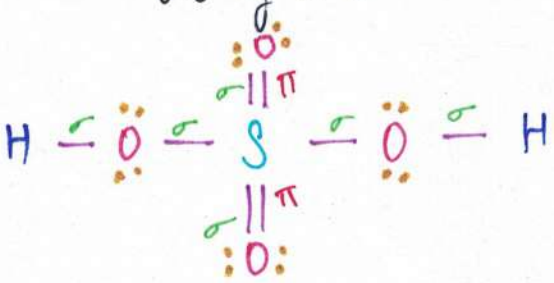
O.S of S = 4

5σ, 1π

1 lone pair on S

6. SULPHURIC ACID (H_2SO_4)

O.S of S = 6



6σ, 2π

0 lone pair on S

8 lone pair on O atoms.

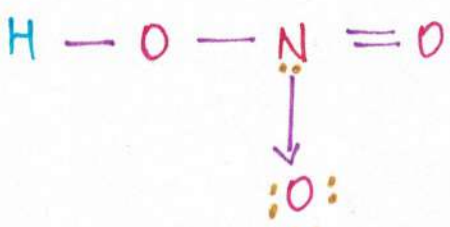
7. NITROUS ACID (HNO_2^-)

O.S of N = +3

3σ, 1π, 1 lp. N

8. NITRIC ACID (HNO_3)

O.S of N = 5



(But N cannot make 5 bonds)

4σ, 1π, 0 lp on N

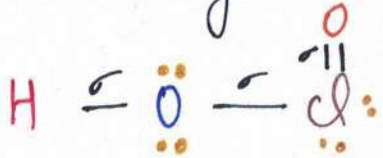
9. HYPOCHLOROUS ACID (HCLO)

3 lp on cl atom, 2 bond.

10 CHLOROUS ACID ($HClO_2$)

O.S of cl = 3

3σ, 1π
2 lp on cl



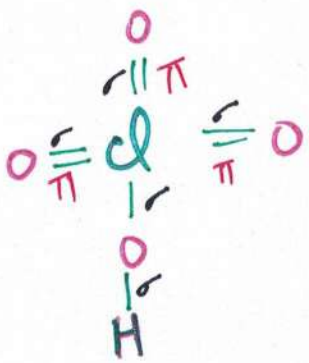
11 CHLORIC ACID ($HClO_3$)

2π 4σ
1 l.p

12. PERCHLORIC ACID ($HClO_4$)

O.S of cl = 7

5σ, 3π 0 l.p.

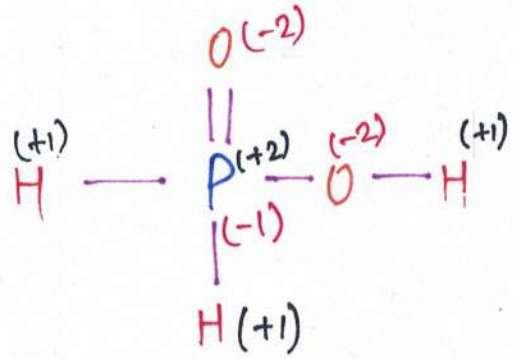


13. HYPHORPHOROUS ACID (H_3PO_2)

O.S of P = 7

in oxyacids P always forms P=O

no. of bonds = 5

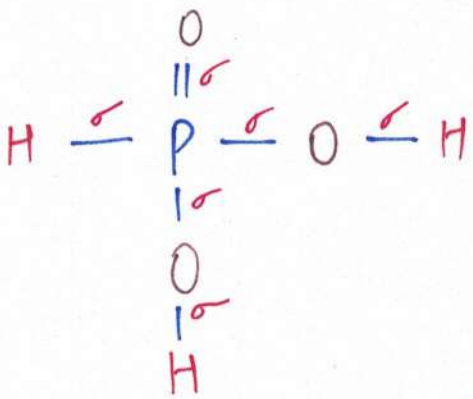


5σ, 1π

0 lone pair on P

in oxiacids of P, P=O is formed & is stable. Generally P. forms 5 bond in oxiacids for P, d pie-p pie bonding is strong.

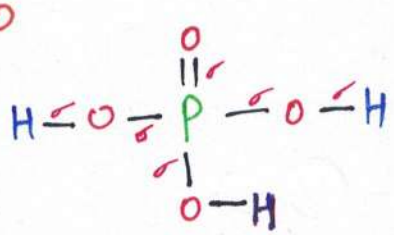
14. O-PHOSPHOROUS ACID (H_3PO_3)



6σ, 1π

0 l.p

15. PHOSPHORIC ACID (H_3PO_4)



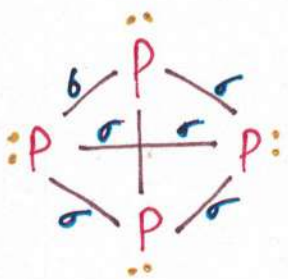
7σ, 1π

0 lp

Hydrogen attached with O atom or more E.N element is acidic H. Hydrogen directly attached to element is called reducing H.

III When O.S of central atom is less than its group. no. X-X linkage present in a compound. when more than one central atom present in a compound.

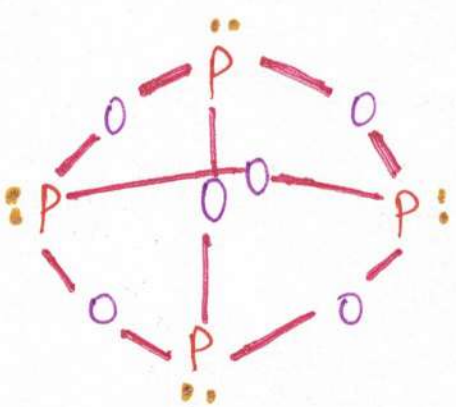
1. P_4 WHITE PHOSPHORUS



6 σ bond
σ P-P linkage
& total 4 lp on P.

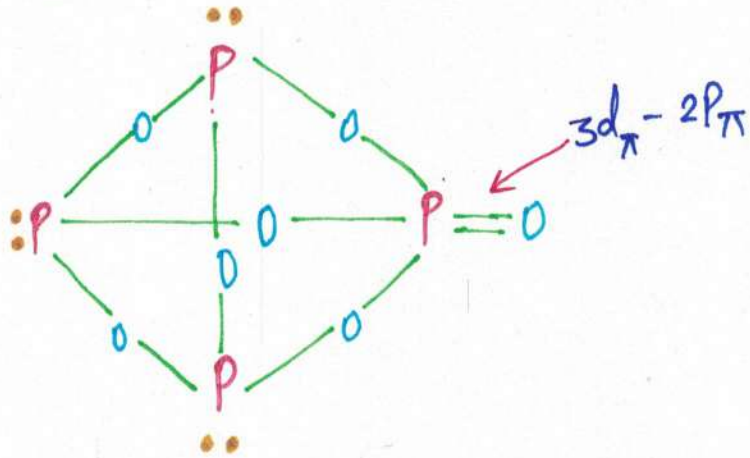
2. P_4O_6 PHOSPHORUS TRIOXIDE

O.S of P is 3 (less than g no.) but P-P linkage not present Instead P-O-P linkage is present.



6 P-O-P linkage
P-P linkage = 0
4 lp on P

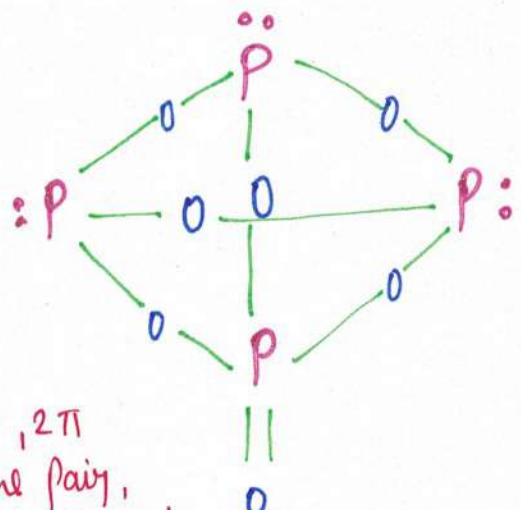
3. P_4O_7 PHOSPHORUS OXIDE



3 lp on P

6 P-O-P linkage

4. P_4O_8 TETRAPHOSPHORUS OCTAOXIDE

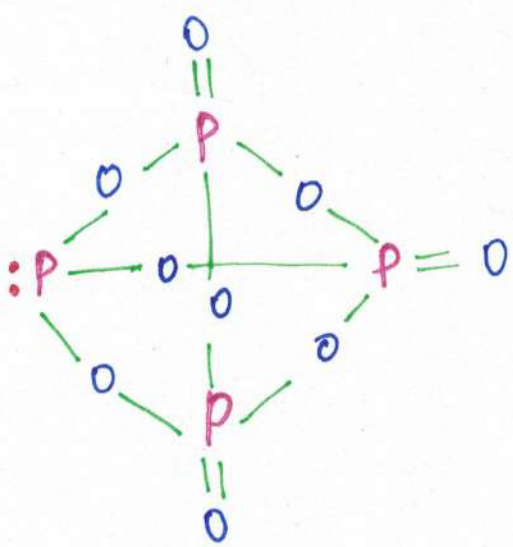


15 σ, 2 π

2 lone pair,

two $d\pi - p\pi$ bond

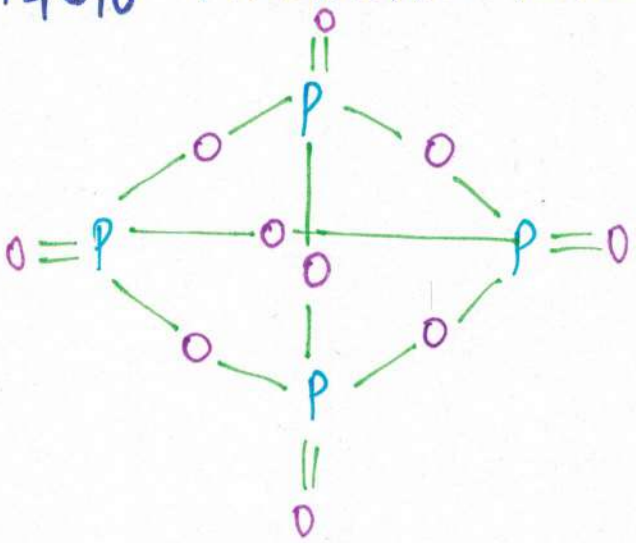
5. P_4O_9 TETRAPHOSPHOROUS NONOXIDE



$3d\pi - p\pi$ bond

1 lone pair

6. P_4O_{10} PHOSPHORUS PENTOXIDE



4 $d_\pi - P_\pi$ bond
Zero lone pair

$$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$$

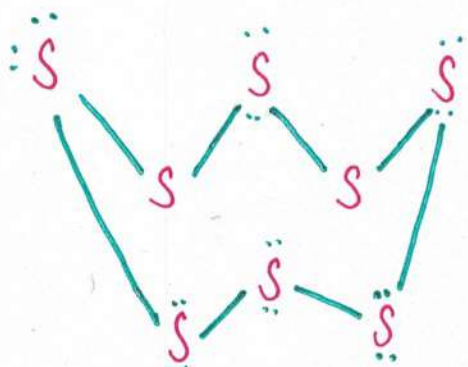
$$P_4O_7 + 6H_2O \longrightarrow 3H_3PO_3 + H_3PO_4$$

$$P_4O_8 + 6H_2O \longrightarrow 2H_3PO_3 + 2H_3PO_4$$

$$P_4O_9 + 6H_2O \longrightarrow 3H_3PO_4 + H_3PO_3$$

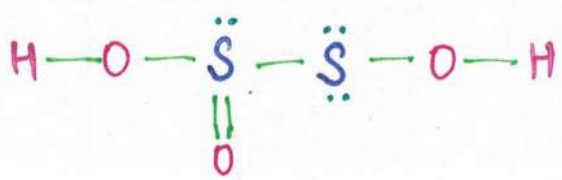
$$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$$

7. S_8 SULPHUR



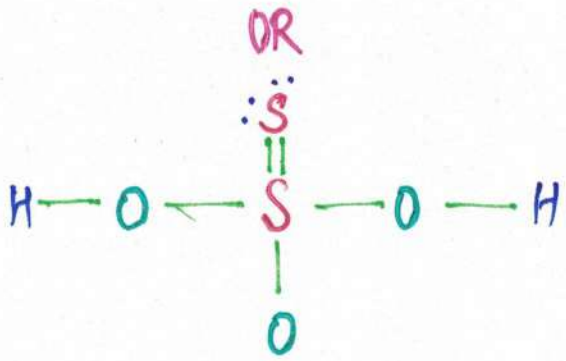
8 S-S bond
16 lone pair

8. $H_2S_2O_3$ THIOSULFURIC ACID



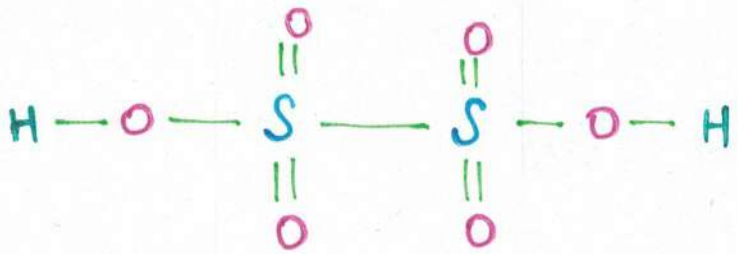
average O.S of $S = \frac{4+0}{2} = 2$

OR



Thionic Series :- $H_2S_xO_6$ where $x \geq 2$

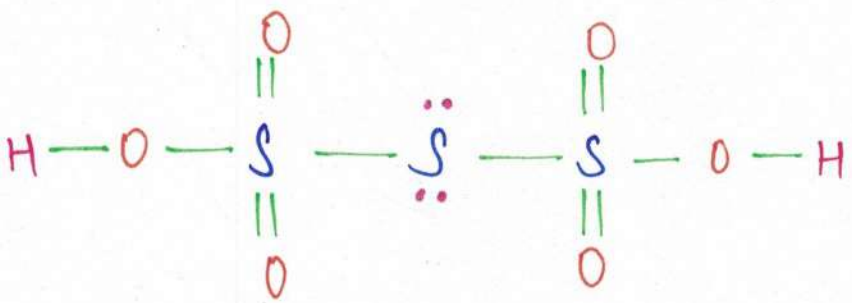
9. $H_2S_2O_6$ DITHIONIC ACID



9σ 4π
Zero lone pair on S atom

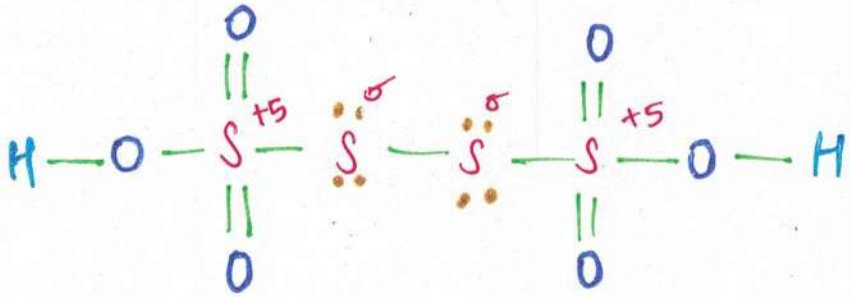
In Polythionic acid, oxygen atom attached with terminal S-atom

10 $H_2S_3O_6$ TRITHIONIC ACID



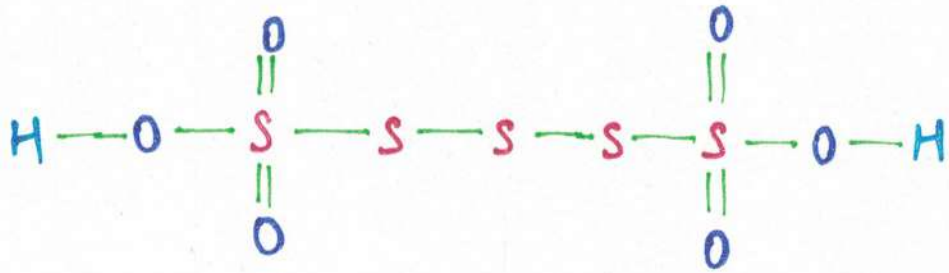
10σ, 4π
2 lone pair on S atom

11. $H_2S_4O_6$ TETRATHEONIC ACID



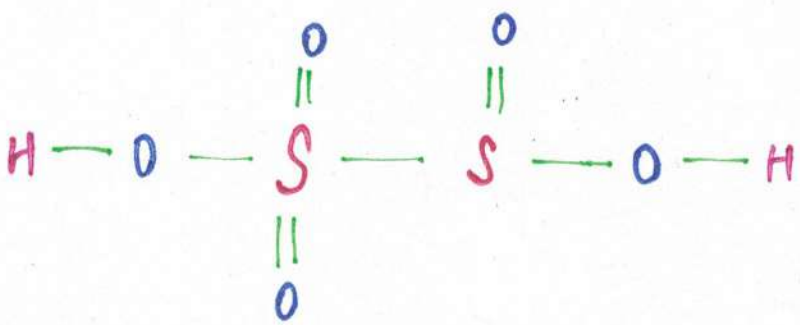
11σ 4π bond
4 lp
no. of S-S linkage = 4

12. $H_2S_5O_6$ PENTATHIONIC ACID



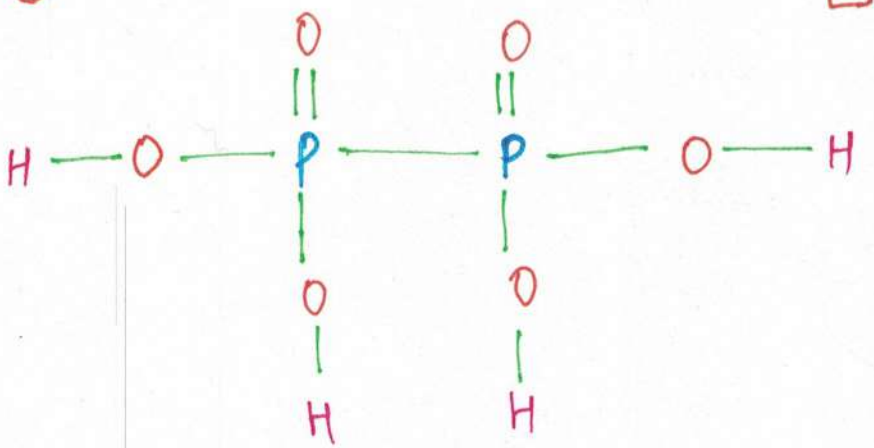
12σ, 4π bond
6 lp
no. of S-S linkage = 5

13. $H_2S_2O_5$ PYROSULPHUROUS ACID



8σ, 3π bond
1 lone pair

14. $H_4P_2O_6$ HYPOPHOSPHORIC ACID

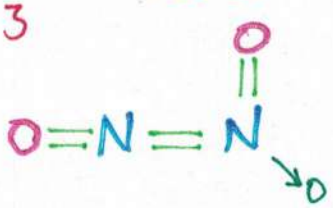


11σ 2π
0 lone pair
no. of acidic H = 4

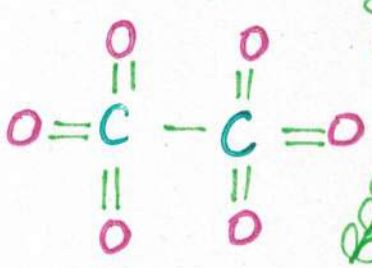
IV When oxidation state of central atom is equal to its group no. X-O-X linkage is present in a compound.

Rull no:- 3

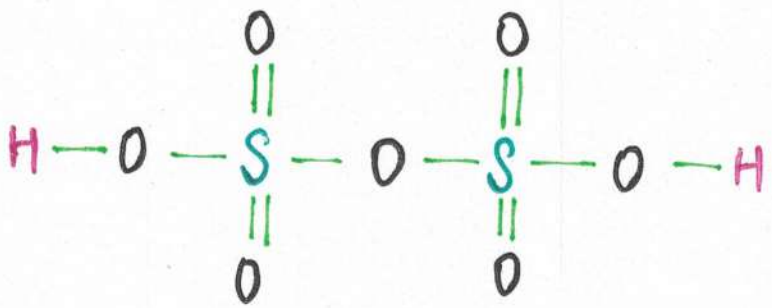
N_2O_3



Cl_2O_6

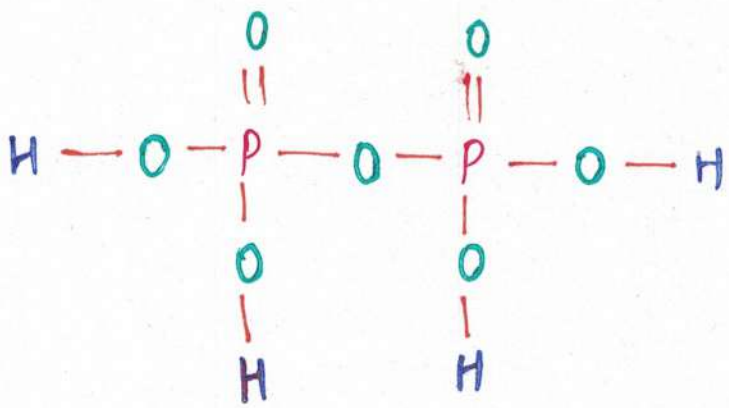


1. $H_2S_2O_7$ Pyrosulphuric acid (oleum)



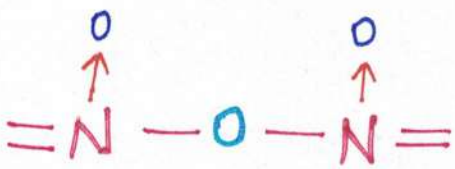
$4 d_\pi - p_\pi$ bond
10 σ bond
1 S-O-S linkage.

2. $H_4P_2O_7$ Pyrophosphoric acid



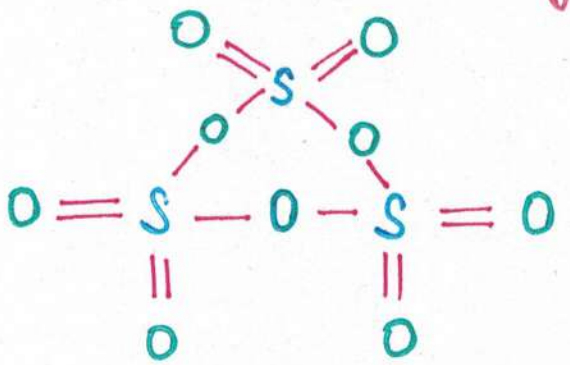
12 σ, $2 d_\pi - p_\pi$ bond
0 σ bond -
1 P-O-P linkage

3. N_2O_5 Nitrogen Pentaoxide



6 σ $12 d_\pi - p_\pi$
one N-O-N linkage

4. SO_3 Solid S_3O_9 Trimer of sulphur trioxide

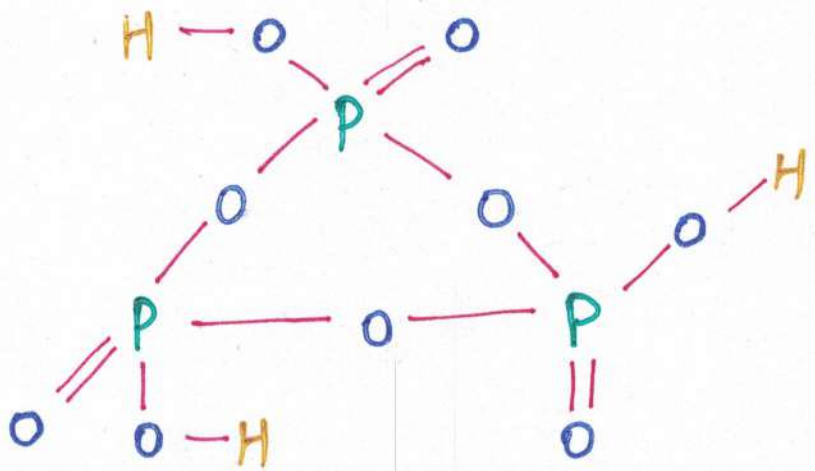


3 S-O-S linkage

6 $d\pi - p\pi$, 12 σ zero lp.

5. Meta phosphoric acid doesn't exist in monomer form. It exist in polymer form

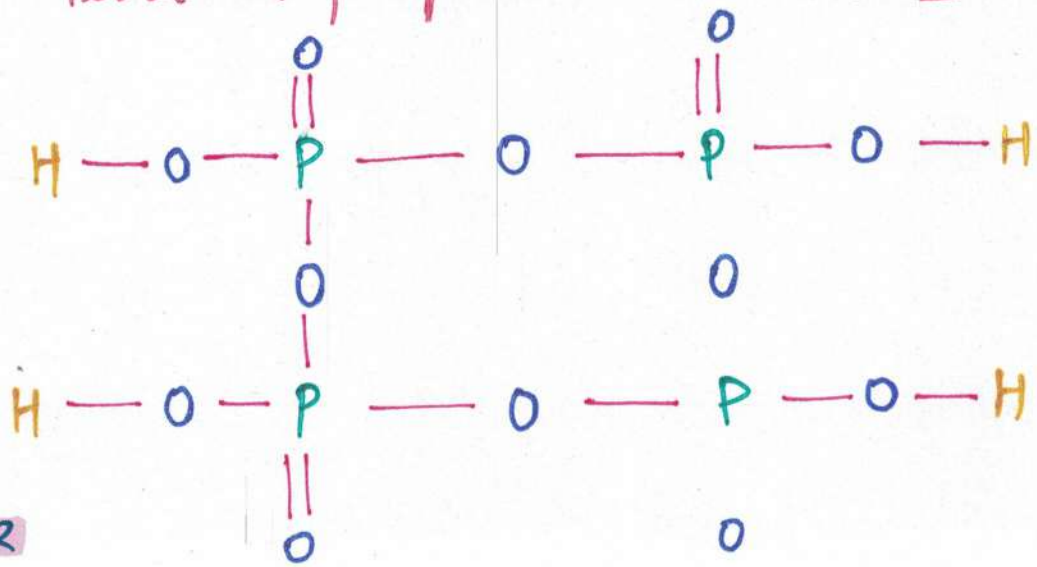
cyclic trimeta phosphoric acid $[H_3P_3O_9]$



15 σ, 3 $d\pi - p\pi$

3 P-OP linkage

Tetrameta phosphoric acid $[H_4P_4O_{12}]$



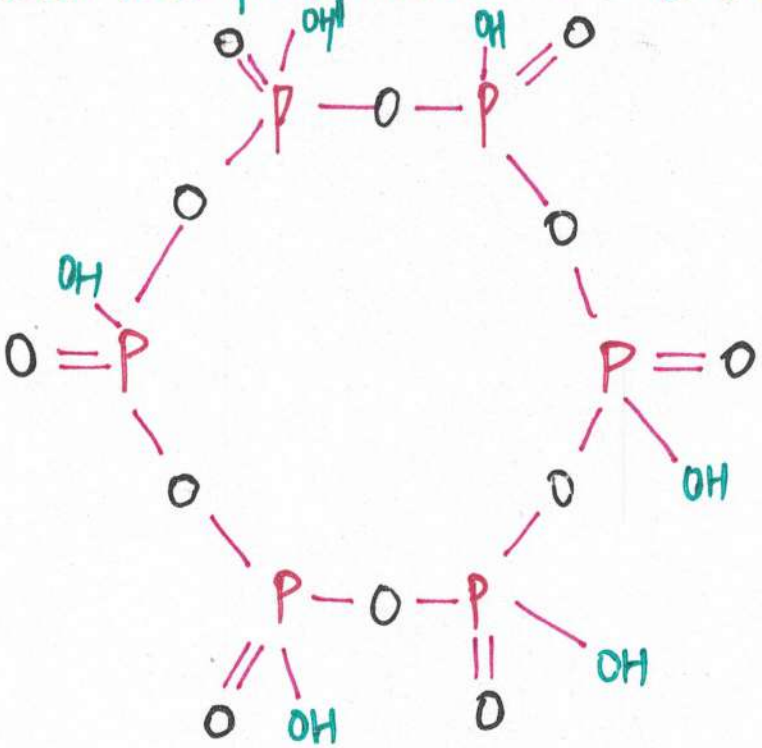
20 σ

4 $d\pi - p\pi$ bond

4 P-O-P linkage

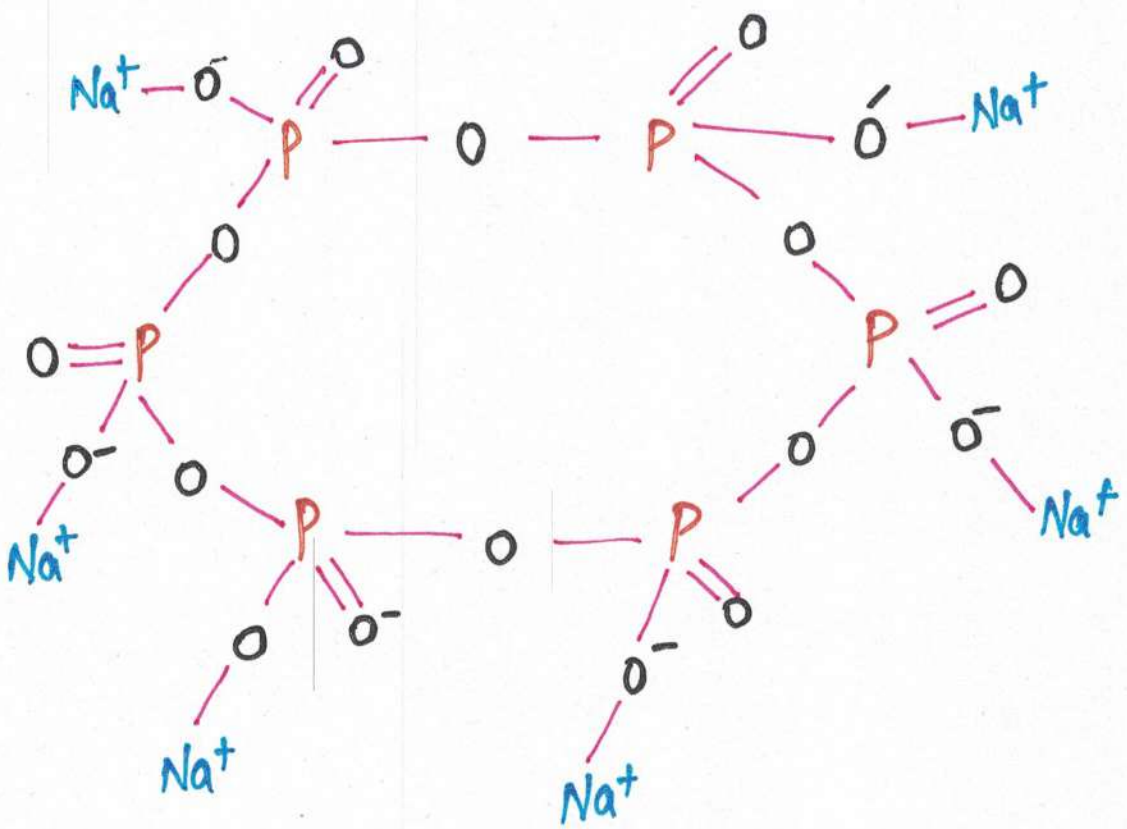
Zero lp.

Hexametaphosphoric acid $[H_6P_6O_{18}]$



30 σ
6 dπ - pπ bond
6 P-O-P linkage

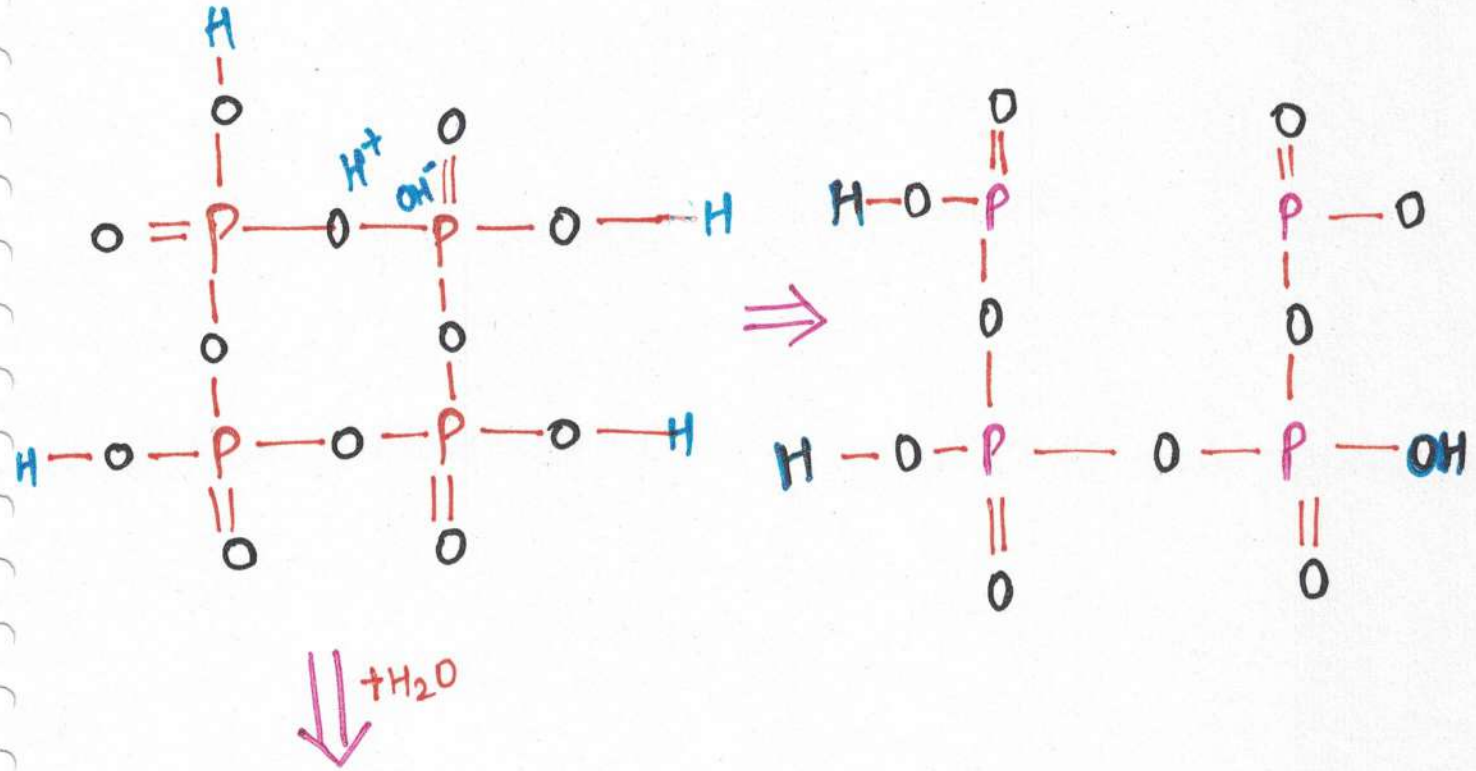
Calogen (Water softner) $[Na_6(PO_3)_6]$
Sodium HexaMeta phosphate



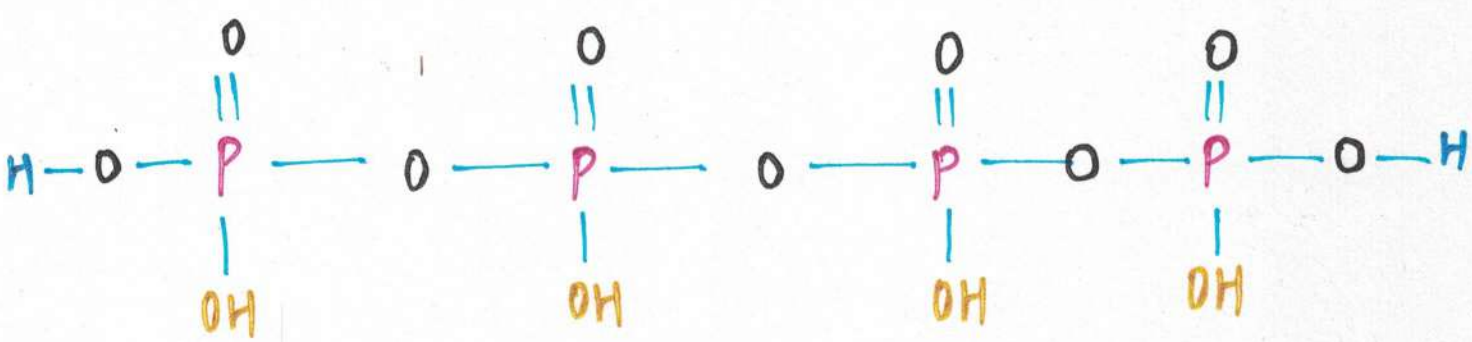
24 σ
6 dπ - pπ bond
6 P-O-P linkage

Tetrameta phosphoric acid $(HPO_3)_4$

$(HPO_3)_4$ $\xrightarrow{\text{Hydrolysis}}$ Product/s is/are :-



$\xrightarrow{+H_2O}$



$H_6P_4O_{13}$

Cyclic $\rightarrow$ $H_nP_nO_{3n}$

linear polymer $\rightarrow$ $H_{n+2}P_nO_{3n+1}$

$(HPO_3)_4$ $\xrightarrow{+2H_2O}$ 2 mole $H_4P_2O_7$ pyrophosphoric acid

$(HPO_3)_4$ $\xrightarrow{+4H_2O}$ 4 mole H_3PO_4 o-phosphoric acid

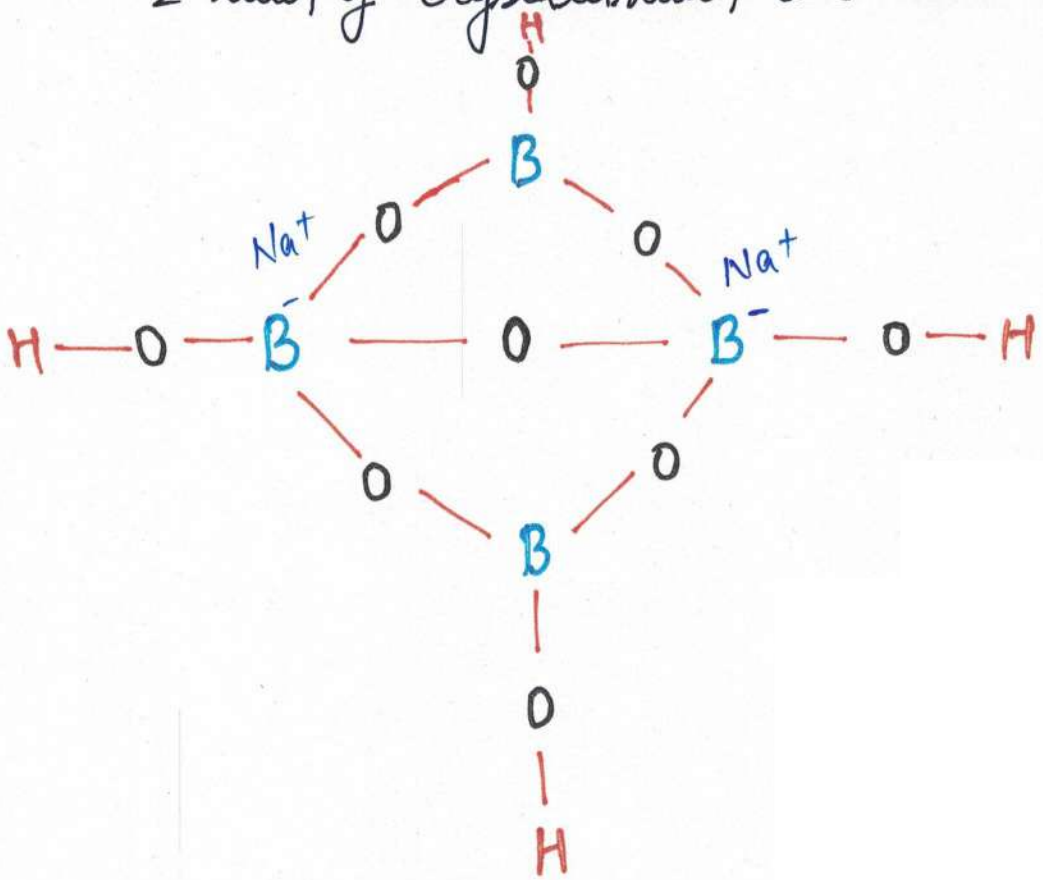
Products are $H_6P_4O_{13}$ → linear polymer

(chain polymer of O- phosphoric acid)

$H_4P_2O_7$ and H_3PO_4

Borax :- $Na_2B_4O_7 \cdot 10H_2O$

2 water of crystallisation are added in its structure.



5 B-O-B bond, 4 B-O-H bond, 8 crystalline H_2O

2B (three bond) 2B (four bond)

V. If O.S of central atom is more than its group no. perosy linkage is present in a compound.

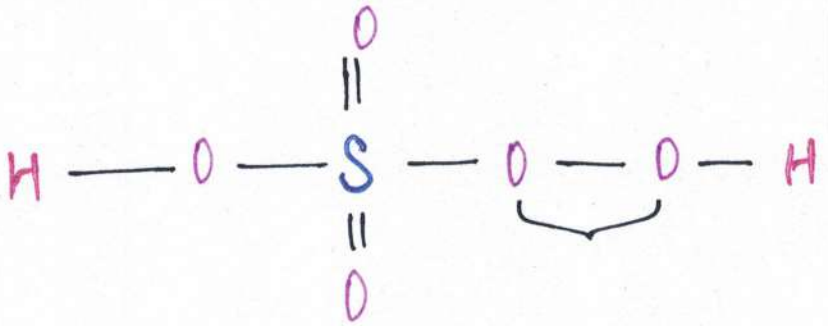
1. Hydrogen Peroxide H_2O_2

H — O — O — H

no. of proxy linkage = $\frac{\text{Calculated O.S} - \text{group no.}}{2}$

2. Caro's acid H_2SO_5

no. of proxy linkage = $\frac{8-6}{2} = \frac{2}{2} = 1$

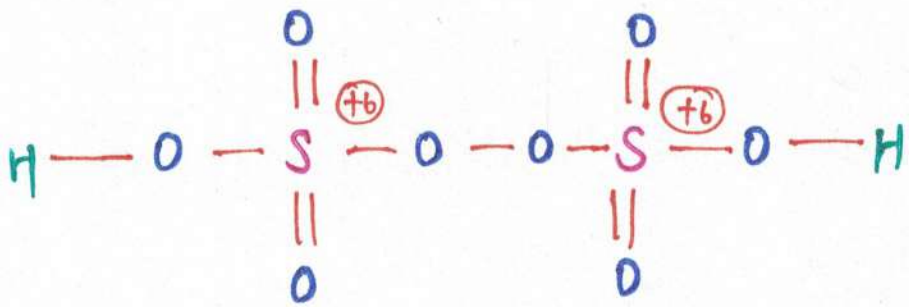


Peroxy meno sulphuric acid

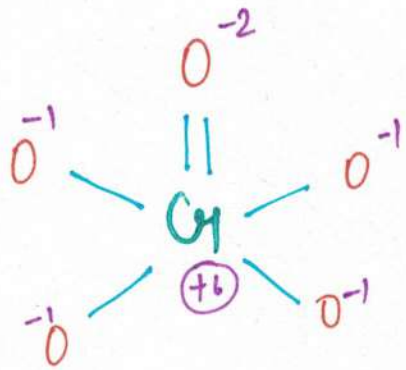
3. Peroxy disulphuric acid $H_2S_2O_8$

[Marshall's acid]

(+7)



4. Chromium pentaoxide CrO_5

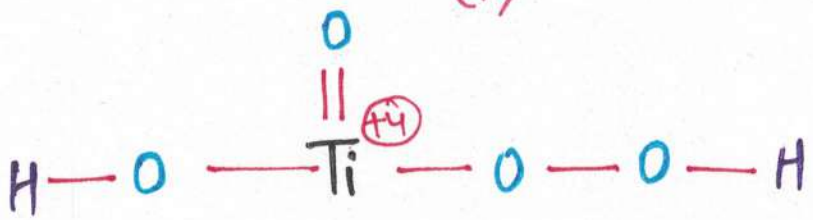


2 proxy linkage

(Butterfly structure)

5. Pertitanic acid H_2TiO_4

(+6)

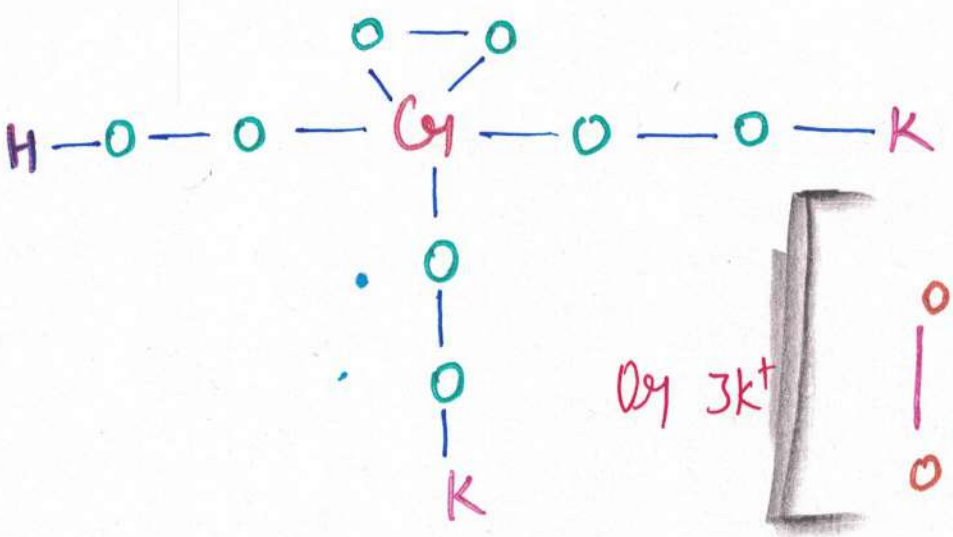


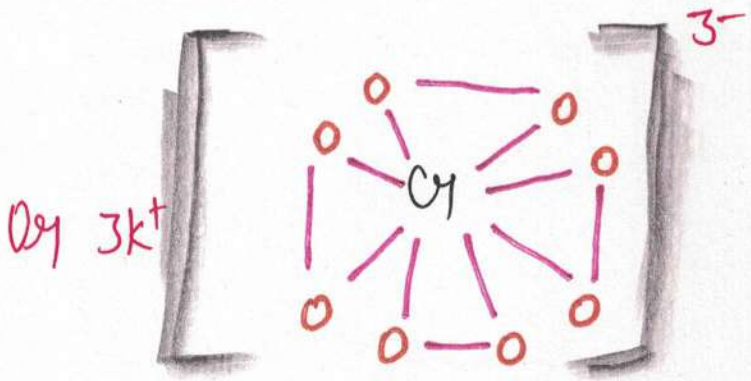
6. $K_2Cr_4O_8$

O.S of Cr is +5

C O.S = 13

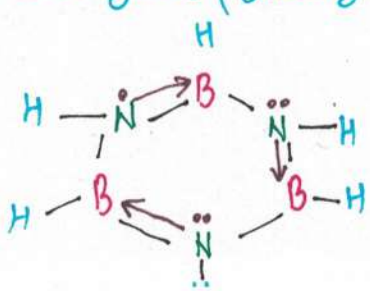
peroxy link = $\frac{13-5}{2} = 4$





General Example

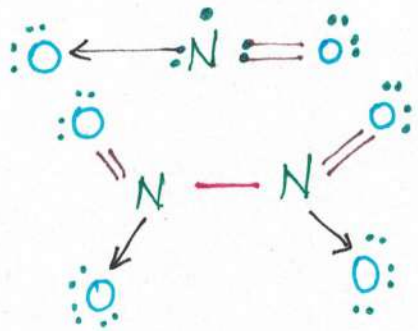
1. Inorganic Benzene (Borazine / {Borazole})



12 σ bond

3 co-ordinate bond

2. NO_2 Nitrogen dioxide



(monomer)
(dimer)

CORDINETE BOND

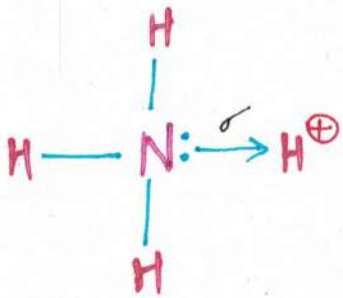
It is formed by unequal sharing of e^-'s between 2 atoms.

Conditions :-

- One atom should have vacant orbital and another atom should have lone pair of e^-.
- It is represented by an arrow [from donor to accepter]

Example :-

1. Ammonium ion NH_4^+



donor → N
accepter → H

2. Ozone O_3

3. Carbon monoxide

$:C \leftarrow\!\!\equiv O$ (π and σ, π)

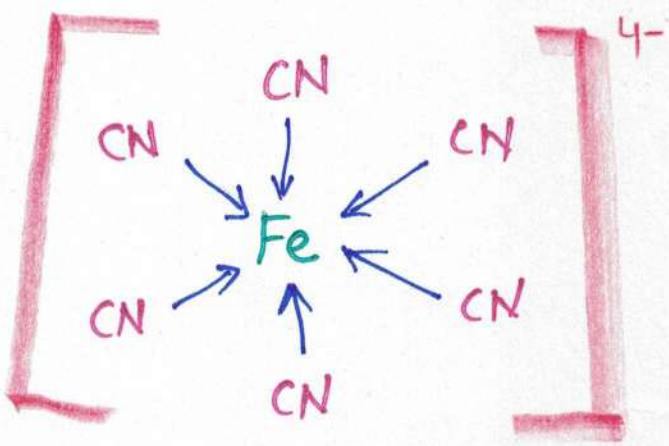
4. Co-ordination compounds

$K_4[Fe(CN)_6]$ Potassium Ferrocyanide

$K_4[Fe(CN)_6] \longrightarrow 4K^+ + [Fe(CN)_6]^{4-}$

$Fe \leftarrow :\overset{\ominus}{C} \equiv N:$

acceptor $\leftarrow$ donor



PROPERTIES

1. Melting point and boiling point of co-ordinate compounds are higher than covalent compounds and lower than ionic compounds.
2. Co-ordinate compounds are soluble in H_2O. They conduct electricity.
3. They show isomersion. because co-ordinate bond has directional character.

BACK BONDING

After sigma bond formation, if co-ordinate bond is formed between 2 atoms, bond is known as back bond and phenomenon is known as **Back Bonding**.

Conditions :—

- One atom should have vacant orbitals.
- Another atom should have lone pair of e⁻.

Type's of Back Bonding

- $p\pi - p\pi$
- $p\pi - d\pi$
- $d\pi - d\pi$

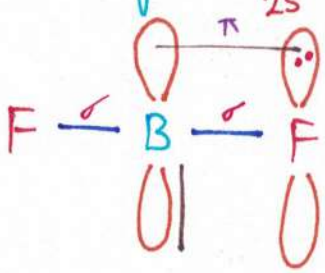
Eg :- Boron trihalides $[BX_3]$

X = F, Cl, Br, I

B_{gs} = [↑↓] [1] [ ] [ ]
2s 2p

B_{es} = [1] [1] [1] [ ]

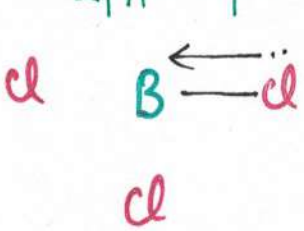
X = $ns^2\ np_x^2\ np_y^2\ np_z^2$



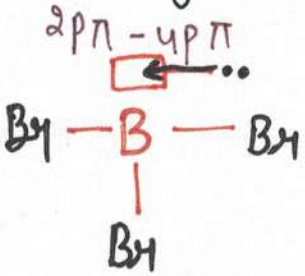
one F will donate its lone pair to vacant p orbital of B.

Bonding will be back bonding as bond is formed second time b/w B and F. This bond is strong as principal quantum no. of F and B is same.

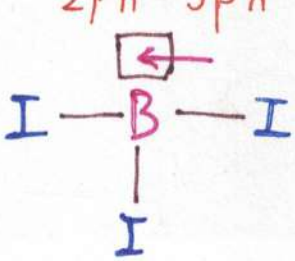
$2p\pi - 3p\pi$



$2p\pi - 4p\pi$



$2p\pi - 5p\pi$



Strength of Backbonding = $BF_3 > BCl_3 > BBr_3 > BI_3$

LEWIS ACID CHARACTER

Electron deficient compounds are known as Lewis acid. Generally IIIA group element forms electron deficient compounds.

$BF_3 < BCl_3 < BBr_3 < BI_3$

Lewis acid character $\propto \frac{1}{\text{BackBonding}}$

Other example are:-

Carbon monoxide
metal carbonile (CO)

Question:- BH_3 does not exist in monomer form. It exist in dimer form whereas Boron trihalides exist in monomer form.

Answer:- B has vacant orbitals but no lone pair present on valence shell of H atom. Therefore, backbonding is not possible between B and H whereas all halogens have 3 lone pair of electrons.

3 centered 2- electron bond

(a) Central atom should have vacant orbitals. (incomplete octet)

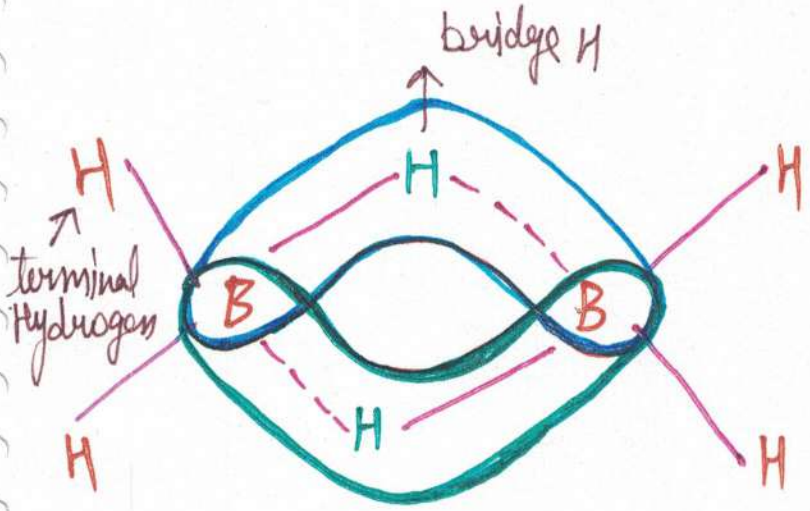
(b) bridge atom should not have lone pair of e^- (complete due octet configuration).

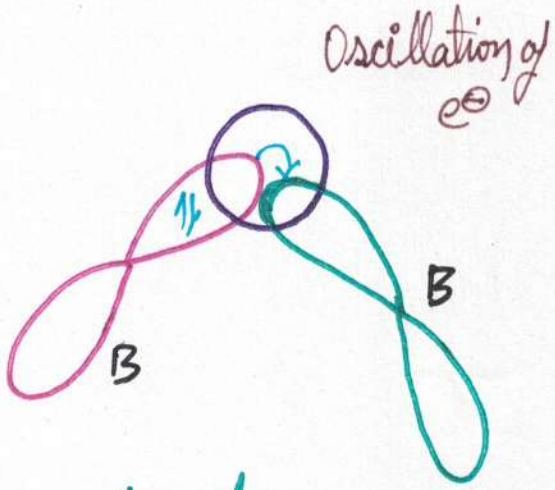
e.g:- B_2H_6 [diborane]

(1) 4 terminal B-H bond.

(2.) 2 Bridge B-H bond - one H lie above the plane and other H lie below the plane.

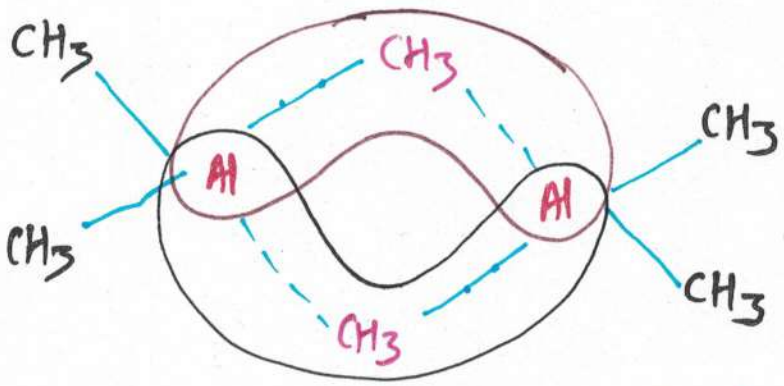
(3.) Bridge B-H bond is longer & stronger than terminal B-H bond.





Bridge bond or Banana bond

e.g:- $Al_2(CH_3)_6$

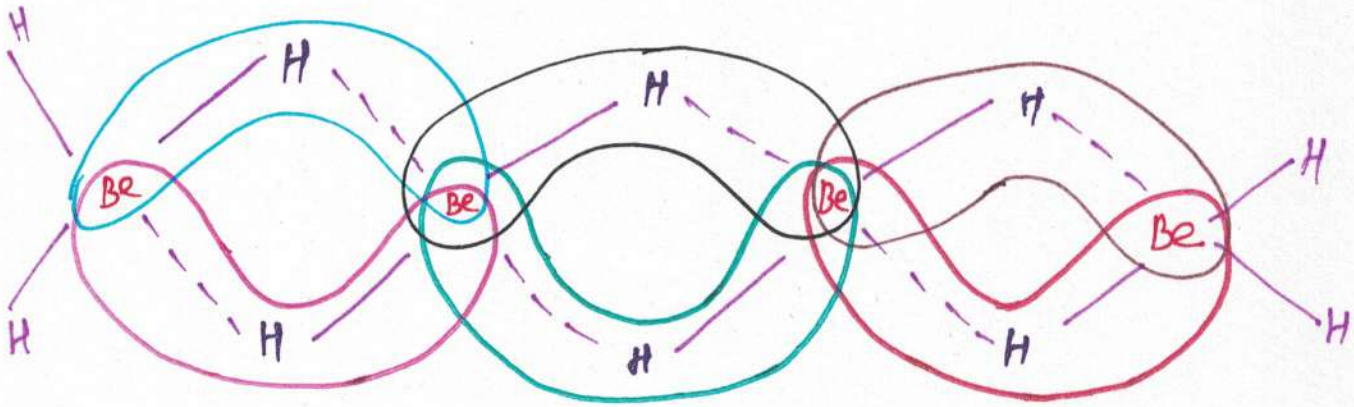


3c - $2e^-$ Bridge bond

e.g:- BeH_2

It exist in polymer form.

It's a unique compound in which all bonds are banana bonds.

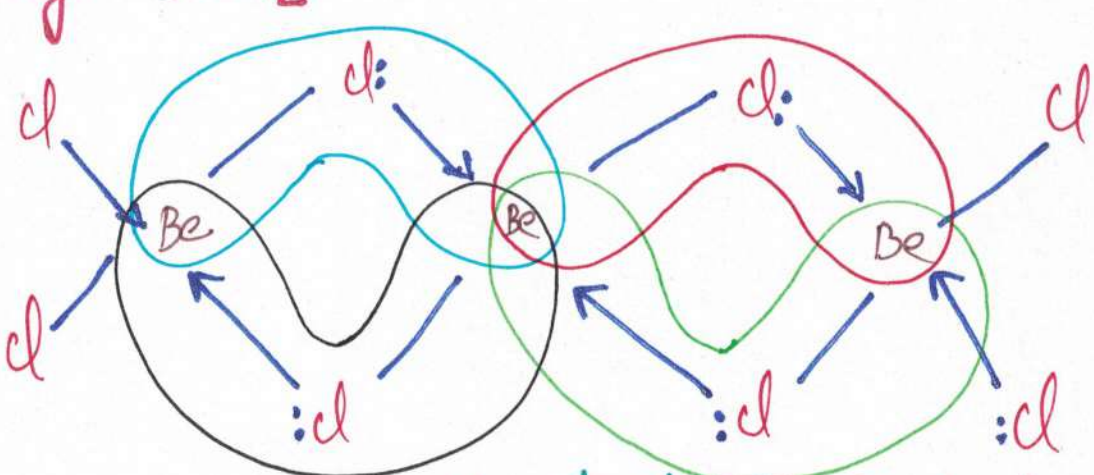


BeH_2

3 Centered 4- electron bond

1. One atom should have lone pair of e⁻'s.
2. one atom should have vacant orbitals.

e.g:- $BeCl_2$

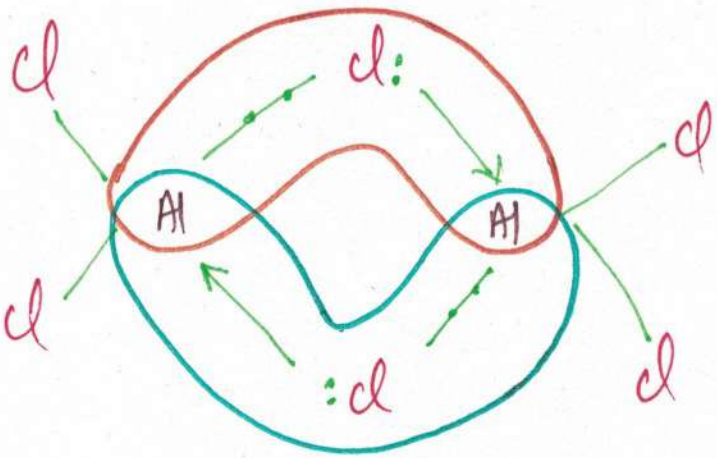


3 centered 4- electron bond

AlF_3 is ionic compound

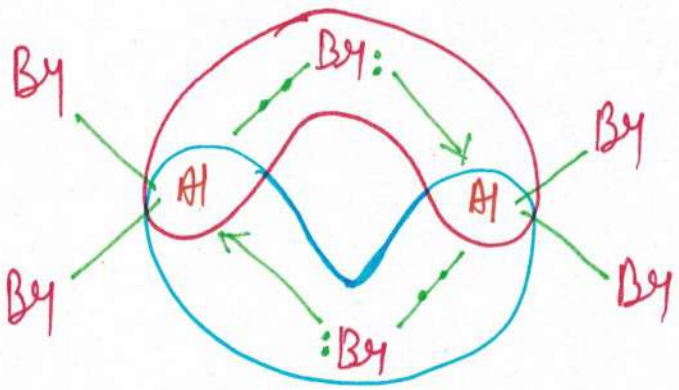
Al^{+3} 3F

$AlCl_3$ exist in diner form in solid state.



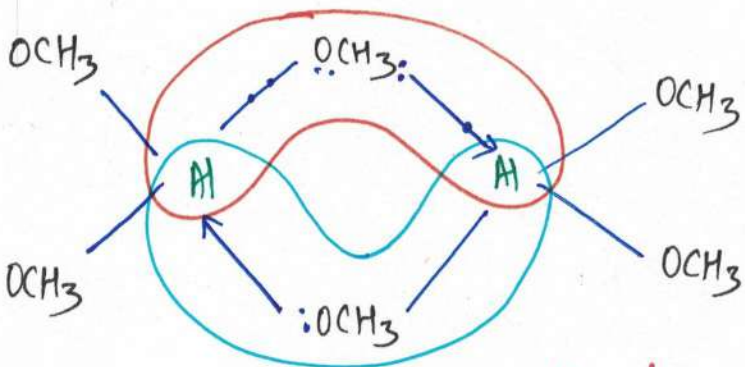
$AlCl_3$ - 3 centered 4-e⁻ bond

$AlBr_2$ exist in dimer form.



AlI_3 doesn't exist
Size of I is very large.

$Al_2(OCH_3)_6$



3 c 4e⁻ bond

HYBRIDISATION

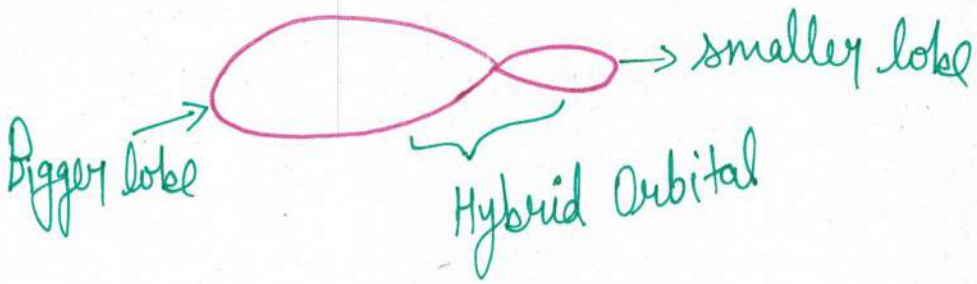
It is a hypothetical phenomenon which is used to explain geometry of molecule and bond parameters like bond length bond energy and bond angle.

HYBRIDISATION :- A phenomenon in which intermixing of atomy orbitals of almost same size takes place. The new orbitals formed are called hybrid orbitals.

- no. of pure atomic orbitals participating in hybridisation is equal to the no. of hybrid orbitals formed.

eg:- one S + one p = two new sp hybrid orbitals
one S + two p = three new sp^2
one s + three p = four new sp^3
one S + three p + one d = five new sp^3d
one s + three p + two d = six new sp^3d^2

- One hybrid orbital is represented by one bigger lobe & other smaller lobe. Always, the bigger lobe participate in bonding.



- Hybrid Orbitals form only sigma bond. Unhybrid (pure) orbitals form pie (π) bond.
- Completly filled half filled, empty orbitals. participate in hybridisation.

TYPES OF HYBRIDISATION

I. SP hybridisation

steric number (SN) = no. of σ bond + no. of lone pair of e⁻'s

SN = 2 ⇒ 2 σ bond + 0 lone pair = sp

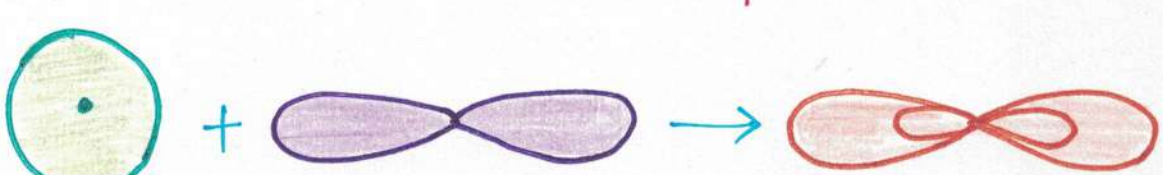
SN = 3 ⇒ 3 σ bond ⟶ sp^2

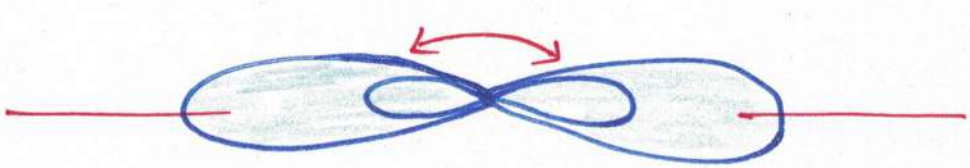
SN = 4 ⇒ sp^3

SN = 5 ⇒ sp^3d^1

SN = 6 ⇒ sp^3d^2

Sp has stric no. 2 ⇒ 2 σ bond + 0 lp





Linear geometry

example :-

A. $BeCl_2$

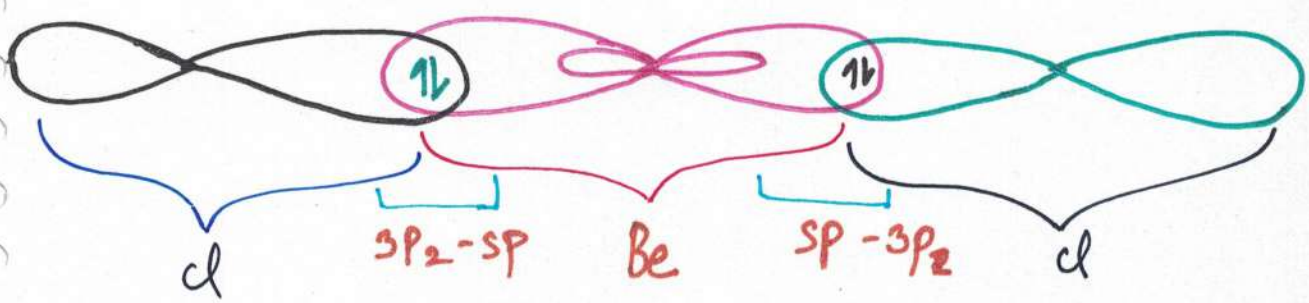
Be ground state = 1↓ | | | |

Be excited state = 1 (2s) | 1 (2p) | | |

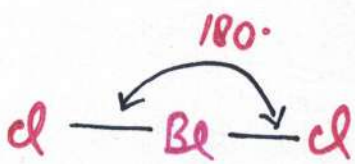
After hybridisation Be → 1 1 (sp hybrid) | (unhybrid)

Be - Sp Hybrid Orbitals

now, 2 cl atom can bond with this Be



- 2 σ bond [sp - $3p_z$]
- Bond angle = 180°
- geometry / shape = Linear



$SN = \frac{1}{2}$ [no. of valence shell e⁻ of central atom + monovalent atom attached with CA. − cation − anion]

B. Carbon dioxide [CO_2]

$SN = \frac{1}{2}[4+0-0+0] = 2$

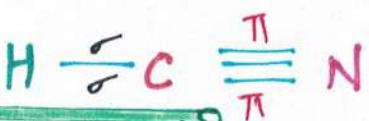
$\Rightarrow$ SP hybridisation

SP Linear

BA = 180°

C. Hydrogen cyanide [HCN]

$SN = \frac{1}{2}[4+0-0-0] = 2$



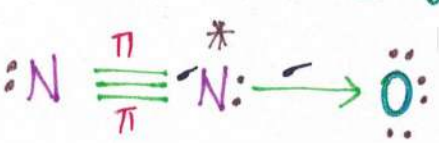
SP Linear

BA = 180°

D. Cyanogen $[CN]_2$

Linear, BA = 180°

E. Nitrous oxide [N_2O]

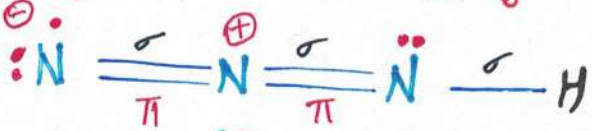


$2\sigma \longrightarrow Sp$

Geometry Linear

BA = 180°

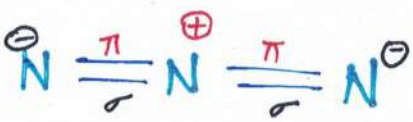
F. Hydrazoic acid [N_3H]



SP, Linear BA = 180°

G. Azide [N_3^{-1}]

SP



SP, Linear BA = 180°

H. Nitronium ion $[NO_2^+]$

$SN = \frac{1}{2}[5 + 0 - 1 + 0] = 2$ — SP hybridisation

$:\ddot{O} \overset{\sigma}{=} \overset{\oplus}{N} \overset{\sigma}{=} \ddot{O}:$ — SP, Linear, BA = 180°

I. $[HgCl]$ Mercuric chloride

$Cl \overset{\sigma}{-} Hg \overset{\sigma}{-} Cl$ — SP, Linear, BA = 180°

$Hg = 54[Xe]\ 4p^{14}\ 5d^{10}\ 6s^2\ 6p^0$

Hg^{+2} in es $= 54[Xe]\ 4f^{14}\ 5d^{10}\ 6s^1\ 6pz^1$

Hg after Hybridization: $6s^1$ $6pz^1$ → | 1 | 1 | (Cl Cl)

J. $[Ag(NH_3)_2]^{+1}$

O.S of Ag $= x + 2 \times 0 = +1$

$x = +1$

$_{47}Ag$ gs $= 4d^{10}\ 5s^0\ 5pz^0\ 5px^0\ 5py^0$

$Ag^{+1} =$ | | | → empty orbital

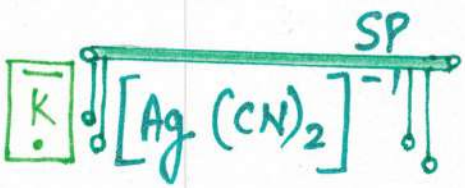
$\ddot{N}H_3$ $\ddot{N}H_3$ → lone pair

+

|| co-ordinate bond

$\left[{}_3HN: \overset{\sigma}{\longrightarrow} Ag \overset{\sigma}{\longleftarrow} :NH_3\right]^{+1}$

Vacant orbitals are participating in hybridisation.



$x + 2x - 1 = 1$

$x = 1$

$Ag^{+1} = 36\ [Kr]\ 4d^{10}\ 5s^0\ 5pz^0\ 5px^0\ 5py^0$

Ag^+ After hybridisation =

cyanide → $[\bar{C} \equiv N]$

SP, Linear, BA = 180°

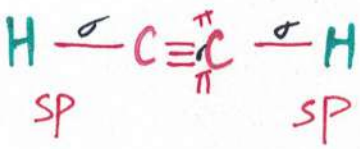
in d-block elements, vacant orbitals participate in Hybridisation.

SN = 2 ⇒ 1 σ + 1 lone pair

1. Carbon monoxide [Co]

1 lp + 1 σ = sp Linear geometry

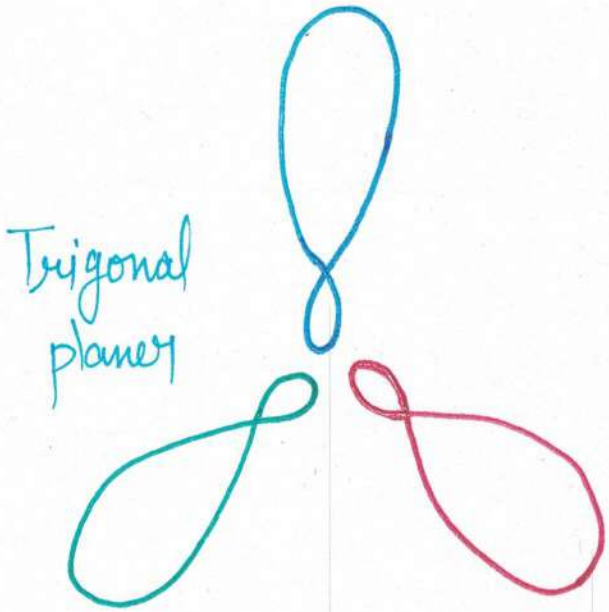
2. Ethyne

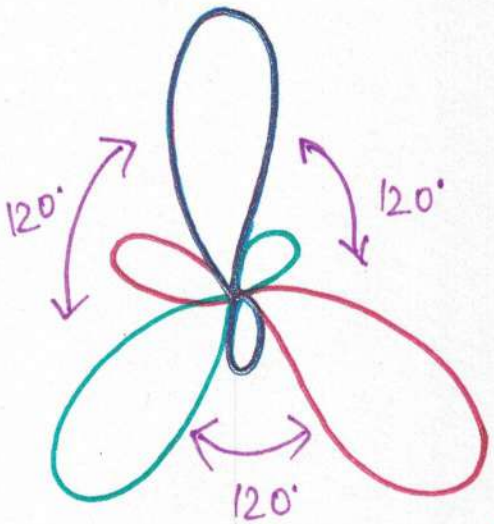


II. SP^2 Hybridisation

one s + two p orbitals = three sp^2 hybrid orbitals

ns + npz + npx or npy ⟶ three sp^2 hybrid orbitals





sp^2 hybrid orbital

$SN = 3 = 3\sigma + 0\ lp$

Example :-

Boron trihalides (BX_3, X = F, Cl, Br, I)

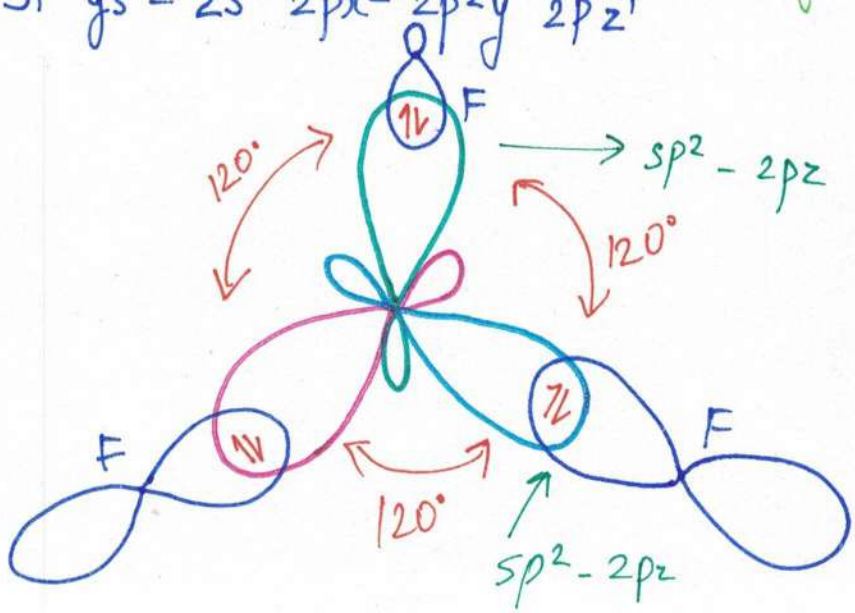
➢ BF_3

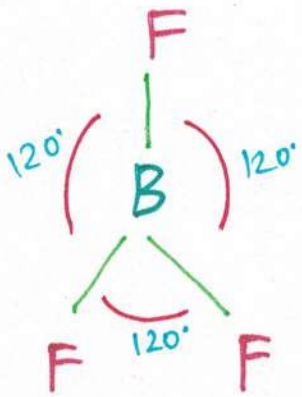
B gs = [1 (2s)] [1 | | | (2p)]

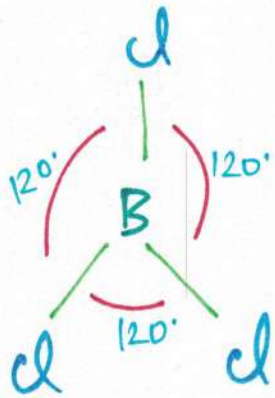
B es = [1 (2s)] [1 | 1 | | (p_x p_y p_z)]

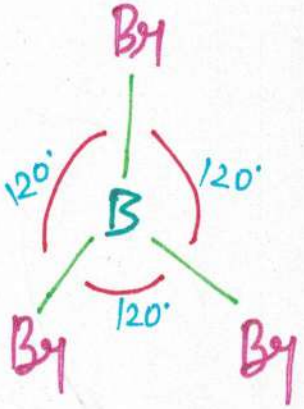
B after Hybridisation = [1 | 1 | 1 (sp^2)] [ ] unhybrid

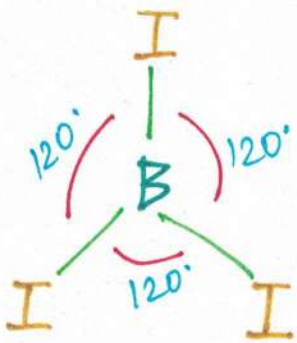
3F gs = $2s^2\ 2p_x^2\ 2p_y^2\ 2p_z^1$









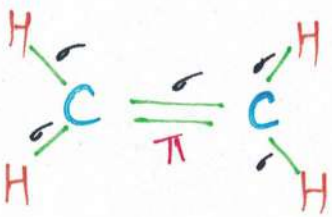


3 σ bond → sp^2 hybridisation

Bond angle → 120°

Geometry → trigonal - planar

2. C_2H_4 Ethene

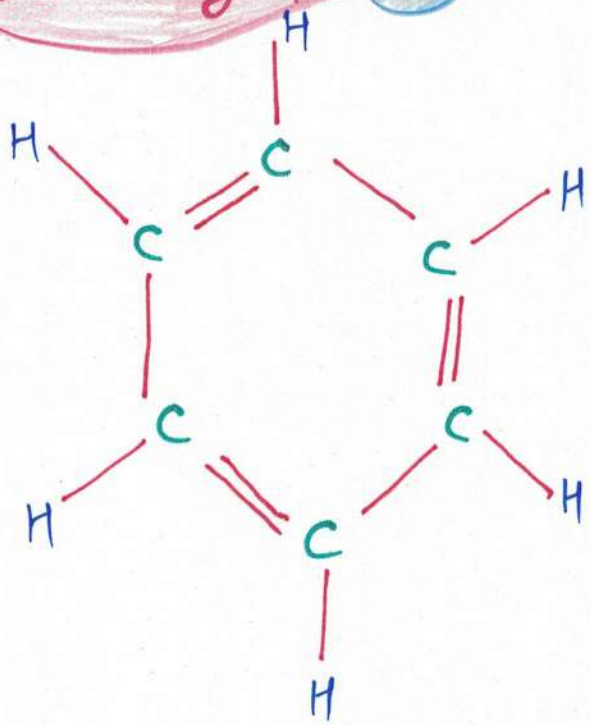


sp^2

Bond angle = 120°

Geometry = trigonal planar

3. C_6H_6 Benzene

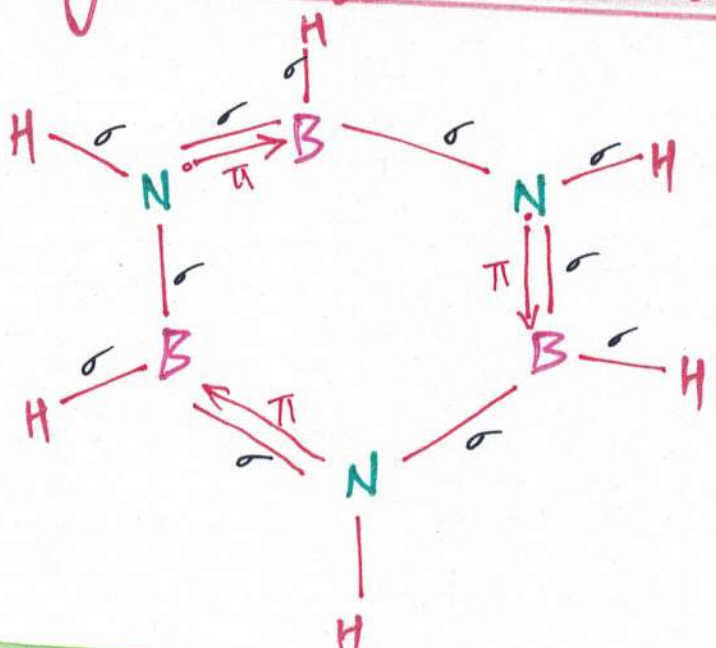


every C has sp^2 hybridisation

BA = 120°

geometry → trigonal planar

4. Inorganic Benzene $B_3N_3H_6$ (Borazine)



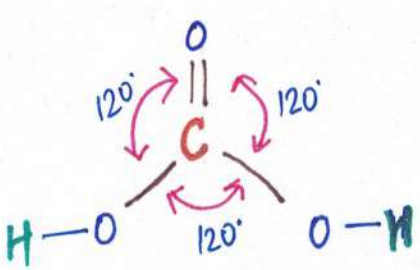
N and B have sp^2 hybridisation geometry

↓

trigonal planer.

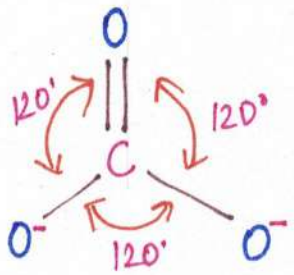
BA = 120°

5. Carbonic acid H_2CO_3



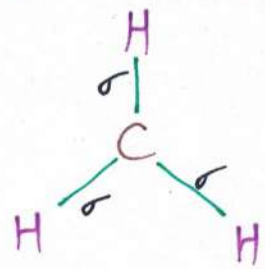
Sp2

6. Carbonate ion CO_3^{2-}



geometry → trigonal planar

7. Methyl Carbocation CH_3^+



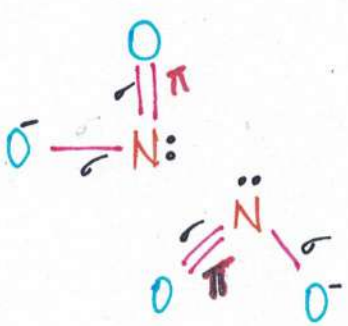
sp^2 hybridisation

BA = 120°

trigonal planar

when SN = 3 ⇒ 2σ + 1lp

Nitric (NO_2^{-1})



2σ + 1lp = 3

sp^2

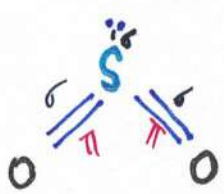
Geometry → trigonal planar

Shape → Bent / angular

BA < 120°

8. SO_2 Sulphur dioxide

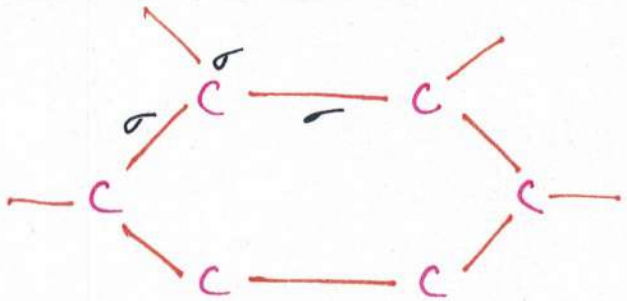
$SN = \frac{1}{2}[6+0-0+0] = 3 \Rightarrow sp^2$



BA < 120°

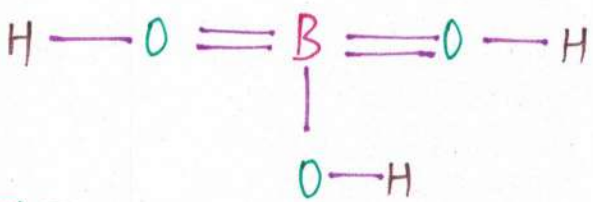
shape → bend on angular

9. **Graphite**



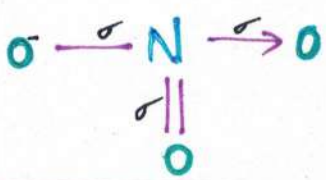
3σ bond → sp^2
geometry → trigonal planar
BA = 120°

10. **Boric acid (H_3BO_3)**



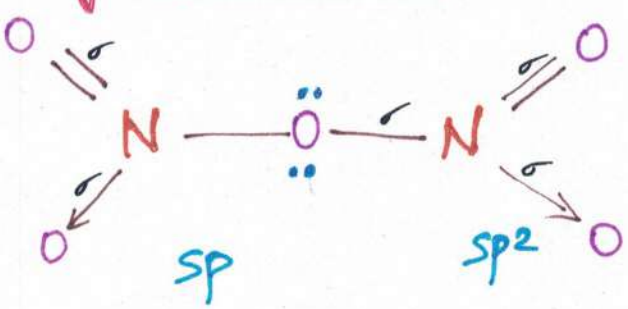
3σ bond → sp^2
geometry → trigonal planar
BA = 120°

11. **Nitrate NO_3^-**



3σ bond → sp^2 hybridization

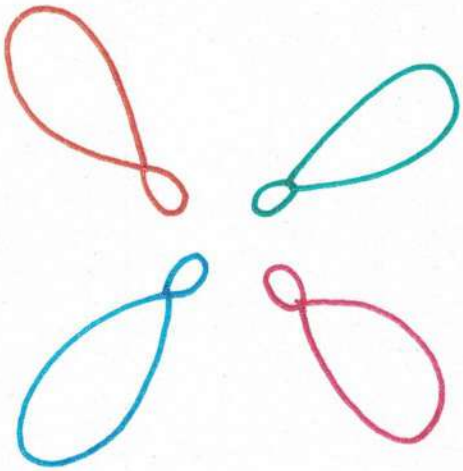
12. **Nitrogen pentaoxide**

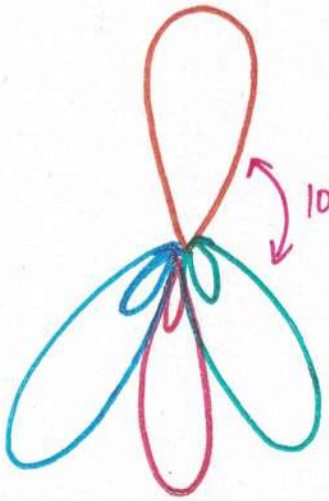


trigonal planar
BA ≃ 120°

III. SP^3 Hybridization

one S + three p = four sp^3 hybrid orbitals
ns + npx + npy + npz → four sp^2 hybrid orbitals





Tetrahedral

sp^3 hybrid orbitals

SN=4 = 4σ 0lp

example:- CH_4, SiH_4, CCl_4, $SiCl_4$, CHX_3

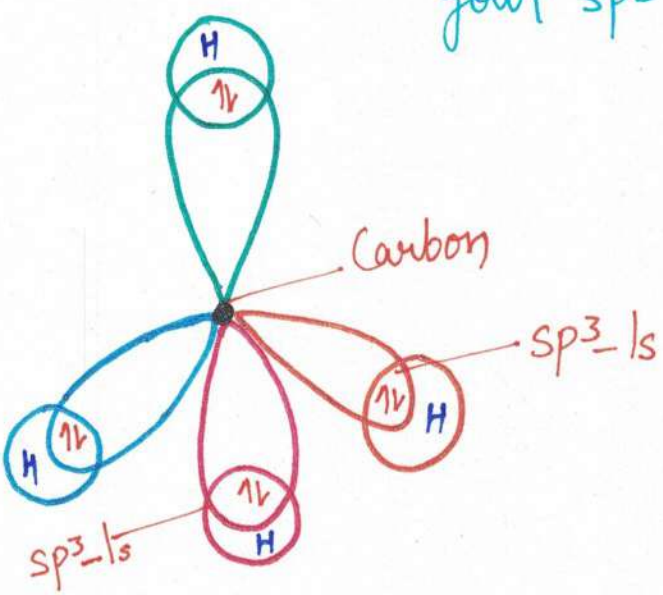
CH_4

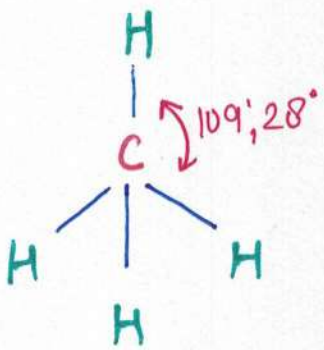
C in g.s = 1↓ | 1 | 1 | _

C in e.s = 1 | 1 | 1 | 1

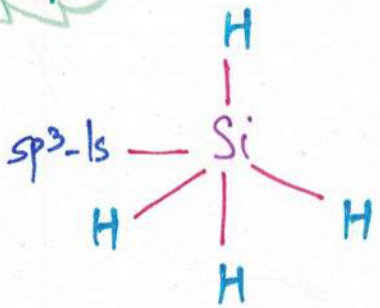
C after Hybridisation = 1 | 1 | 1 | 1

four sp^3 hybrid orbitals



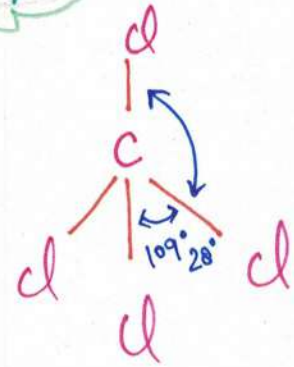


2. SiH_4



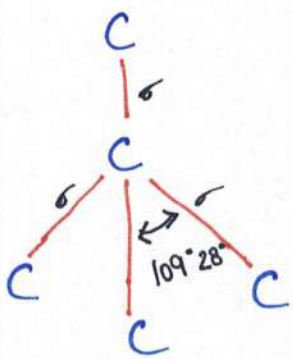
4σ → sp³
tetrahedral
109° 28'

3. CCl_4



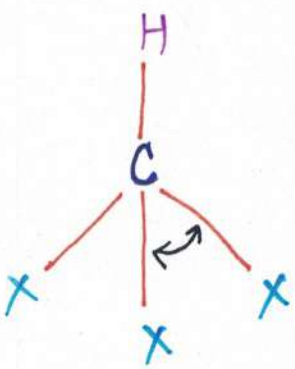
4σ → sp³
tetrahedral
109° 28°

4. Diamond



4σ = sp³, tetrahedral
BA = 109° 28°

5. CHX_3



BA = 109° 28°
4σ ⇒ sp³, tetrahedral

SN = 3 σ, 1 lp

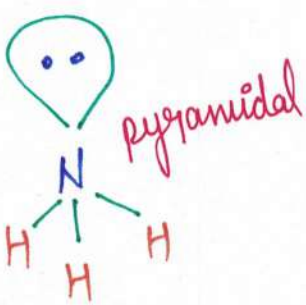
Example:-

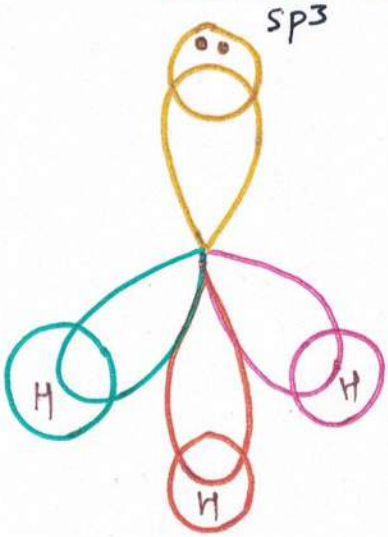
NH_3, NF_3, NCl_3, NX_3, PX_3, XeO_3

6. NH_3

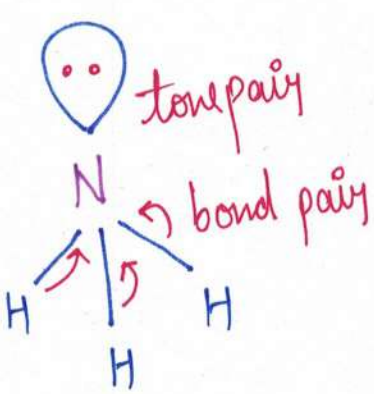
N gs → [↿⇂] [1] [1] [1]

N after hybridisation = [↿⇂] [1] [1] [1]
sp3





Sp^3-s



lp - bp repulsion decreases bond angle.

bp - bp repulsion increases bond angle.

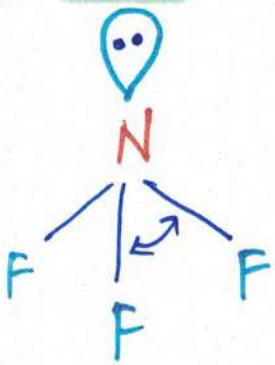
lp - bp > bp - bp

⇒ BA is less than 109° 28'

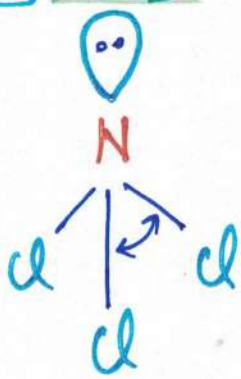
BA = 107.5°

Shape : Pyramidal

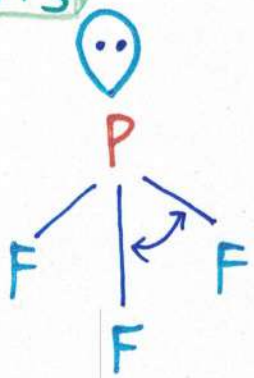
7. NF_3



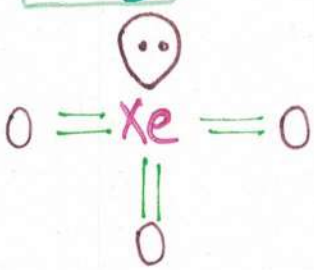
8. NCl_3



9. PF_3



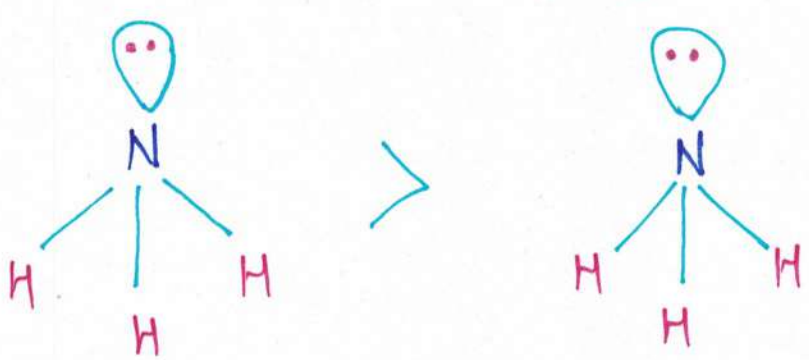
10. XeO_3



Pyramidal shape

BA $< 109°28'$

Question:- Comment on bond angle of $F\hat{N}F$ (NF_3) and $H\hat{N}F$ (NH_3)



$X > Y$

Bond angle of NH_3 > bond angle of NF_3

Bond angle depends on electronegativity of bonded atom.

BA of $NF_3 \simeq 105°$

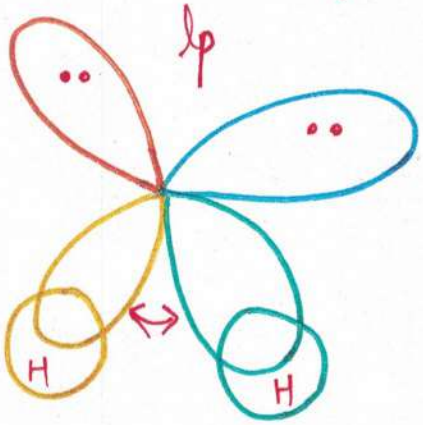
As EN of bonded atom increases, bond angle decreases

SN = 4 = 2σ, 2lp

11. H_2O

O[hybrid] = | 1b | 1b | 1 | 1 |

four sp^3 hybrid orbital



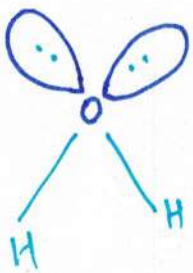
BA = 104.5°

lp - lp > lp - bp > bp - bp

BA < 107.5°

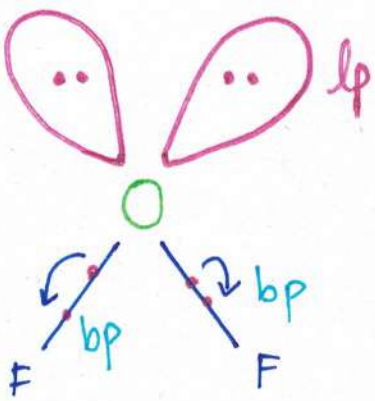
or

BA < 109°28'



Shape → bent/angular

12. OF_2



Shape :- bent/angular

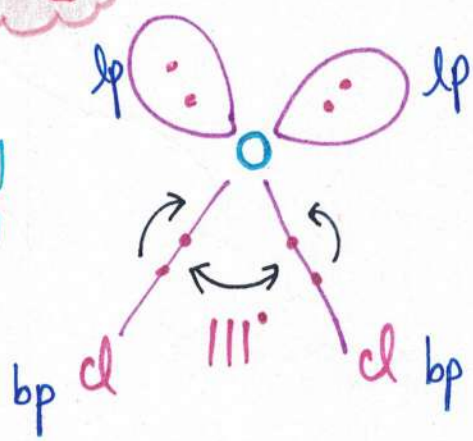
OF2 and H_2O have some hybridisation and same central atom.

BA of OF_2 < 109°28'

⇒ BA of H_2O > BA of OF_2

Electronegativity of bonded atom $\propto \frac{1}{BA}$

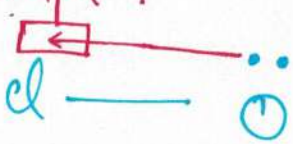
13. OCl_2



exception (BA = 111°)

BA = 111°

Triple 1



Chlorine is III period element, vacant d-orbital is available in cl atom. lone pair of e^- present on oxygen atom,

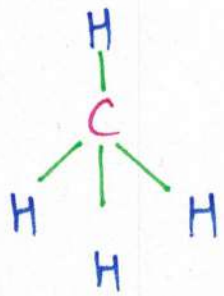
Backbonding takes place between O & cl and some pie character develops in OCl_2 and π-orbitals occupy more space. More bp-bp increases bond angle. BA of OCl_2 is 111°, which is an exception to original bond angle (109°28')

Question:- Correct increasing order of BA of OCl_2, OF_2 and OH_2.

Answer:- $\underset{104°}{OF_2} < \underset{}{OH_2} < \underset{111°}{OCl_2}$

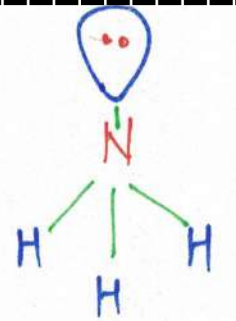
Question:- Why BA of CH_4 is more than NH_3 and NH_3 is more than H_2O.

Answer:- $CH_4 > NH_3 > H_2O$



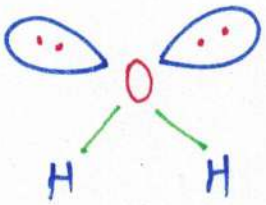
0 lp

Tetrahedral



1 lp

Pyramidal



2 lp

Bent/Angular

DRAGO'S RULE

This rule is applicable on sp^3 hybridisation. According to this rule, when electronegativity of central atom is less than or equal to 2.5 and bonded atom is hydrogen, central atom does not undergoes hybridisation only pure-p-orbital participate in bonding. Therefore, bond angle remains almost equal to 90°.

Applicable on 8 molecules

PH_3 H_2S

AsH_3 H_2Se

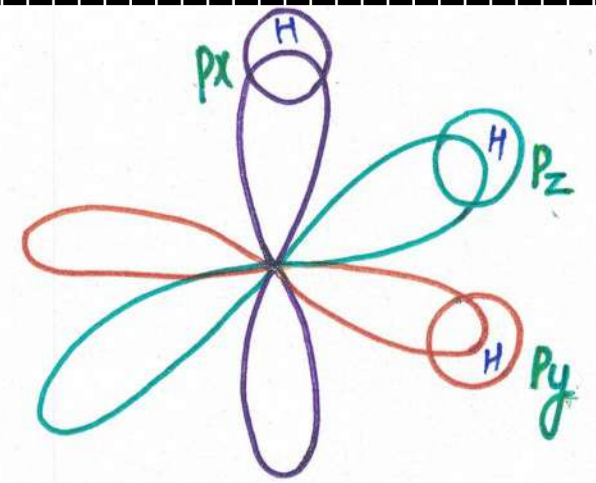
SbH_3 H_2Te

BiH_3 H_2Po

— Hybridisation is not possible

BA = 90°

These atoms does not prepare themselves for hybridisation for a small atom like Hydrogen.



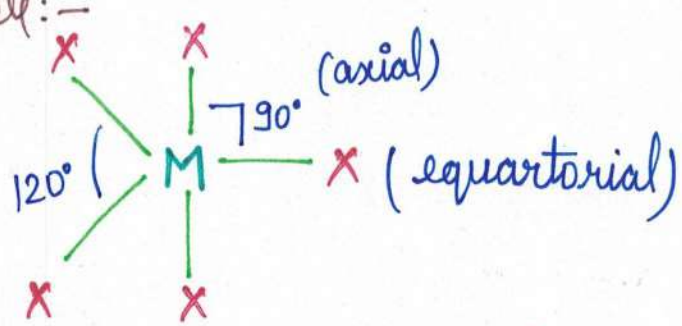
Bonding takes place like this in PH_3, AsH_3, SbH_3, BiH_3

sp^3d Hybridisation

one s + px + py + pz + (dz^2) = Five sp^3d hybrid orbital

shape → trigonal bipyramidal

example:-

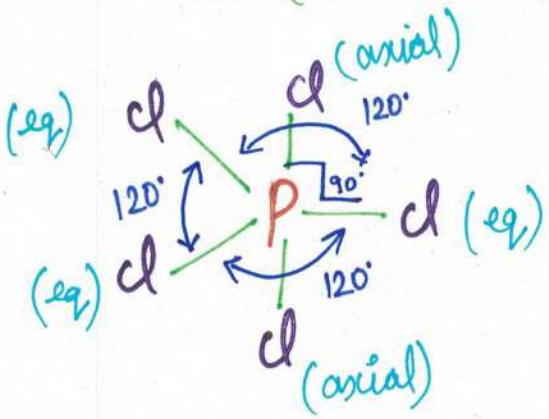


6 → 90° angles

SN = 5 ⟶ 5σ + 0lp

sp^3d hybridisation, BA = 120°, 90°

example:- PX_5 (X = F, Cl, Br, I)



$Cl - \hat{P} - Cl_a = 90°$

$Cl - \hat{P} - Cl_{eq} = 120°$

Geometry → trigonal bipyramidal

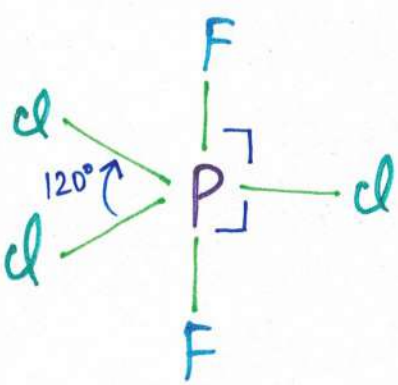
Phosphorous and Cl(eq) lie in the same plane.
Cl(axial) is above and below the plane.

More EN atom goes to axial position

BENT'S RULE

According to Bent's Rule if lone pair of electron present on central atom and more electronegative element is also attached with C.A, then lone pair goes to equatorial position and bond pair goes to axial position to minimise lp - bp and bp - bp repulsion.

example: - PCl_3F_2



F is more E.N. element. Hence, it will goto axial position and Cl goes to equatorial position.

SN = 5 $4\sigma + 1lp$

example:- Sulphur Tetrafluoride

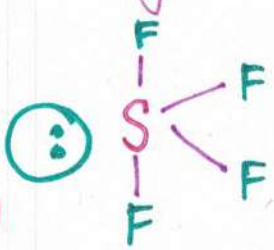
S gs = $\uparrow\downarrow$ | $\uparrow\downarrow$ $\uparrow$ $\uparrow$ | | | | |

S es = $\uparrow\downarrow$ (3s) | 1 1 1 (3p) | 1 (3d)

3p

After hybridisation: 3d: $\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ — F F F F F

Geometry → TBP (Trigonal Bipyramidal)



lone pair goes to eq. position

shape:- distorted tetrahedral or see saw

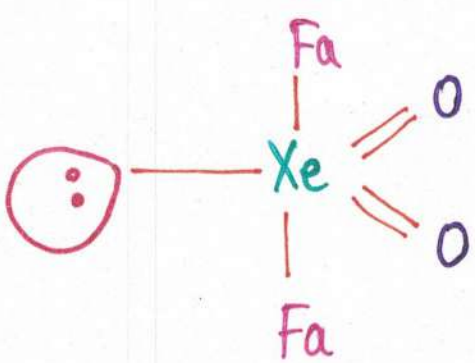
BA < 90°

BA < 120°

eq:- SeF_4, TeF_4

example:- XeO_2F_2

$SN = \frac{1}{2}[8+2-0+0] = 5 \longrightarrow sp^3d$



BA < 120°

BA < 90°

shape:- see saw

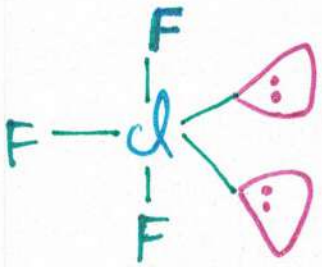
lone pair goes to eq. position. F (more EN) goes to axial position.

SN=5 ⟶ 3σ + 2lp

example:- ClF_3, BrF_3, $XeOF_2$

1. ClF_3

$SN = \frac{1}{2}[7+3] = 5 \longrightarrow sp^3d$



BA < 90°

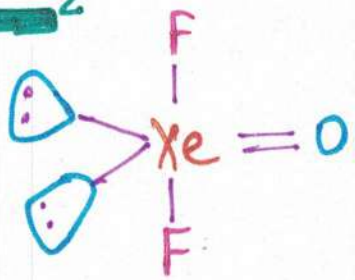
120° angle is not here.

lp - lp repulsion will increase

Hence BA < 90°

shape :- bent T-shape

2. $XeOF_2$



Bent T-shape

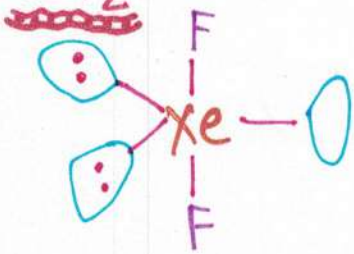
BA < 90°

3σ + 2lp

SN = 5 2σ + 3lp

example :- XeF_2, ICl_2^-, I_3^-, IBr_2^-, ICl_4^-

1. XeF_2

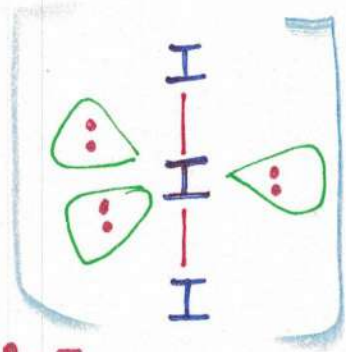


$F \hat{Xe} F = 180°$

shape → Linear

2. I_3^-

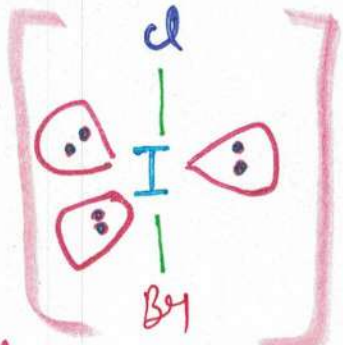
$SN = \frac{1}{2}[7+2-0+1] = 5 \longrightarrow sp^3d$



Shape → Linear

BA = 180°

3. $ICl Br^-$

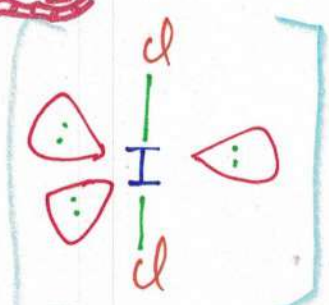


Linear

180°

Resultant of lone Pair will be 0.

4. ICl_2



Linear

180°

sp^3d

SN=5

(a) 5σ + 0lp = TBP, 90°, 120°

(b) 4σ + 1lp = see saw, BA < 90° BA < 120°

(c) 3σ + 2lp = bent T-shape, BA < 90°

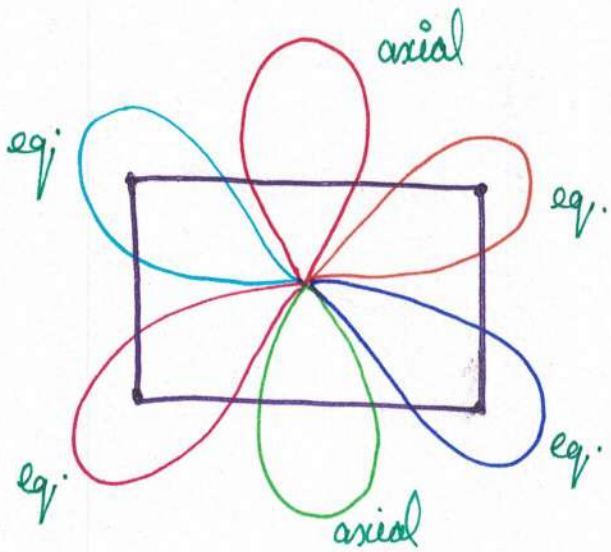
(d) 2σ + 3lp = Linear BA = 180°

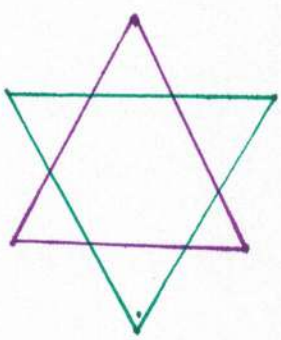
V sp^3d^2 Hybridisation

$ns + npx + npy + npz + d_{z^2} + d_{x^2-y^2}$

(dx^2-y^2)

= six sp^3d^2 hybrid orbitals





A. $6\sigma + 0lp$

Example:-

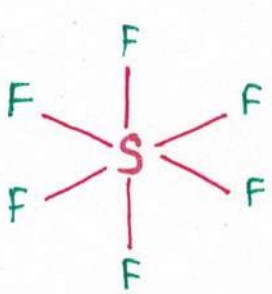
I. SF_6

S_{gs} = 3s | 3p | 3d

S_{es} = 3s | 3p | 3d

S hybridised = | | | | | | |

sp^3d^2 hybrid orbitals



All bond angles are 90°

geometry ⇒ octahedral

Other example ⇒ SeF_6, TeF_6

Q. $5\sigma + 1lp$

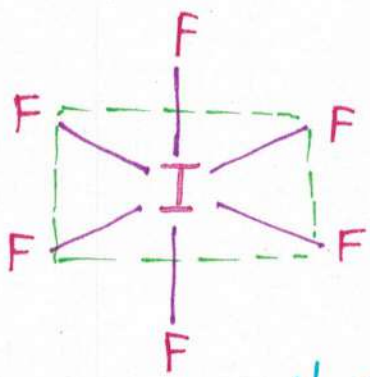
example:- $XeOF_4$, BrF_5, IF_5 etc

I gs = [1↓] [1↓ 1↓ 1] [ ]

I cs = [1↓] [1 1 1] [1 1]

5s 5p 5d

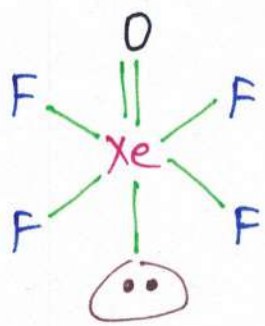
I hybridised



lone pair goes to axial position so that min. repulsion occurs.

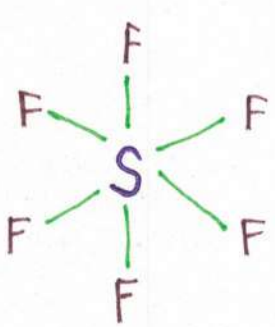
shape - square pyramidal BA < 90°

1. $XeOF_4$



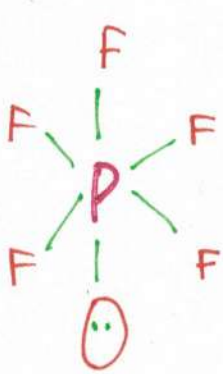
Square pyramid

BA < 90°



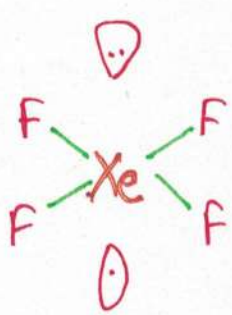
0 lp

Octahedral



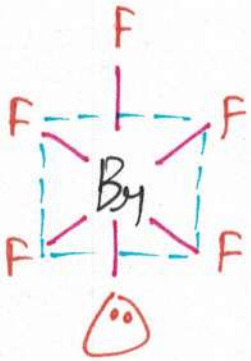
1 lp

square Pyramidal



2 lp

Square planer



2. BrF_5

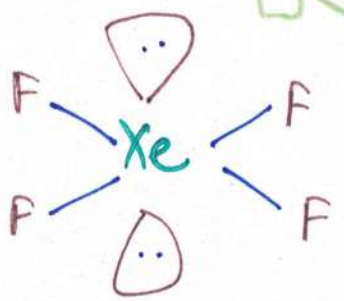
Square pyramidal

BA < 90°

c. $4\sigma + 2\,lp \longrightarrow sp^3d^3$

example:-

3. XeF_4



Square planar

BA = 90°

both lp - will cancel out each other's effect

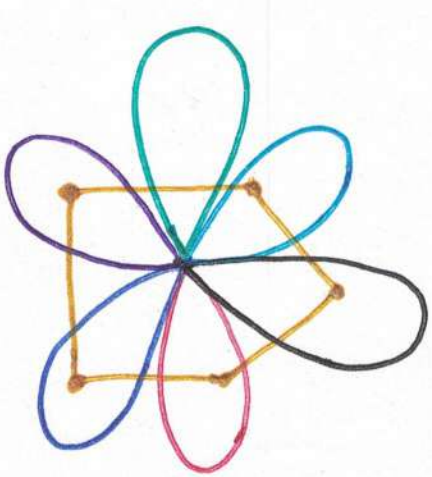
Sp^3d^3 Hybridisation

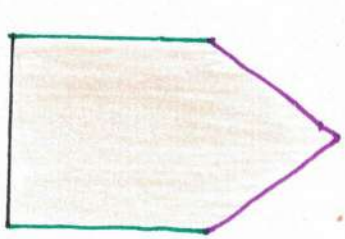
$ns + npx + npy + npz + dz^2 + dx^2-y^2 / dy-z + dzx$

$\Rightarrow$ seven sp^3d^3 hybrid orbitals

geometry → pentagonal bipyramidal

BA = 90° BA = 72°





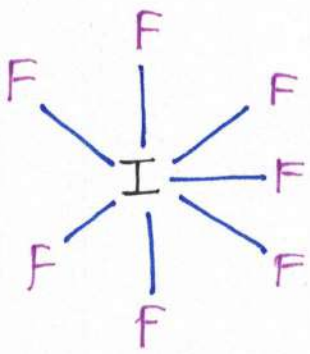
1. IF_7

I_{gs} = 5s: ↑↓ | 5p: ↑↓ ↑ ↑ | 5d: ( ) ( ) ( ) ( ) ( )

I_{es} = 5s: 1 | 5p: 1 1 1 | 5d: 1 1 1 ( ) ( )

I_{hyb} = 1 1 1 1 1 1 1

seven sp^3d^3 hybrid orbitals



BA = 90° (above the plane / below the plane)

BA = 72°

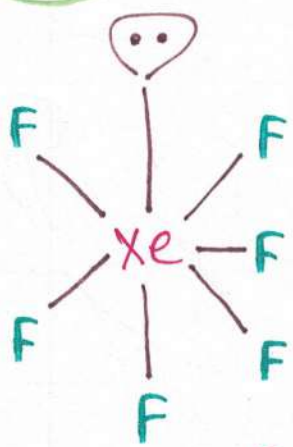
geometry ⇒ pentagonal bipyramidal

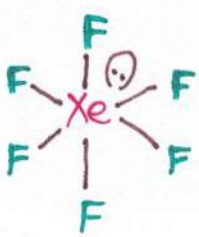
10 angles of 90°

D. 6σ + 1lp

example:-

1. XeF_6





lp - bp repulsion decrease bond angle

BA < 90°

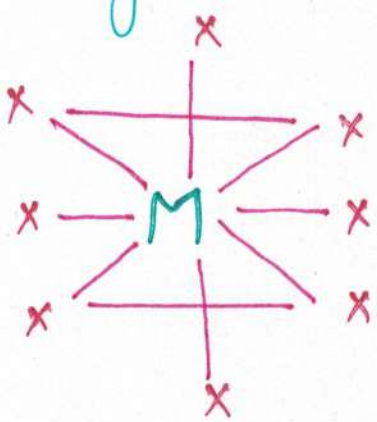
BA < 72°

Shape:- 1) Pentagonal pyramided
2) Distorted Octahedral or caped Octahedral

SP^3d^4 Hybridisation

$ns + np + npy + npz + dz^2 + dzx + dxy / dyz + dx^2 - y^2$

Geometry → dodecahedron



dodecahedron

1. $Cs^+(XeF_7)^-$

Sp^3 →
- CH_4 Tetrahedral
- NH_3 Pyramidal
- H_2O Bent / Angular

SP → Linear

sp^2 → trigonal planar

Sp^3d →
- PCl_5 Trigonal bipyramided
- SF_4 See Saw
- PCl_3 Bent T shape
- I_3^- Linear

Sp^3d^2 →
- SF_6 Octahidral
- IF_5 Square pyramidal
- XeF_4 Square planar

Sp^3d^3 → Pentagonal bipyramidal (IF_7)

Sp^3d^4 → Dodecahedral

INNER D-ORBITAL HYBRIDISATION

CORDINATION COMPOUNDS

Transition metal forms inner d- orbital complexes. because in transition metal d- electron participate in bonding.

1. dsp^2 Hybridization

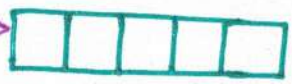
$(n-1)\,d_{x^2-y^2} + ns + np_z + np_x / np_y$

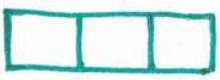
⟶ four dsp^2 hybrid orbitals.

Geometry ⟶ Square planar.

BA = 90°

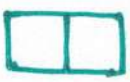
d-orbital →





t_{2g}

$(d_{xy} / d_{yz} / d_{zx})$



e_g

$(d_{x^2-y^2}, d_{z^2})$

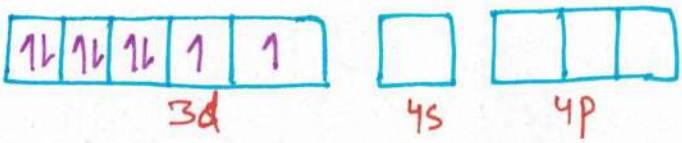
In transition elements, if strong field ligand (group of atom carrying +ve or -ve charge and having lone pair) is attached with C.A pairing of d-e^- takes place, therfore vacant d-orbitals available for hybridisation. If weak field ligands (O^{2-}, F^-, Cl^-, Br^-, I^-) attached with C.A. pairing of e^- does not take place. Other orbitals participates in hybridisation.

CN^-, CO are strong field ligand strong field ligand [less electro-negative element] weak field ligand [more electronegative-element]

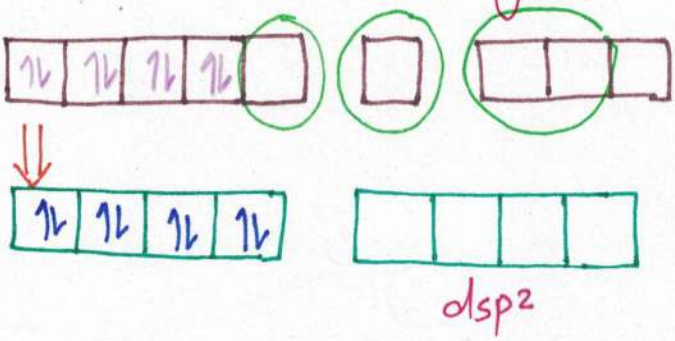
Example

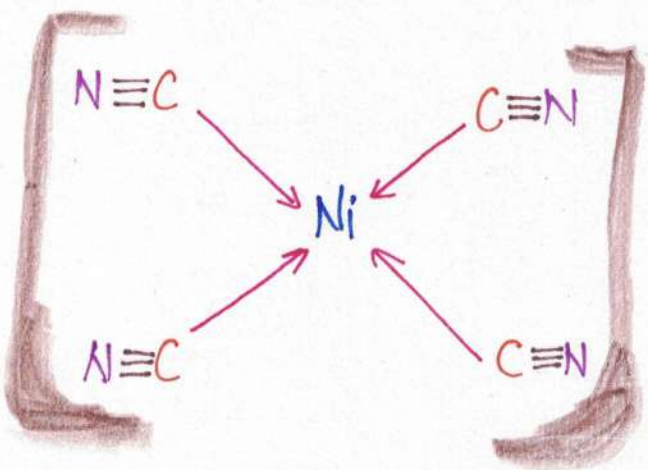
1. $[Ni(CN)_4]^{2-}$

Nickel is in +2 O.S

Ni^{+2} gs = 

$:C \equiv N:$ (strong field ligand)



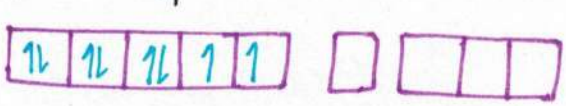


shape / geometry → Squard planar

BA = 90°

2. $[NiCl_4]^{2-}$

O.S of Ni = +2

Ni^{+2} gs = 

Cl is weak field ligand, pairing of e^-s don't takes place

BA = 109°, 28'

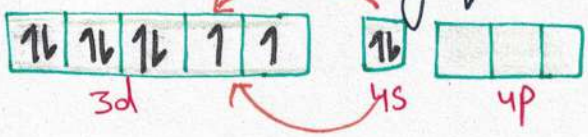
geometry → Tetrahedral

e.g:- $Ni(CO)_4$

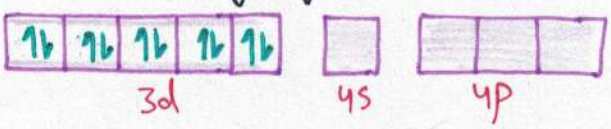
$x + 4 \times 0 = 0 \Rightarrow x = 0$

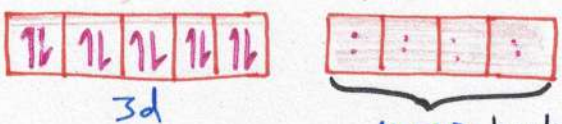
O.S of Ni = Zero

C≡O is strong field ligand

Ni = 

CO is SFL. pairing of e^- occurs.





d^3s Hybridisation

SN = 4 = 4σ + 0lp

$(n-1)d_{xy} + (n-1)d_{yz} + (n-1)d_{zx} + ns$ or $(n-1)t_{2g} + ns$

Four d^3s hybrid orbitals

Geometry and shape → Tetrahedral

BA = 109°28

Conditions

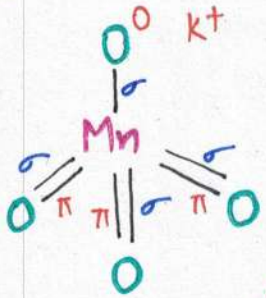
Transition metal

highest O.S

SN = 4

Example:- MnO_4^-, CrO_4^{2-}, $Cl_2O_7^{2-}$

$KMnO_4 \longrightarrow K^+ + MnO_4^-$



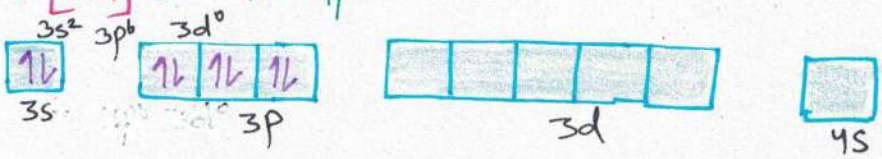
4σ bond

O.S of manganese = +7

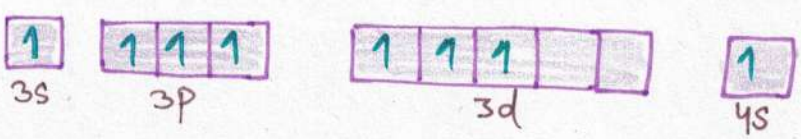
d^3s hybridisation

Geometry ⟶ Tetrahedral

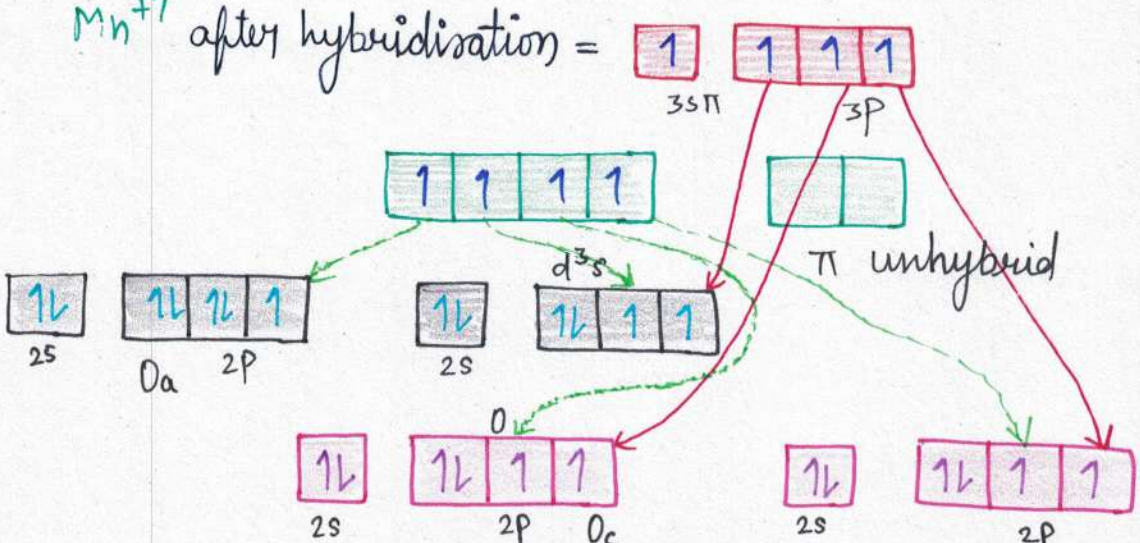
Mn^{+7} = 18 [Ar] $3d^0\ 4s^0\ 4p^0$



Mn^{+7} es =

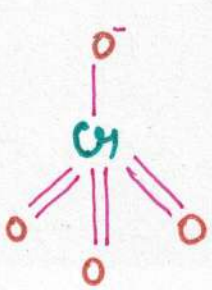


Mn^{+7} after hybridisation =



Mn is a transition element and here, it is in ^{0d} its highest O.S +7. Its SN= , Hence it will have d^3s hybridisation MnO_4 shows colour as O^- come closer to Mn and charge transfer takes place giving out colour.

CrO_4^{2-}

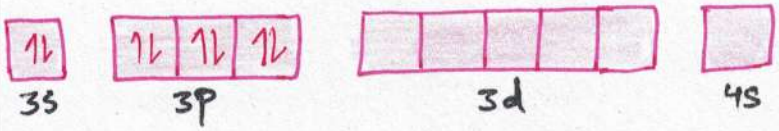


dichromate ion

4σ bond

BA = 109°28'

Geometry ⟶ Tetrahedral

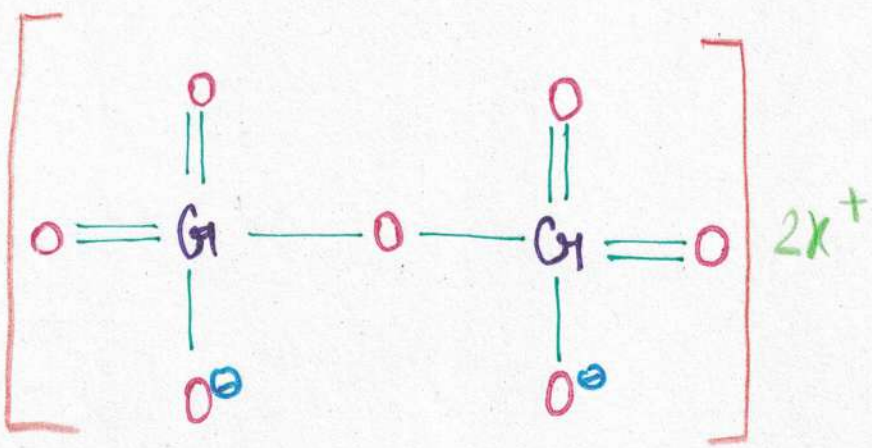
Cr^{+6} = 

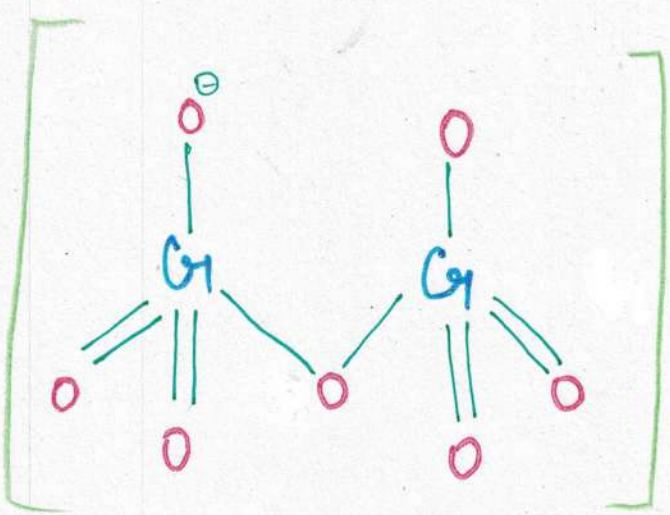
Cr^{+6} es =

Same story as Mn

$K_2Cr_2O_7$

O.S of Cr is 6. O.S = group no.





d^3s hybridisation

Question:- What is the hybridisation of Mn in $KMnO_4$?

(a) d^3s (b) dsp^2 (c) sp^3 (d) dsp^3 (e) d^2sp^3

Ans:- (a) d^3s

Question:- What is hybridisation of Mn in $KMnO_4$?

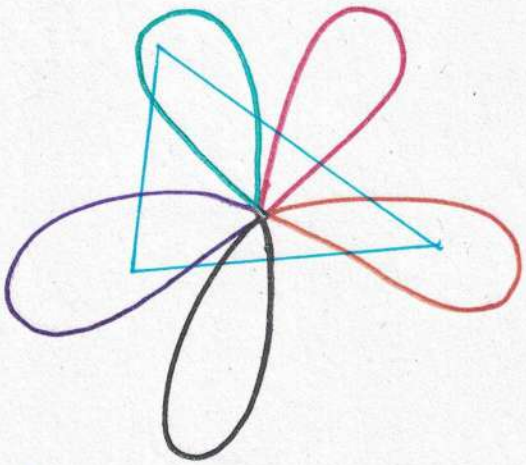
(a) sp^3 (b) dsp^2 (c) dsp^3 (d) d^2sp^3

Ans:- (a) sp^3

dsp^3 hybridisation

inner d orbital is used

$(n-1)d_z + ns + np_x + np_y + np_z \longrightarrow$ Five dsp^3 hybridisation orbitals



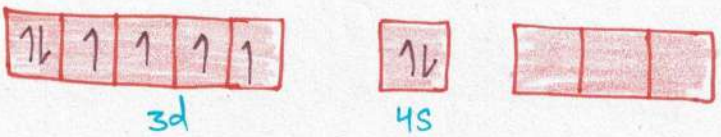
Geometry → trigonal bipyramidal

BA = 90° and 120°

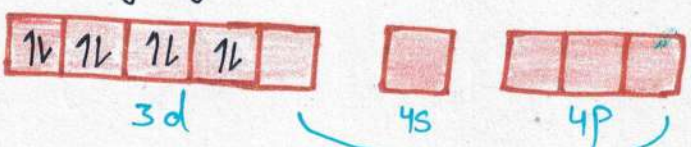
Example:-

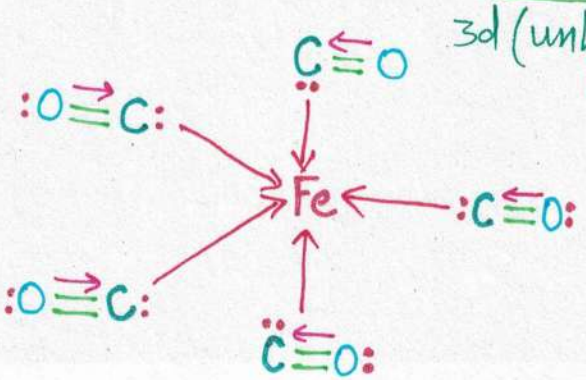
$Fe(Co)_5$

O.S of Fe is 0

Fe gs = 

carbonite is sfL. pairing of e^- occurs.

Fe after hybridisation = 



Trigonal bipyramidal

90°, 120°

Sp^3d

d^2sp^3 hybridisation

$(n-1)\,dz^2 + (n-1)\,dx^2y^2 + ns + npx + npy + npz$

→ six d^2sp^3 hybrid orbitals

geo/shape ⟶ Octahedral

Example:—

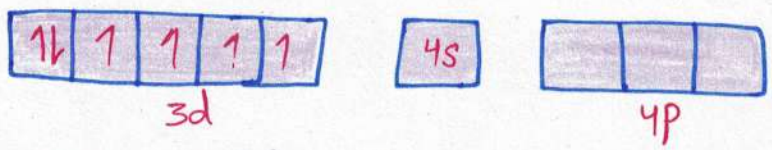
$$K_4[Fe(CN)_6] \longrightarrow 4K^+ + [Fe(CN)_6]^{+}$$

Ferrocyanide

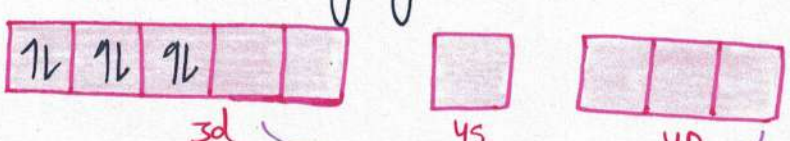
$[Fe(CN)_6]^{4-}$

O.S of Fe = +2

Fe^{+2} gs =



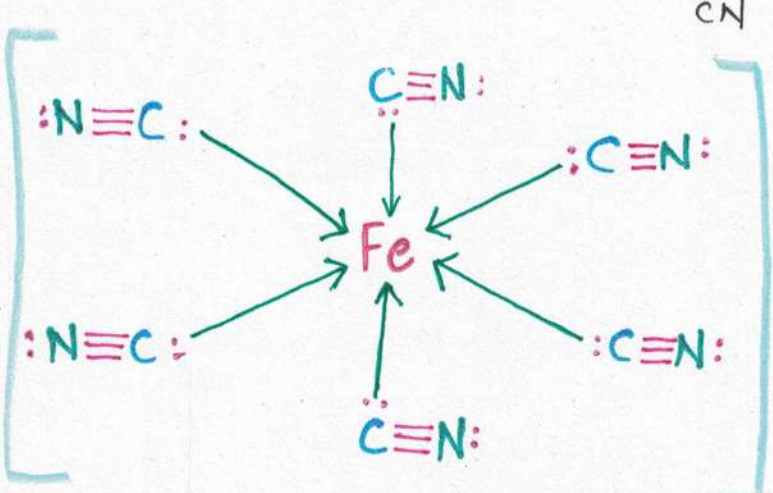
CN is SFL. pairing of e^- occurs



d^2sp^3

CN

d^2sp^3



d^2sp^3, Octahedral

All Bond angles are equal to 90°.

HYBRIDISATION IN SOLID AND LIQUID STATE

PF_5 - Phosphorus pentafloride

1. $PF_5(g) \rightarrow sp^3d$

Geometry → Trigonal bipyramidal

BA → 90°, 120°

PCl_5 (solid) → $[PCl_4]^+ [PCl_6]^-$

2. $[PCl_4]^+$ SN = 4

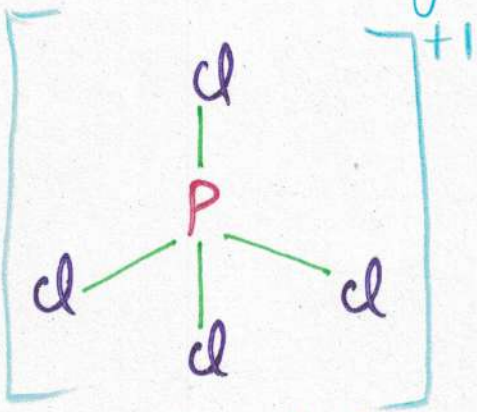
sp^3 hybridisation

Geometry → Tetrahedral

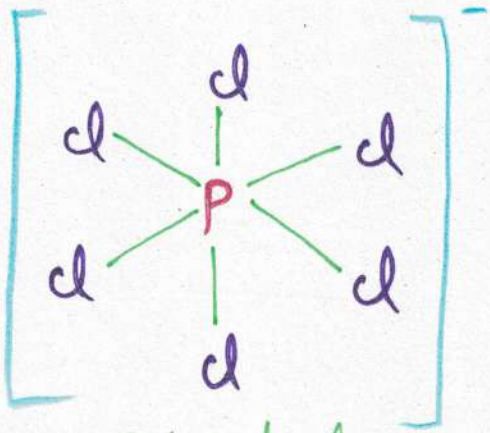
$[PCl_6]^-$ SN = 6

sp^3d^2 hybridisation

Geometry → Octahedral



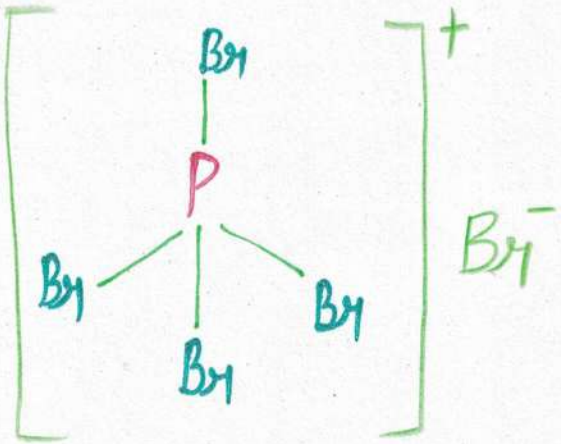
Tetrahedral



Octahedral

$[PCl_5] = [PCl_6]^- [PCl_4]^+$

3. PBr_5 (solid) $\longrightarrow [PBr_4]^+ + Br^-$

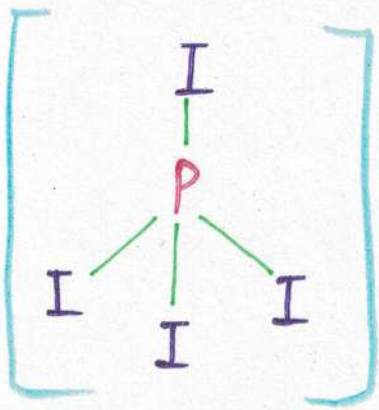


SN = 4

Tetrahedral BA ≅ 109° 28'

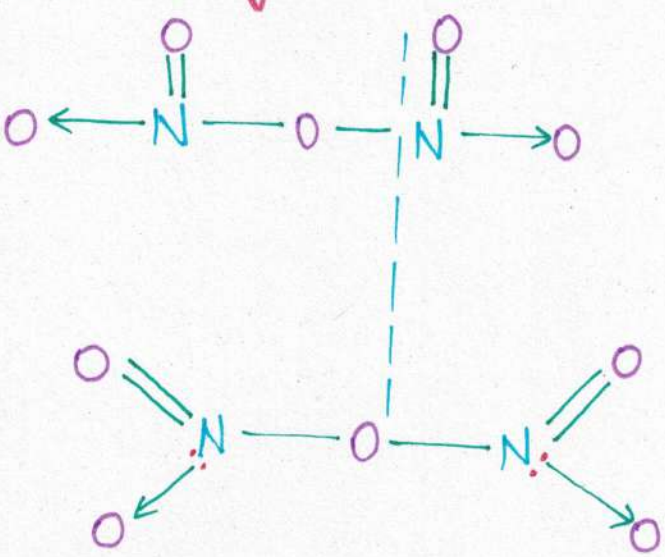
4. PI_5 (Liquid) $\longrightarrow [PI_4]^+ I^-$

PI_5 does not exist in solid state



sp^3 hybridisation

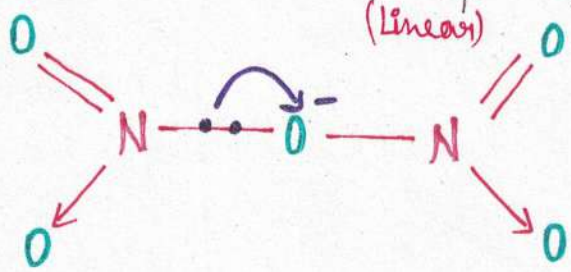
5. N_2O_5 Nitrogen pentaoxide



[gas phase]

ionic phase :- $[NO_2^+]$ $[NO_3^{-1}]$

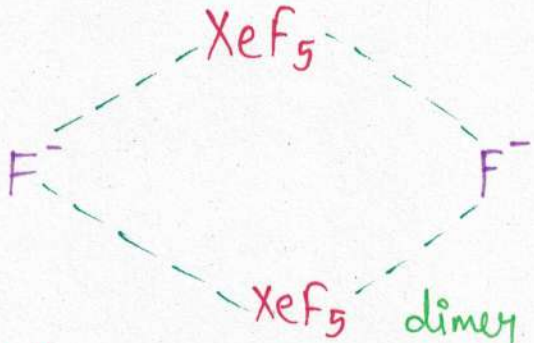
sp (Linear) sp2 (trigonal planar)



6. $XeF_5 \rightarrow [XeF_5]^+ F^-$ (ionic phase)

$$SN = \frac{8+5-1}{2} = \frac{12}{2} = 6$$

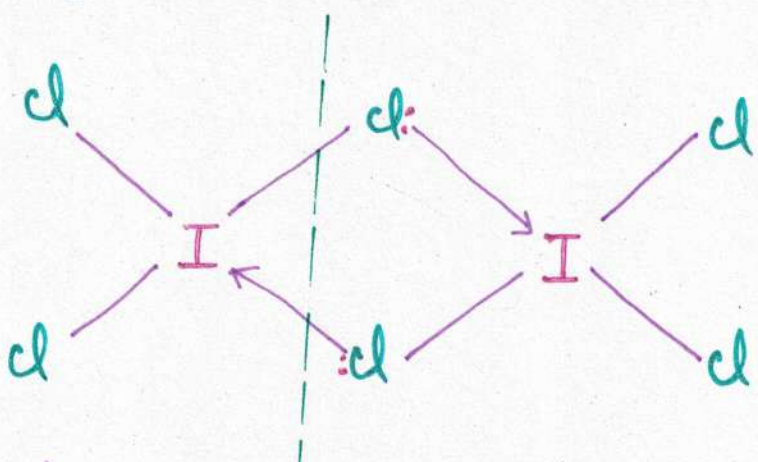
sp^3d^2



dimer

7. I_2Cl_6 (liq)

ICl_3 does not exist in nature. It exist as dimer I_2Cl_3.

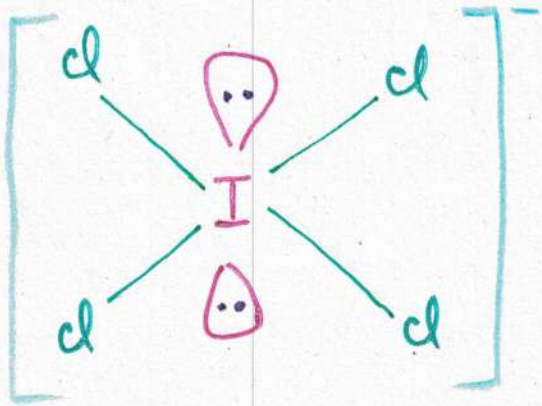


$ICl_2^+ \rightarrow \frac{1}{2}[7+2-1] = 4$

sp^3d

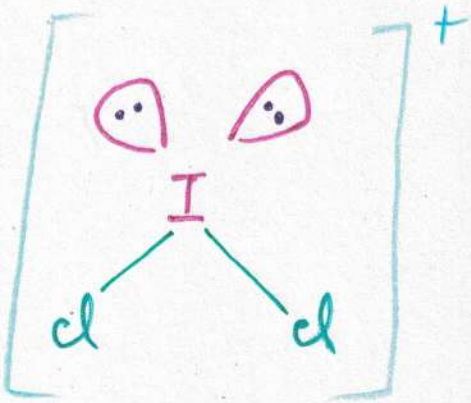
$I_2Cl_6 \rightarrow [ICl_4]^-$ $[ICl_2]^+$

sp^3d^2 sp^3d



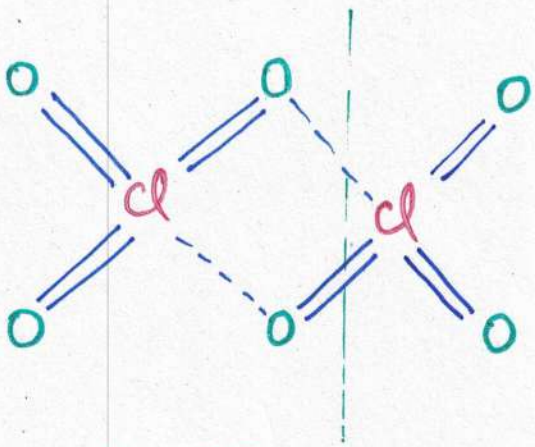
Geo → Octahedral

Shape → Square planar



Bent shape

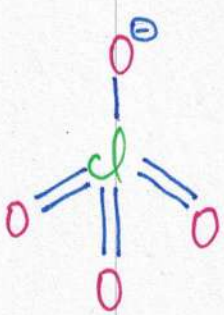
Cl_2O_6

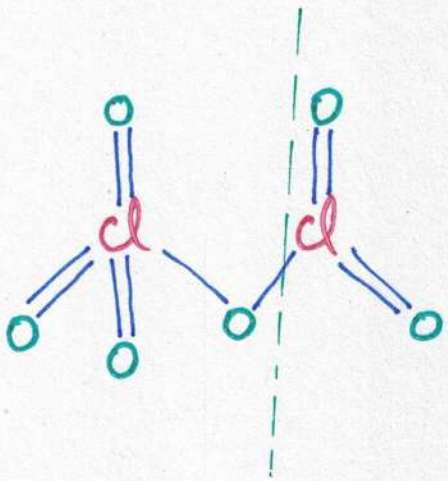


$[ClO_4]^-$

$\frac{1}{2}[7+1] = 4$

sp^3

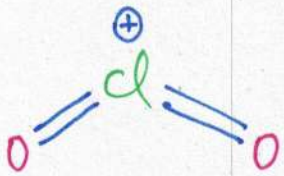




$[ClO_2]^+$

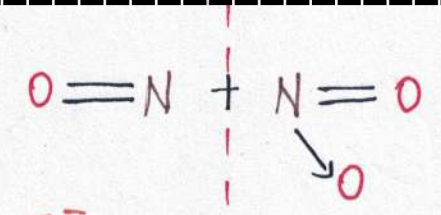
$\frac{1}{2}[7-1] = \frac{6}{2} = 3$

sp^2



Bent / angular

N_2O_3



$[NO]^+$ $[NO_2]^-$

$\frac{1}{2}[5-1]$ $\frac{1}{2}[5+1]$

$= 4/2 = 2$ $= 6/2 = 3$

SP sp^2

$2PCl_5(s) = [PCl_4]^+ [PCl_6]^-$

$PBr_5(s) = [PBr_4]^+ [Br]^-$

$PI_5(l) = [PI_4]^+ [I]^-$

$N_2O_5 = [NO_2]^+ [NO_3]^-$

$XeF_6 = [XeF_5]^+ [F]^-$

$I_2Cl_6 = [ICl_2]^+ [ICl_4]^-$

$Cl_2O_6 = [ClO_2]^+ [ClO_4]^-$

$N_2O_3 = [NO]^+ [NO_2]^-$

Valence Shell Electron Pair Repulsion Theory

VSEPR

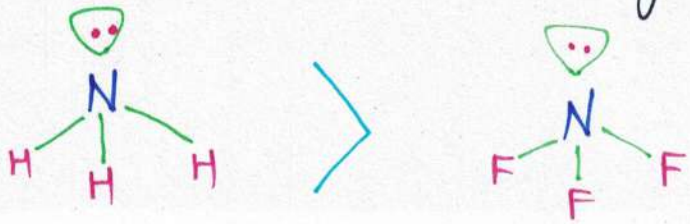
According to valence shell electron pair repulsion theory, every molecule acquire those shape in space in which lone-pair - lone pair repulsion, lone pair - bond pair repulsion and bond - pair - bond pair repulsion is minimum.

If central atom has no lone pair, it has only bond pair electron, geometry of molecule will be regular.

a. 2σ bond sp hyb ; linear geo ; BA = 180°
b. 3σ bond sp^2 hyb ; trigonal planar ; BA = 120°
c. 4σ bond sp^3 hyb ; tetrahedral ; BA = 109°28'
d. 5σ bond sp^3d hyb ; trigonal bipyramidal BA = 120°, 90°
e. 6σ bond sp^3d^2 ; Octahedral , BA = 90°
f. 7σ bond sp^3d^3 ; pentagonal, bipyramidal BA = 72° BA = 90°

If Electronegativity of bonded atom increases central atom remains same, bond angle decreases.

a.



bp – bp rep ↑ B.A

lp – bp rep ↓ B.A

BA = 107.5°

lp – bp >> bp – bp

BA = 105°

b.

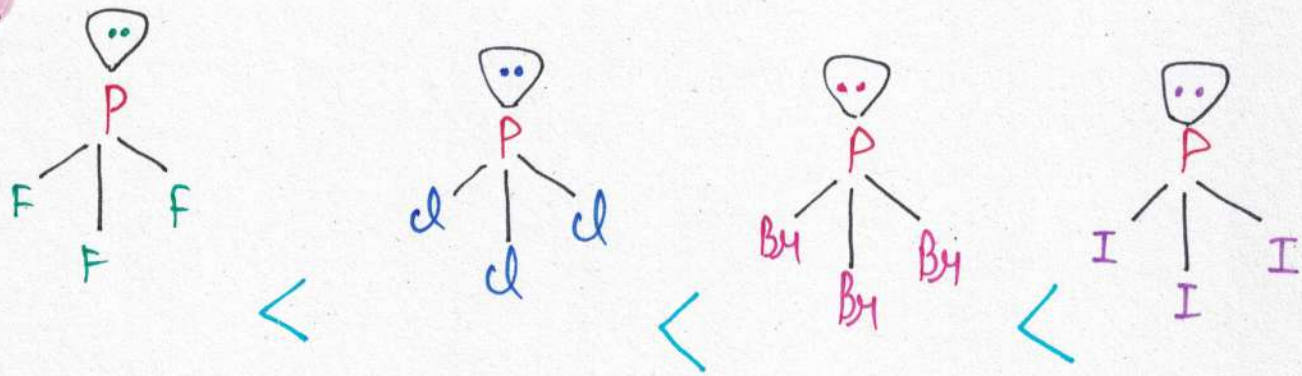
EN ∝ 1/BA

Bonded atom

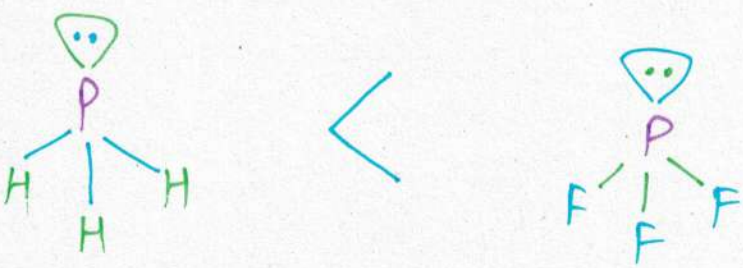
c.

Backbonding

d.



e.



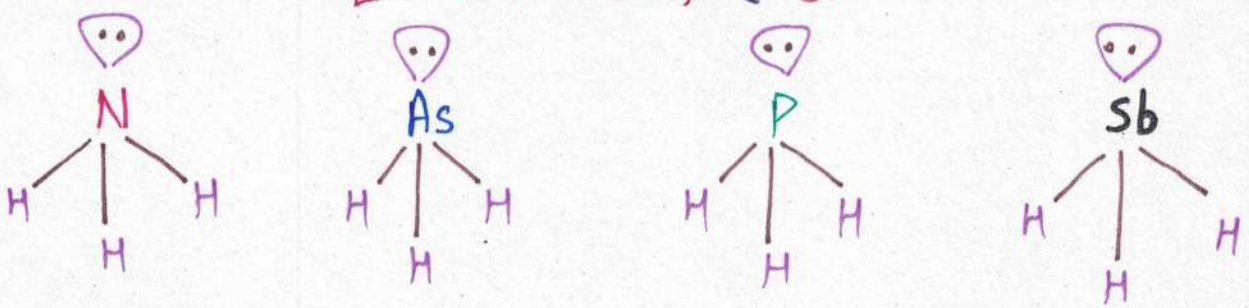
BA = 90°

pure p orbitals participate in bonding

sp^3 hybridisation

If electronegativity of central atom decreases bonded atom remains same, bond angle decreases.

EN central atom $\propto$ BA

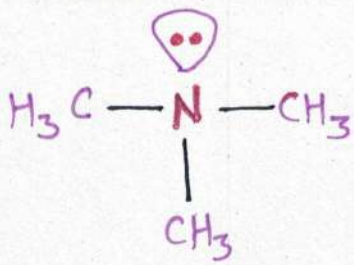


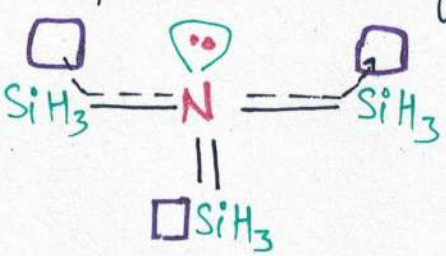
$NH_3 \gg PH_3 > AsH_3 > SbH_3$

107.5° 91° 90° 90°

down the group BA decreases

If central atom have lone pair of e^- and bonded have vacant orbitals lone pair does not participates in hybridisation.





BA < 109°28'

$3\sigma + 1lp \rightarrow 4$

sp^3 hybridisation

pyramidal shape

Backbonding Occurs

$3\sigma \rightarrow sp^2$

trigonal planar

(Si has vacant d-orbitals)

BOND PARAMETERS

BOND LENGTH

Internuclear distance between 2 bonded atoms in a molecule is k/as bond length.

$$d_{A-B} = r_A + r_B - 0.09\,|X_A - X_B|$$

r_A = radius of atom A
r_B = radius of atom B
X_A = EN of atom A
X_B = EN of atom B

Factors affecting Bond Length :-

1. ELECTRONEGATIVITY

$$\text{Bond Length} \propto \frac{1}{\Delta EN}$$

As electronegativity diff. increase BL decrease.

e.g:-

1. CF < C-Cl < C-Br < C-I
2. H-F < H-Cl < H-Br < H-I
3. N-H > O-H > F-H

BL

2. ATOMIC SIZE (Size of Atom)

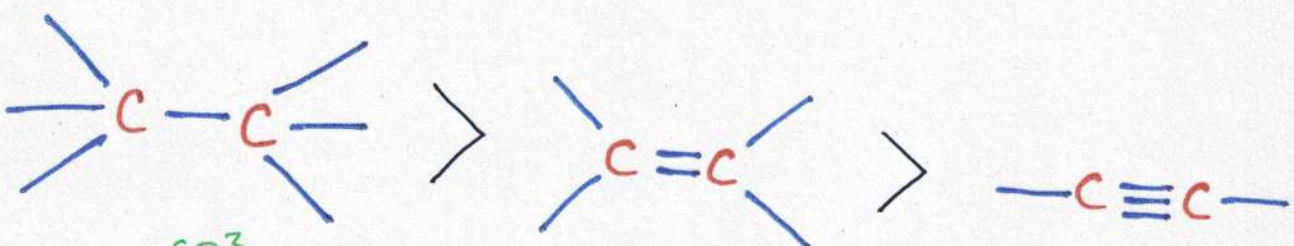
Bond Length ∝ Atomic Size

1. C-C < Si-Si < Ge-Ge < Sn-Sn
2. O-O < S-S < Se-Se < Te-Te
3. N-N < P-P < As-As < Sb-Sb
4. F-F < Cl-Cl < Br-Br < I-I

3. % S CHARACTER IN HYBRID ORBITALS

$$BL \propto \frac{1}{\%\ S\ character\ in\ hybrid\ orbitals}$$

$sp^3 > sp^2 > sp$ [B.L]



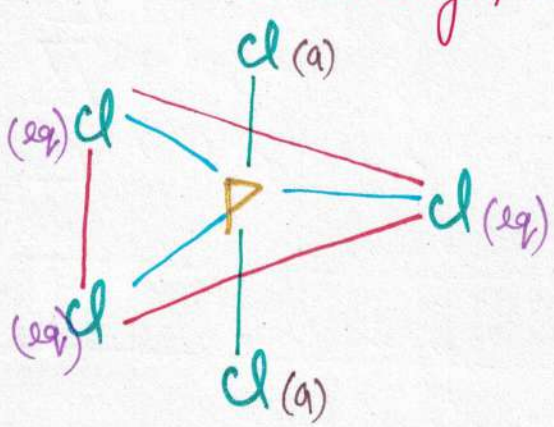
$CH_3-\overset{b}{C}\equiv CH_2$ $\quad$ $CH_3-CH_2-\overset{a}{C}H_3$ $\quad$ $CH_3-\overset{c}{C}\equiv C-H$

$a > b > c$

$sp^3 > sp^2 > sp$ [B.L]

4. BOND ANGLE

$$Bond\ Length \propto \frac{1}{Bond\text{-}angle}$$

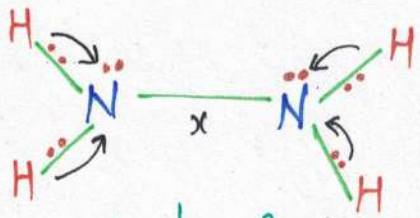


$P\text{-}Cl_{eq} > PCl_{ax}$ — Bond angle

$P\text{-}Cl_{eq} < P\text{-}Cl_{axes}$ Bond angle

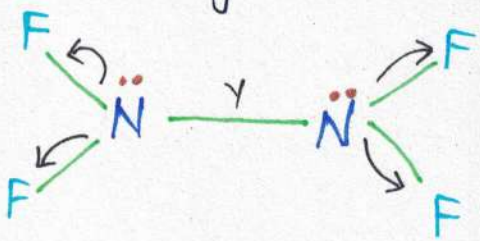
5. If lone pair present on central atom, lone pair – lone pair repulsion occurs Therefore, bond-Length increases.

1.



Hydrazine

BA will be increase

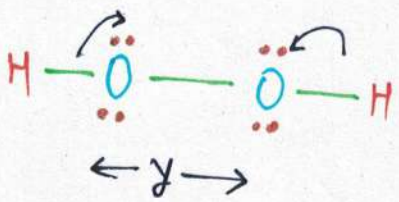


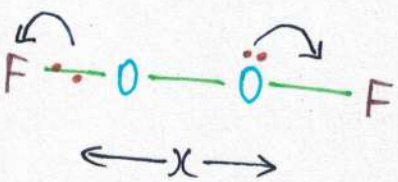
BA will be decrease

$x < y$

BL of N_2H_4 < BL of N_2F_4

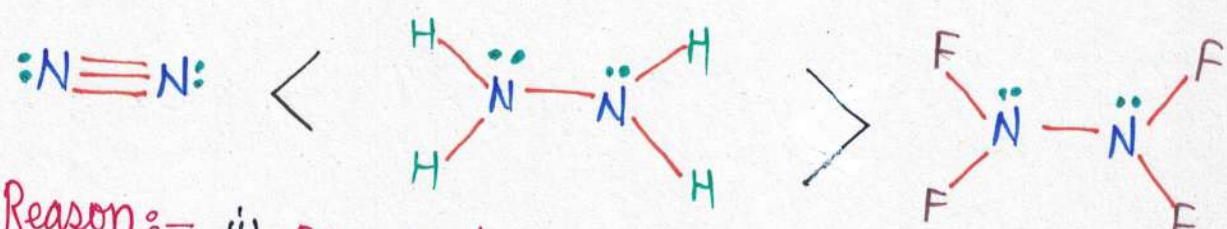
2.





BL $y > x$

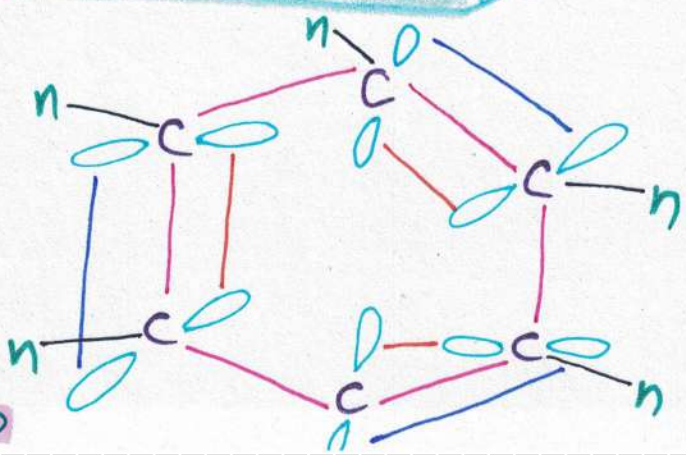
3. N_2 , N_2H_4 , N_2F_4

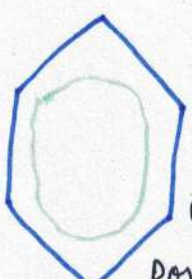


Reason :– (i) BL $\propto \frac{1}{\text{Bond Length}}$

(i) lp-lp , bp-bp repulsion ↑↑ ses BL

6. RESONANCE





Benzene

only π e⁻ change position in reasonable structure

If Molecule shows Resonance, BZ changes BL of σ bond / single bond decreases because some pie character develops in between 2 atoms whereas BL of double bond increases because pie charact decreases

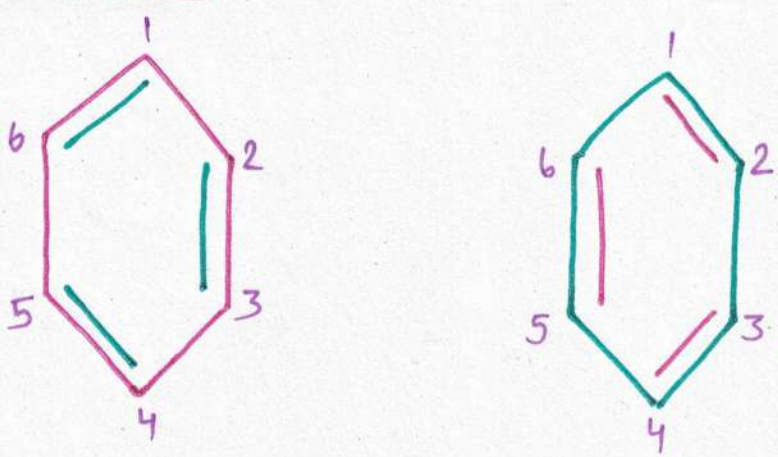
e.g:- in **Benzene** all bond length are equal and all bonds are identicle.

$$BL \propto \frac{1}{\text{Bond Order}}$$

$$\text{Border Order} = \frac{\text{Total no. of bonds}}{\text{Total no. of resonating structures}}$$

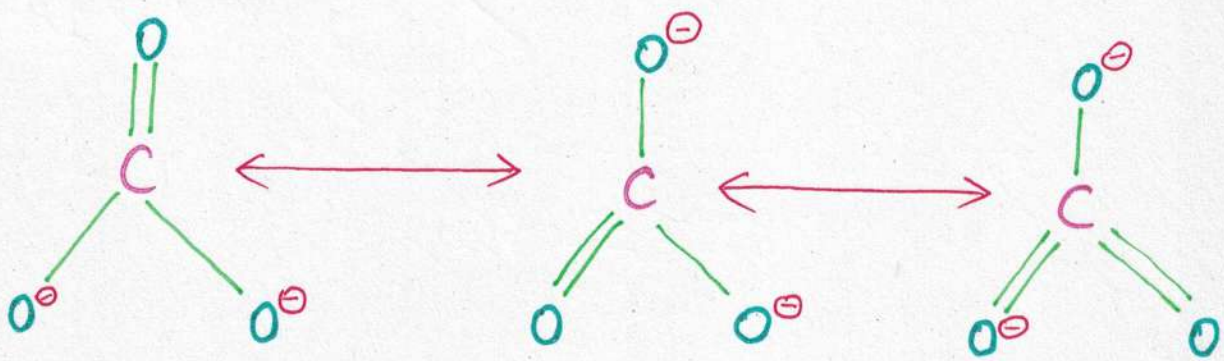
e.g:-

1. BENZENE



1 and 2 $\quad BO = \frac{2+1}{2} = 1.5$

2. CARBONATE



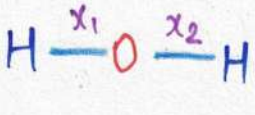
$BO = 4/3$

BOND ENERGY

Energy required to break bond between two bonded atom in a molecule is known as **bond dissociation energy** or energy released when 2 atoms combines and forms molecule is known as **bond formation energy**.

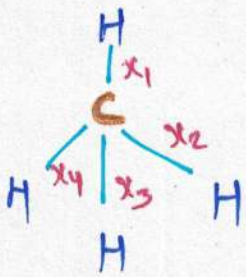
Magⁿ of bond formation energy and bond dissociation energy is same but their sign is opposite.

Bond energy = Bond dissociation energy

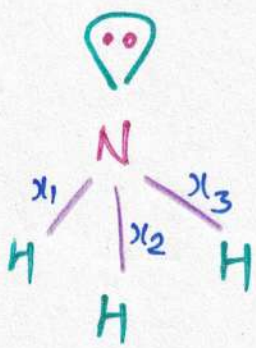


bond energy = av. bond dissociation energy

$$= \frac{\Sigma \text{ bde}}{\text{no. of bonds}}$$



$$be = \frac{x_1 + x_2 + x_3 + x_4}{4}$$



$$be = \frac{x_1 + x_2 + x_3}{3}$$

Factors Affecting bond energy

1. BOND ORDER

Bond energy $\propto$ bond order

eg:- C–C < C=C < C≡C

2. BOND LENGTH

$$B.E \propto \frac{1}{B.L}$$

C-F > C-Cl > C Br > C-I

3. SIZE OF ATOM

As atomic size increases, bond length increases, bond energy decreases.

C-C > Si-Si > Ge-Ge > Sn-Sn

4. If lone pair of e⁻s present on bonded atom of central atom, lp-lp repulsion decreases bond strength.
Less amount of energy is required to break this type of bond.

- $\underset{2}{N-N} < \underset{1}{P-P} > \underset{3}{As-As} > \underset{4}{Sb-Sb}$
- $\underset{2}{O-O} < \underset{1}{S-S} > \underset{3}{Se-Se} > \underset{4}{Te-Te}$
- $\underset{3}{F-F} < \underset{1}{Cl-Cl} > \underset{2}{Br-Br} > \underset{4}{I-I}$
- $Cl_2 > Br_2 > F_2 > I_2$

Question:- What is the correct order of bond energy of F_2, O_2 N_2 & H_2.

$N \equiv N$ (N_2) $\quad O=O$ (O_2) $\quad F-F$ (F_2) $\quad H-H$ (H_2)

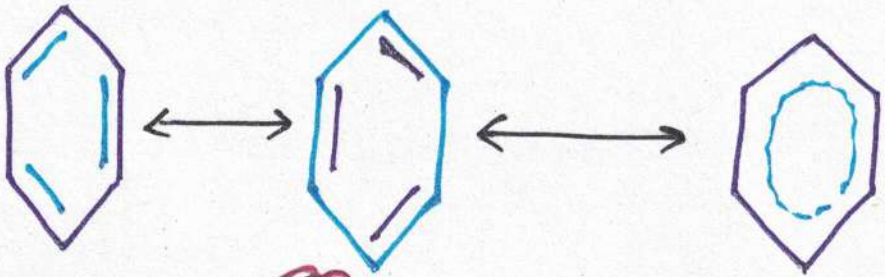
Ans:- $N_2 > O_2 > H_2 > F_2$

5. RESONANCE

Due to Resonance, generally all bond are identical in molecule, therefore, bond energy is also identical.

$:N \overset{x}{\equiv} N: \overset{y}{\rightarrow} O$ BE $x > y$

$\ominus N = N = O$ BL $x < y$



BE is same.

Angle between 2 covalent bond is known as bond-angle.

Factor Affecting Bond angle

1. HYBRIDISATION

As % S-character increase in hybridisation BA increases.

$sp \rightarrow 180°$ $sp^2 \rightarrow 120°$ $sp^3 \rightarrow 109°28'$

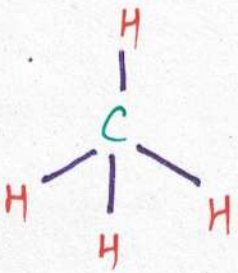
2.

If hybridisation of central atom, remains same, as no. of lone pair increases on central atom, BA decreases.

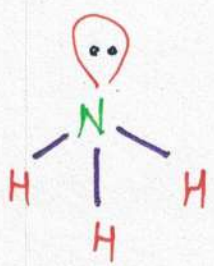
lp-lp > lp-bp > bp-bp repulsion

sp^3

CH_4 > NH_3 > H_2O

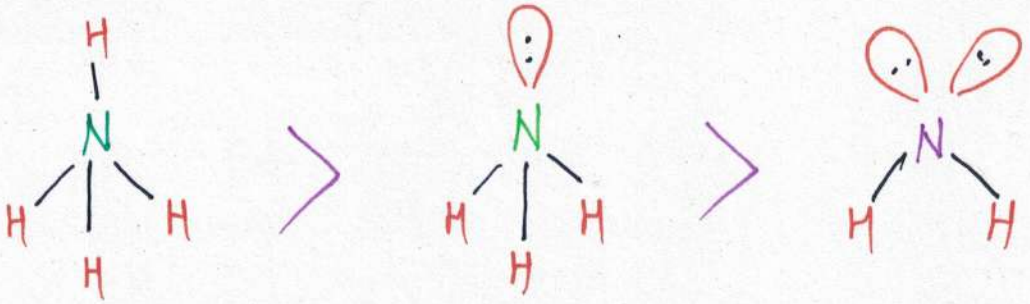


bp-bp



lp-bp > bp-bp

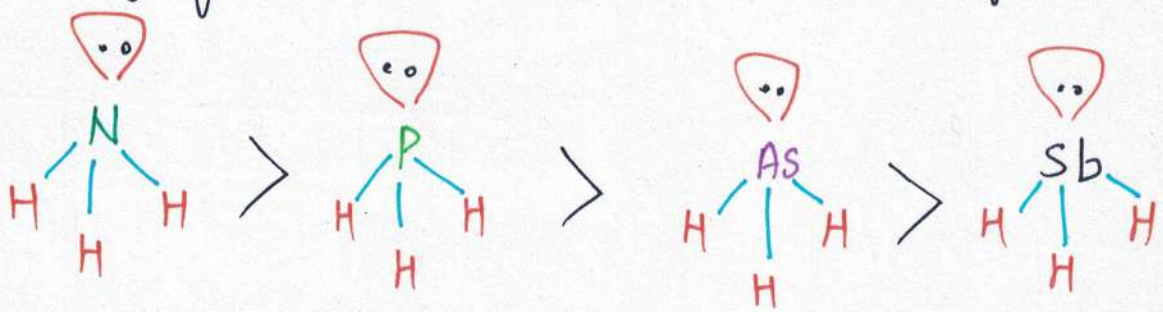
lp-lp > lp-bp > bp-bp



Ammonium ion | Ammonia | Amide

109°28' | 107.5° | 104.5°

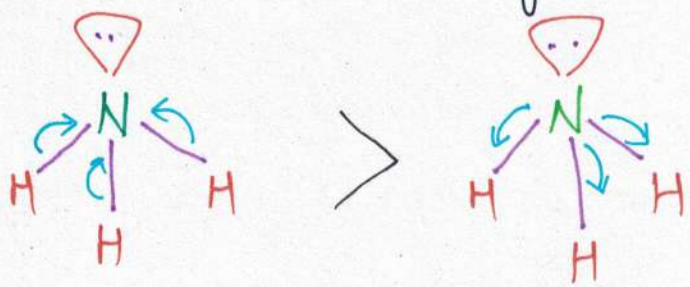
3. Hybridisation of central atom remains same no. of lone pair present on central atom also remain same. as electronegativity of central atom decreases, bond angle abo decreases.



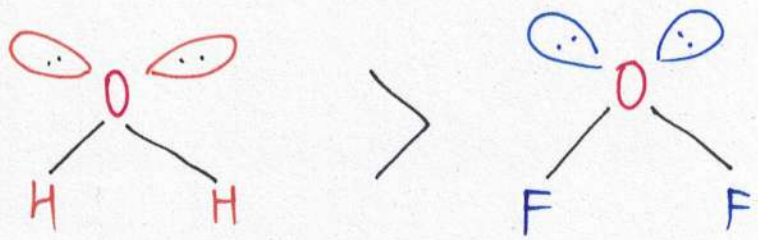
BA = 107.5° | BA = 91° | BA = 90° | BA = 89°

4. As hybridisation of central atom remain same, same no. of lone pair present on C.A. as electronegativity of bonded atom increases, bond angle decreases.

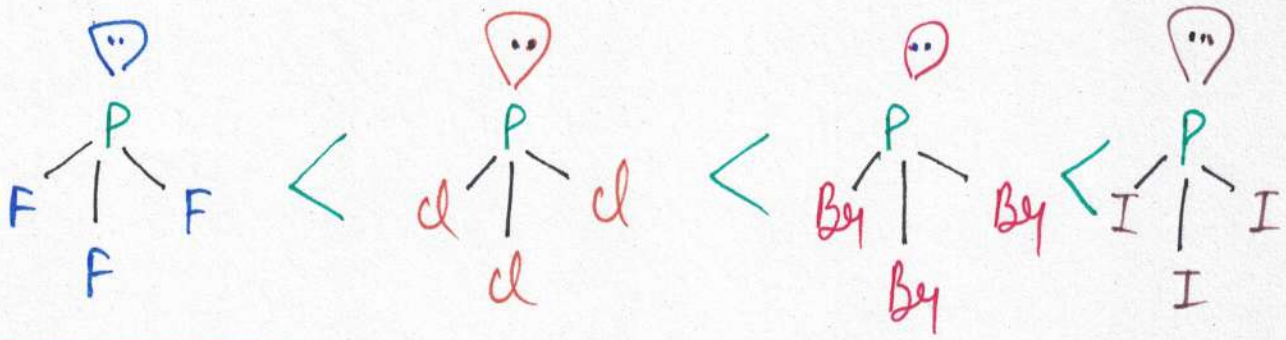
1.



2.



3.

4. 

5. SIZE OF BONDED ATOM

As size of bonded atom increases bond angle increases

PF_3 PCl_3 PBr_3 PI_3

$(x_1 = x_2) > x_3$

$\pi bp - \pi bp > \pi bp - \sigma bp > \sigma bp - \sigma bp$

$\theta_4 > \theta_3 > \theta_2 > \theta_1$

MOLECULAR ORBITAL THEORY [MOT]

Some properties of molecule cannot be explained by valence bond theory (VBT). Like oxygen is paramagnetic in nature and blue coloured liquid. According to VBT oxygen is diamagnetic in nature.

1. MOT was proposed by Mulliken.

2. It is based on Quantum Mechanics.

3. According to this theory, each electron participates in bond-ing.

4. Number of atomic orbitals participating in bonding is equal to no. of molecular orbitals

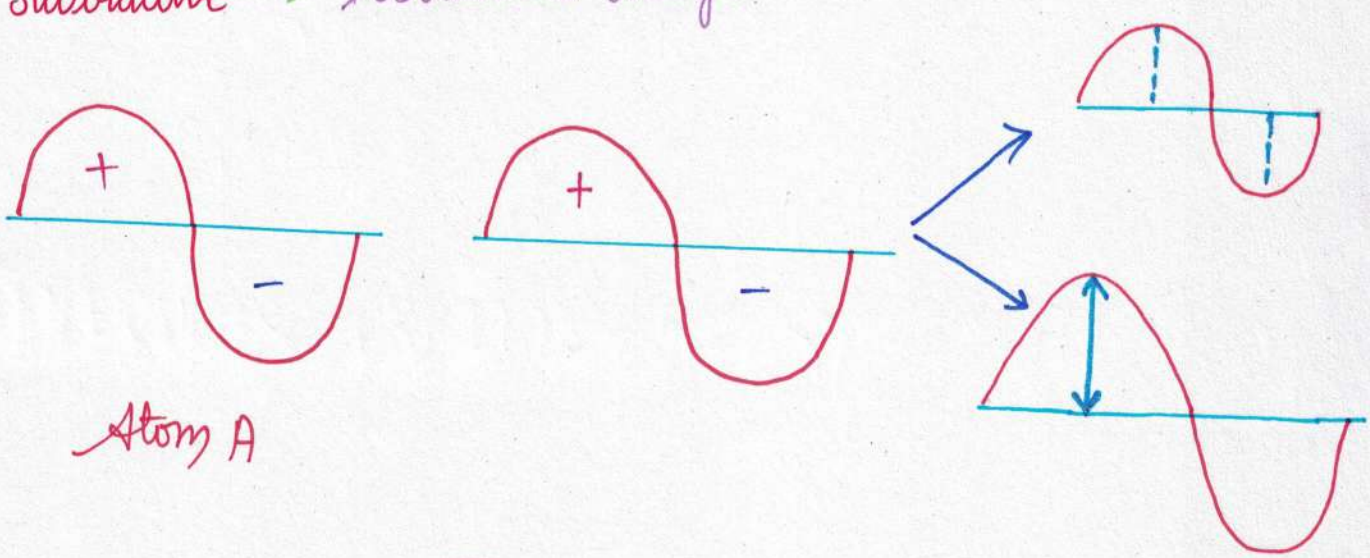
5. Half of the atomic orbitals forms bonding molecular orbital. energy of bonding molecular orbital is Lower than atomic orbital.

6. Half of the atomic orbital forms anti-bonding molecular orbi-tal. energy of anti-bonding molecular orbital is more than atomic orbital.

7. When two atomic orbitals overlaps each other at internuclear axis or perpendicular to internuclear axis, 2 types of combi-nation occurs:-

a. Additive combination b. Subtractive combination

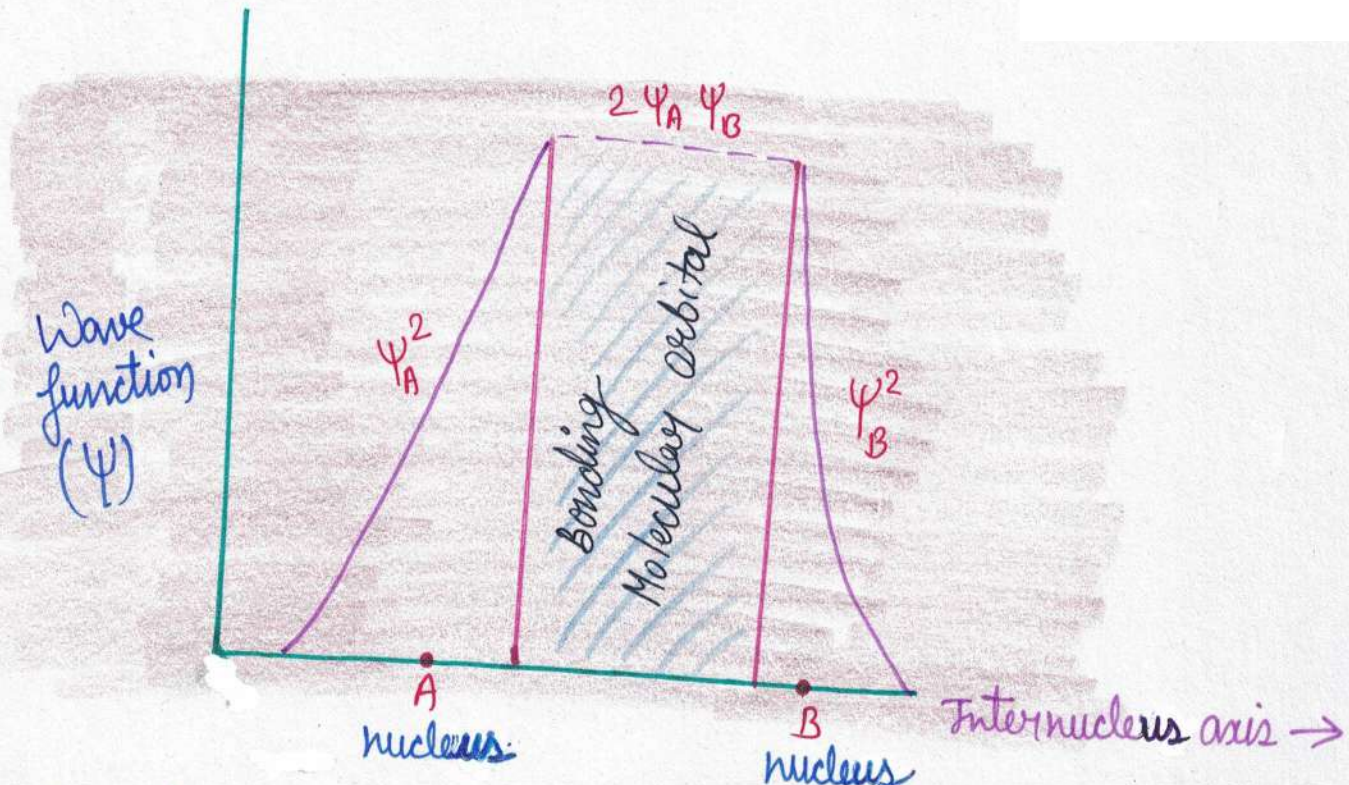
Additive → Constructive Interference
Subtractive → Destructive Interference



Bonding Molecular Orbital

$\Psi \rightarrow$ Psi

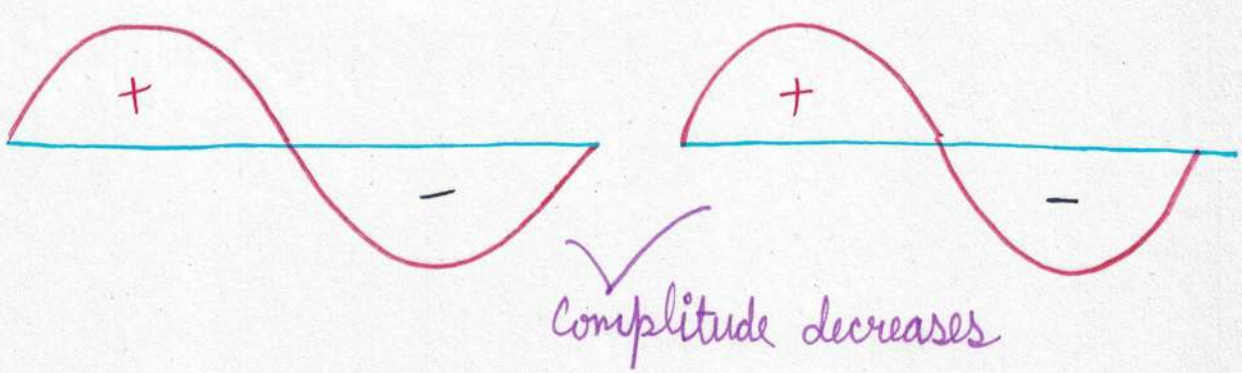
$(\Psi_A + \Psi_B)^2 = \Psi_A^2 + \Psi_B^2 + 2\Psi_A \Psi_B$



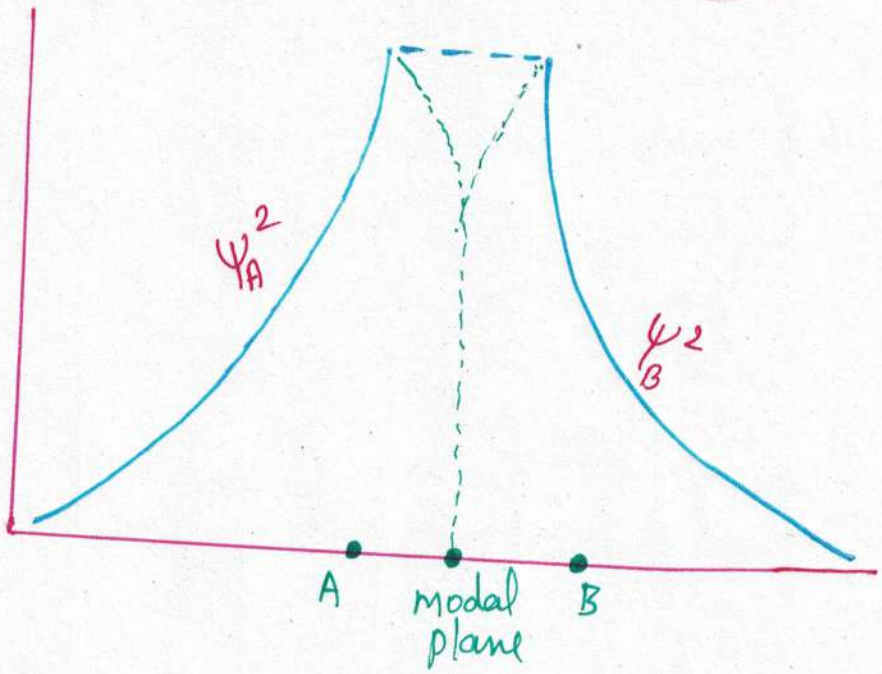
Wave function of atom A → Ψ_A

Wave function of atom B → Ψ_B

SUBTRACTIVE COMBINATION



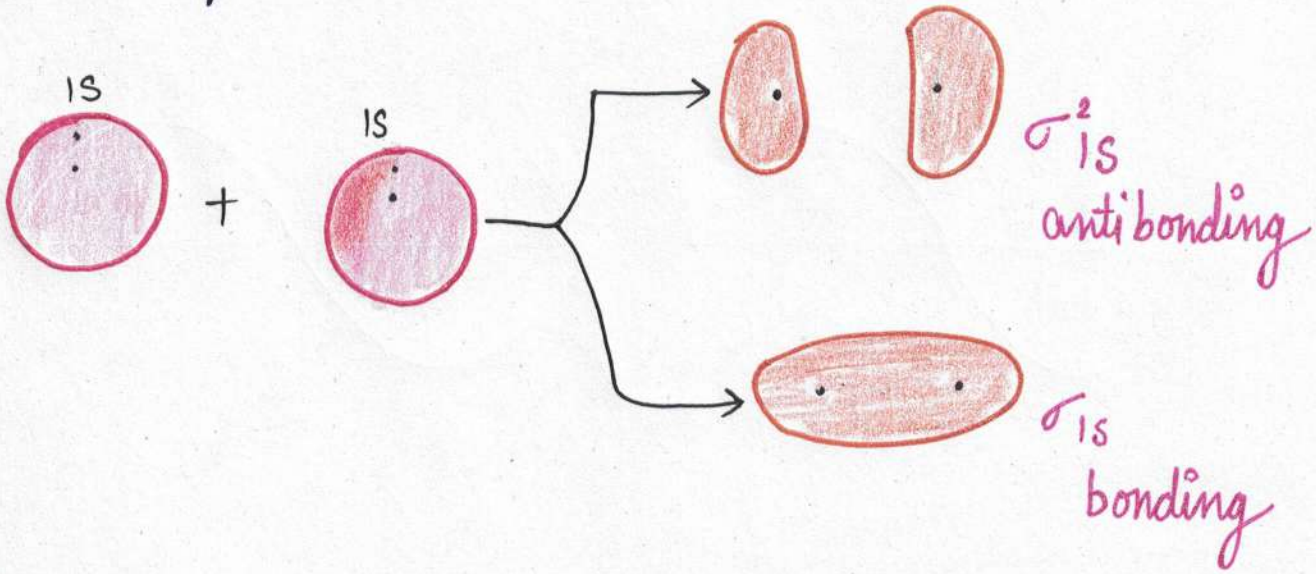
ANTI BONDING MOLECULAR ORBITAL

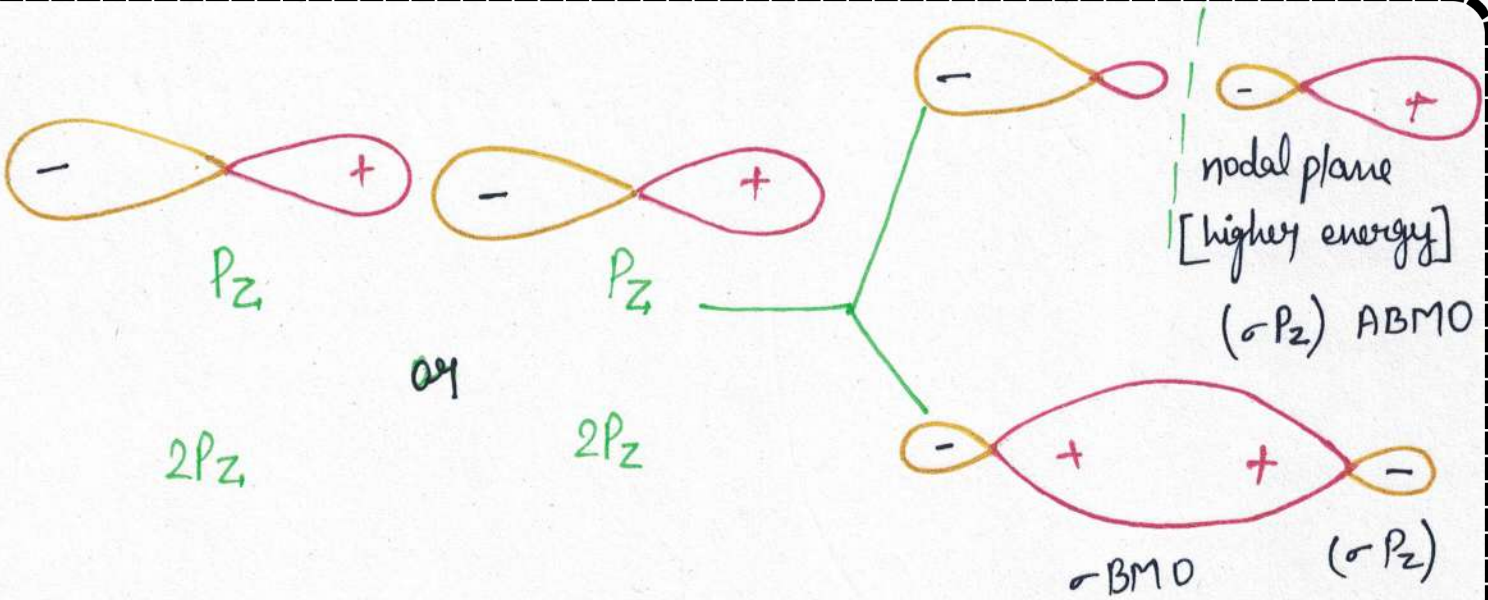


SIGMA MOLECULAR ORBITAL

When 2 half filled atomic orbitals overlap each other at internuclear axis (same phase - crest, crest / trough, trough) sigma bonding molecular / orbital is formed.

When 2 half filled atomic orbitals overlap each other (different-phase) at internuclear axis, sigma antibonding molecular orbital is formed.

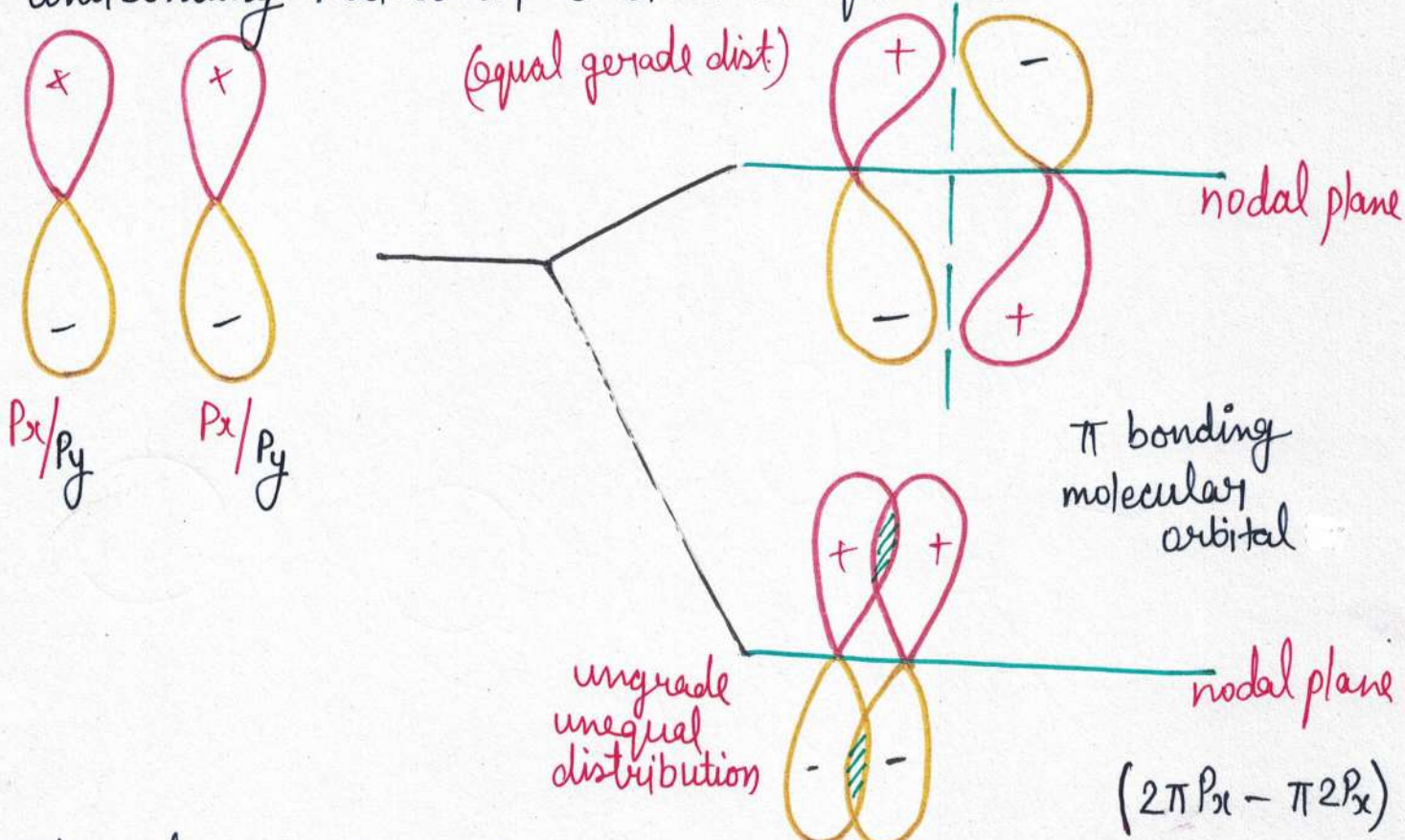




PIA MOLECULAR ORBITALS

When 2 half filled atomic orbitals overlap each other at perpendicular to internuclear axis (same phase) pie bonding molecular orbital is formed.

when 2 half filled atomic orbitals overlap each other at perpendicular to internuclear axis (different phase) pie antibonding molecular orbital is formed.



Internuclear axis is nodal plane in both cases.

ABMO has 2 nodal plane

BMO has 1 nodal plane

π ABMO has higher energy than atomic orbital.

π BMO has lower energy than atomic orbital.

Gerade :- symmetrical / equal distribution

(g) (even).

σ BMO are gerade, π ABMO are gerade.

e.g:- $\sigma 1s$, $\sigma 2s$, $\sigma 2p_z$, $\pi 2p_x$, $\pi 2p_y$.

Ungerade :- asymmetrical / unequal dist.

(ug) (odd)

eg:- σ^*, 1s, $\sigma^* 2s$, $\sigma^* 2p_z$, $\pi 2p_x$, $\pi 2p_y$

ELECTRON FILLING ORDER

Electron first goes to lower energy state and then enters into higher energy state electron filling order follows Afbau's principle.

A. Homoatomic - diatomic molecule

A. MORE THAN 14e⁻ SYSTEM

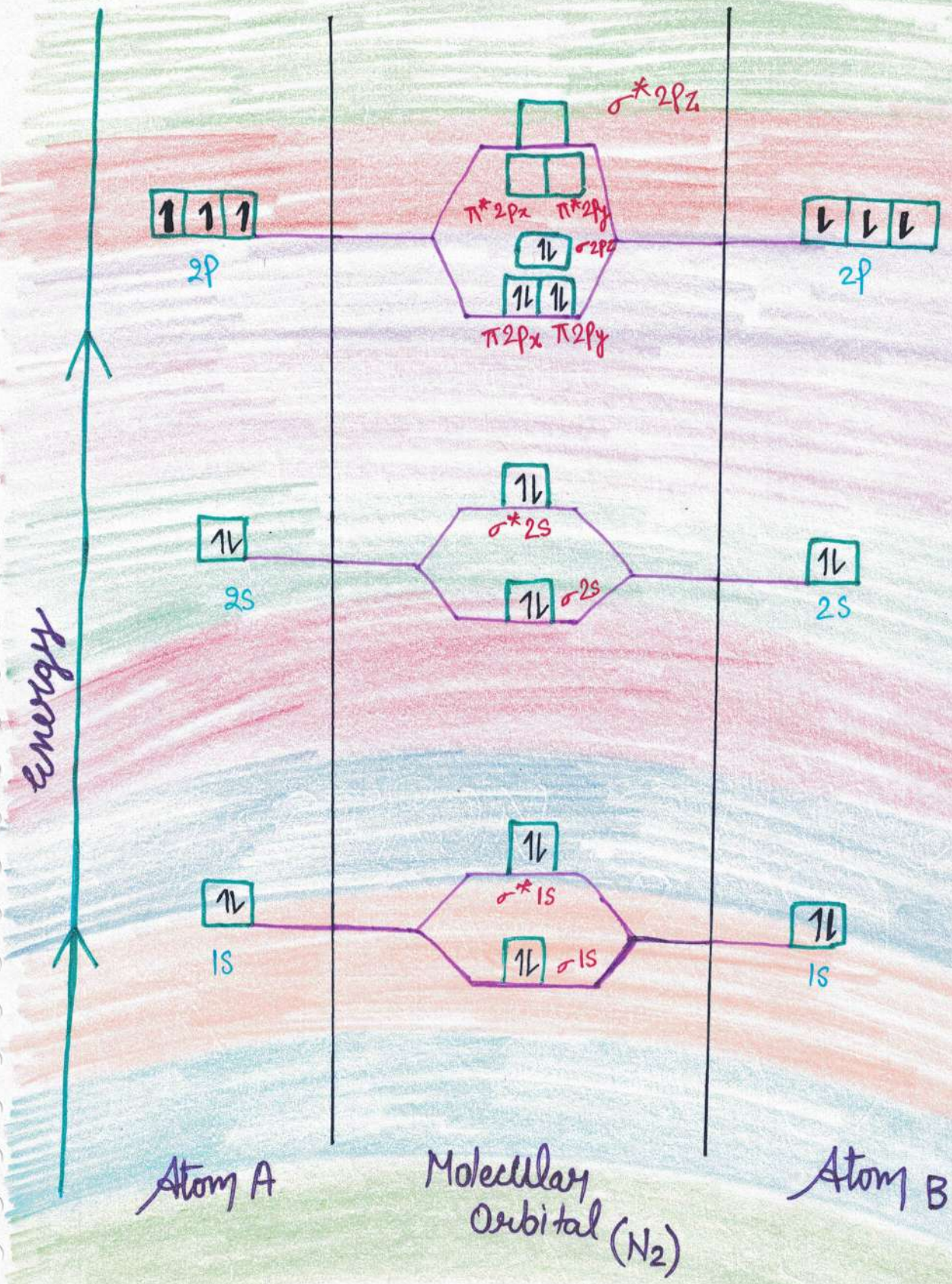
Order :- $\sigma 1s$, $\sigma^* 1s$, $\sigma 2s$, $\sigma^* 2s$, $\sigma 2p_z$, $\pi 2p_x = \pi 2p_y$

$\pi^* 2p_x = \pi^* 2p_y$, $\sigma^* 2p_z$

eg:- (a.) O_2

$= 2 \times 8 = 16\ e^-$

$\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$, $\sigma 2p_z^2$, $\pi 2p_x^2 = \pi 2p_y^2$, $\pi^* 2p_x^1 = \pi^* 2p_y^1$



ENERGY DIAGRAM

Bond Order = $\frac{1}{2}[N_b - N_a]$

$\frac{1}{2}[10-6] = \frac{4}{2} = 2$

(0-0)

O_2 is paramagnetic in nature because 2 unpaired e^- are present in $\pi^* 2p_x$ and $\pi^* 2p_y$ molecular orbital.

$N_b \rightarrow$ no. of bonding orbitals electrons

$N_a \rightarrow$ no. of bonding orbitals electrons

b. O_2^{+1} [Oxygenyl]

no. of $e^- = 15$

B.O $= \frac{1}{2}[10-5] = 2.5$

Order :- $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma^* 2p_x^1$

Paramagnetic

c. O_2^{-1} [Superoxide]

no. of $e^- = 17$

Order :- $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2$

B.O $= \frac{1}{2}[10-7] = \frac{3}{2} = 1.5$

d. O_2^{-2} [Peroxide]

no. of $e^- = 18e^-$

Order:- $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma^* 2p_x^2 = \sigma^* 2p_y^2$

B.O $= \frac{1}{2}[10-8] = 1$

Diamagnetic

A. BORDER ORDER

$O_2^{+1} > O_2 > O_2^{-1} > O_2^{-2}$

2.5 2 1.5 1

Para para para Dia

B. BOND ENERGY/BOND STRENGTH

Bond Energy $\propto$ Bond Order

B.S = $O_2^{+1} > O_2 > O_2^{-1} > O_2^{-2}$

C. BOND LENGTH

Bond Length $\propto \frac{1}{\text{Bond Order}}$

$O_2^{-2} > O_2^{-1} > O_2 > O_2^{+1}$

(e.) F_2 Molecule

$e = 9 \times 2 = 18$

$B.O = \frac{1}{2}[10-8] = \frac{2}{2} = 1$

Order:- $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\sigma 2pz^2$ $\pi 2px^2 = \pi 2py^2$ $\pi^* 2px^2 = \pi^* 2py^2$

Diamagnetic

(f.) Ne_2

20 e^-

$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2,$
$\pi^* 2p_x^2 = \pi^* 2p_y^2, \sigma^* 2p_z^2$

B.O = $\frac{1}{2}[10-10] = 0$

Ne_2 does not exist

B. LESS THAN OR EQUAL TO 14e^- SYSTEM

$\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s, \sigma 2p_z, \pi 2p_x = \pi 2p_y$
$\pi^* 2p_x = \pi^* 2p_y, \sigma^* 2p_z$

$\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s, \pi^2 2p_x = \pi 2p_y, \sigma 2p_z,$
$\pi^* 2p_x = \pi^* 2p_y, \sigma^* 2p_z$

(a.) N_2

14e^-

$\sigma 1s^2 \ \sigma^* 1s^2 \ \sigma 2s^2 \ \sigma^* 2s^2 \ \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_y^2, \sigma 2p_z^2$

B.O = $\frac{1}{2}[10-4] = 3$

Diamagnetic

(b.) N_2^{+1} 13e^- B.O = 2.5

(c.) N_2^{+2} 12e^- B.O = 2

(d.) N_2^{-1} 15e^- B.O = 2.5

(e.) N_2^{-2} 16e^- B.O = 2

in $14e^-$ system, if B.O of e^- increase or Decrease by 1, B.O always decrease by 0.5

$N_2 \rightarrow 14e^-$	:N≡N:	B.O = 3
$N_2^{+1} \rightarrow 13e^-$		B.O = 2.5
$N_2^{+2} \rightarrow 12e^-$		B.O = 2
$N_2^{-2} \rightarrow 16e^-$		B.O = 2

Stability Order :- $N_2 > N_2^{+1} > N_2^{-1} > N_2^{+2} > N_2^{-2}$

as no. of ABMO / $AB e^-$ increase, stability decrease
(if B.O remains same)

as $AB e^-$ are of higher energy.

Bond energy $\propto$ stability $\propto$ bond Order

B.E = $N_2 > N_2^{+1} > N_2^{-1} > N_2^{+2} > N_2^{-2}$

$12e^-$

$\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$, $\pi 2p_x^2 = \pi 2p_y^2$

B.O = $\frac{1}{2}[8-4] = \frac{4}{2} = 2$

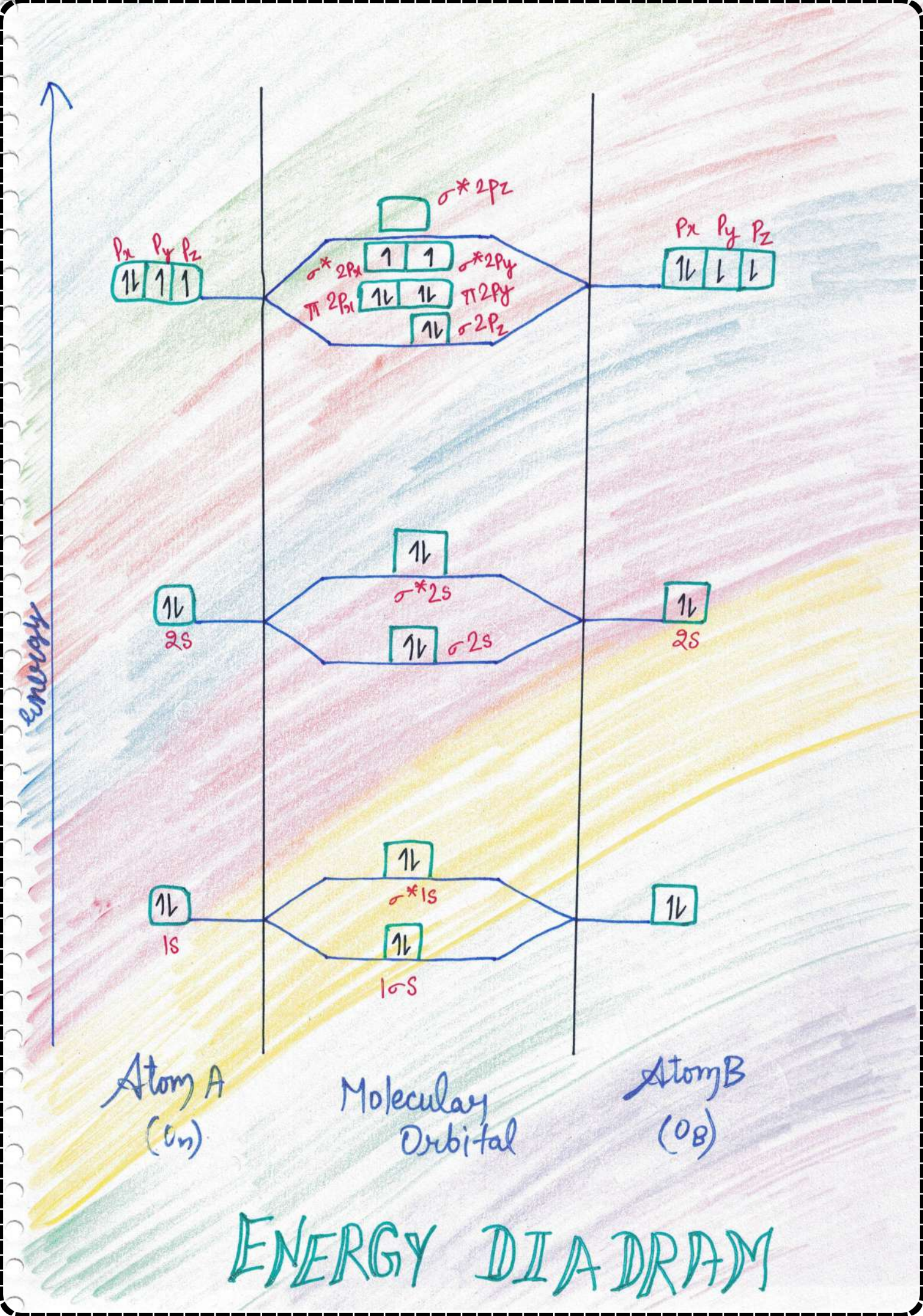
Diamagnetic Nature

$10e^-$

$\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$

B.O = $\frac{1}{2}[4-4] = 0$

does not exist



ENERGY DIADRAM

Li_2

$6e^-$

$1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$

Diamagnetic

$B.O = \frac{1}{2}[4-2] = \frac{2}{2} = 1$

He_2

$4e^-$

$1s^2$, $\sigma^* 1s^2$

Diamagnetic does not exist

$B.O = \frac{1}{2}[2-2] = 0$

Question :- if SP mixing is not valid, what is the B.O and magnetic behaviour of C_2, N_2, and B_2.

C_2

$12e^-$

$\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$, $\sigma 2p_z^2$, $\pi 2p_x' = \pi 2p_y'$

$B.O = \frac{1}{2}[8-4] = \frac{4}{2} = 2$

Nature → Paramagnetic

N_2

$14e^-$

$\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$, $\sigma 2p_z^2$, $\pi 2p_x^2 = \pi 2p_y^2$

$B.O = \frac{1}{2}[10-4] = \frac{6}{2} = 3$

Diamagnetic

B_2

$10e^-$

$\sigma 1s^2$, $\sigma^* 1s^2$, $\sigma 2s^2$, $\sigma^* 2s^2$, $\sigma 2p^2$

$B.O = \frac{1}{2}[6-4] = \frac{2}{2} = 1$ Diamagnetic

QUESTION If SP mixing is not valid, then which of the following occur in paramagnetic nature.

a. B_2 b. C_2 c. N_2 d. Li_2

ANSWER b. C_2

QUESTION Which of the following occurs as paramagnetic in nature?

a. B_2 b. C_2 c. N_2 d. Li_2

H_2

$2e^-$

$\sigma 1s^2$

B.O = $\frac{1}{2}[2] = 1$

Diamagnetic Nature

H_2^{+1}

$2e^- - 1e^- = 1e^-$

$\sigma 1s^1$

B.O = $\frac{1}{2}[1-0] = \frac{1}{2}$

Paramagnetic Nature

H_2^{-1}

$2e^- + 1e^- = 3e^-$

$\sigma 1s^2, \sigma^* 1s^1$

B.O = $\frac{1}{2}[2-2] = \frac{1}{2}$

Paramagnetic Nature

QUESTION Stability Order of H_2, H_2^{-1}, H_2^{+1}.

ANSWER $H_2 > H_2^{+1} > H_2^{-1}$

H_2^{-1} has AB e^-

REASON:- As Antibonding e^- increases stability decreases.

QUESTION Correct increasing order of stability of B_2, Li_2 & H_2 is

Ans. $H_2 > B_2 > Li_2$

	H_2	B_2	Li_2
Be^-	2	6	4
ABe^-	0	4	2

Reason:- As no. of bonding e^- increases stability increases.

First Prefer Bonding & then Antibonding

HeH^+

$e^- = 2e^-$

$B.O = \frac{1}{2}[2] = 1$

HeH^-

$e^- = 4e^-$

$B.O = \frac{1}{2}[2-2] = 0$ does not exist

B. Hetroatomic molecule

1. Cyanide ion (CN^-)

Total no. of $e^- = 6+7+1 = 14e^-$

$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_z^2$

$B.O = \frac{1}{2}[10-4] = \frac{6}{2} = 3$

diamagnetic Nature

2. CO

$e^- = 6+8 = 14e^-$

$\sigma 1s^2 \quad \sigma^* 1s^2 \quad \sigma 2s^2 \quad \sigma^* 2s^2 \quad \pi 2p_x^2 = \pi 2p_y^2, \quad \sigma 2p_z^2$

$B.O = \frac{1}{2}[10-4] = 6/2 = 3$

Diamagnetic Nature

3. CO^+

$e^- = 14 - 1e^- = 13e^-$

$\sigma 1s^2, \quad \sigma^* 1s^2, \quad \sigma 2s^2, \quad \sigma^* 2s^2, \quad \sigma 2p_z^2, \quad \pi 2p_x^2 = 2p_y^2$

$B.O = \frac{1}{2}[9-4] = 5/2 = 2.5$

	CO	CO^+
B.O	3	2.5
B.L	e.O < CO^+	

But experimentally B.L CO^+ < B.L CO

it means B.O CO^+ > B.O CO

CO^+

$\sigma 1s^2 \quad \sigma^* 1s^2 \quad \sigma 2s^2 \quad \pi 2p_x^2 = \pi 2p_y^2 \quad \sigma 2p_z^2, \quad \sigma^* 2s'$

$B.O. = \frac{1}{2}[10-3] = 3.5$

{$\sigma 2s$ & $\sigma^* 2s$}

NO

$e^- = 7+8 = 15e^-$

$\sigma 1s^2, \quad \sigma^* 1s^2, \quad \sigma 2s^2, \quad \sigma^* 2s^2, \quad \pi 2p_x^2 = \pi 2p_y^2, \quad \sigma 2p_z^2, \quad \sigma^* 2p_z'$

$B.O = \frac{1}{2}[10-5] = 2.5$

Paramagnetic Nature

QUESTION IP_{CO} — IP_{CO^+}

ANSWER $IP_{CO} > IP_{CO^+}$

unpaired e^- present in CO^+ ($\sigma^* 2s$)

NO^+

$e^- = 7+8-1 = 14e^-$

$\sigma 1s^2,\ \sigma^* 1s^2,\ \sigma 2s^2,\ \sigma^* 2s^2,\ \pi 2p_x^2 = \pi 2p_y^2,\ \sigma 2p_z^2$

$B.O = \frac{1}{2}[10-4] = 3$ Diamagnetic Nature

Question IP_{N_2} —— IP_{O_2}

Answer $IP_{N_2} > IP_{O_2}$

As O_2 has unpaired e^- in π^* orbitals.

BN

$e^- = 5+7 = 12e^-$

$\sigma 1s^2\ \ \sigma^* 1s^2\ \ \sigma 2s^2\ \ \sigma^* 2s^2\ \ \pi 2p_x^2 = \pi 2p_y^2$

$B.O = \frac{1}{2}[8-4] = \frac{4}{4} = 2$ Diamagnetic Nature

Question Correct increasing order of O-O B.L in O_2, KO_2, K_2O_2 & $O_2[PtF_6]$ is.....

Answer

O_2 $\quad B.O = \frac{1}{2}[4] = 2$

$O_2 > K_2O > K_2O_2 > O_2[PtF_6]$

$O_2 = 16e^-$	$B.O = 2$
$K^+O_2^{-1} = 17e^-$	$B.O = 1.5$
$2K^+O_2^{+2} = 18e^-$	$B.O = 1.0$
$O_2^+[PtF_6] = 15e^-$	$B.O = 2.5$

$B.L \propto \frac{1}{B.O}$

$B.O. = O_2[PtF_6] < O_2 < O_2^- < O_2^{-2}$

$O_2[PtF_6] < O_2 < KO_2 < K_2O_2$

FOR $14e^-$ SYSTEM

For $14e^-$ system → B.O = 3

if One e^- increase or decreas by $14e^-$ system B.O. B.O always decreases by 0.5/e^-

$13e^- \rightarrow 2.5$
$12e^- \rightarrow 2$
$11e^- \rightarrow 1.5$
$10e^- \rightarrow 1$
$9e^- \rightarrow 0.5$
$8e^- \rightarrow 0$

$15e^- \rightarrow 2.5$
$16e^- \rightarrow 2$
$17e^- \rightarrow 1.5$
$18e^- \rightarrow 1$
$19e^- \rightarrow 0.5$
$20e^- \rightarrow 0$

Exception → Co^+ has B.O greater than 3.
i-e 3.5